# Algebraic Geometry—Mumford's Red Book

Summer 2026

# 1 Varieties

## 1.1 Some algebra

**Theorem 1** (Noether's Normalization Lemma)**.** *Let $R$ be an integral domain, finitely generated over a field $k$. If $R$ has transcendence degree $n$ over $k$, then there exist elements $x_1, \ldots, x_n \in R$, algebraically independent over $k$, such that $R$ is integrally dependent on the subring $k[x_1, \ldots, x_n]$ generated by the $x_i$'s.*

**Theorem 2** (Going-up theorem of Cohen-Seidenberg)**.** *Let $R$ be a ring (commutative) and $S \subseteq R$ a subring such that $R$ is integrally dependent on $S$. For all prime ideals $P \subseteq S$, there exist prime ideals $P' \subseteq R$ such that $P' \cap S = P$.*

**Lemma 3.** *Let $R$ be a field and $S \subseteq R$ a subring such that $R$ is integrally dependent on $S$. Then $S$ is a field.*

**Theorem 4** (Weak Nullstellensatz)**.** *Let $k$ be an algebraically closed field. Then the maximal ideals in the ring $k[X_1, \ldots, X_n]$ are the ideals*

$$(X_1 - a_1, \ldots, X_n - a_n)$$

*where $a_1, \ldots, a_n \in k$.*

**Example 5.** *Let $A = k[x]$. Then:*

- *$B_1 = A[y]/(y^2 - x)$ is integral over $A$*
- *$B_2 = A[x^{-1}]$ is not integral over $A$ (but is algebraic over $A$)*
- *$B_3 = A[y]$ is not integral over $A$ (not even algebraic)*

*Geometrically, $A$ is the affine coordinate ring of the affine line ($= x - axis$).*

- *$B_1$ is a parabola*
- *$B_2$ is a hyperbola; $A[x^{-1}] \cong k[x, y]/(xy - 1)$ via $y \mapsto x^{-1}$*
  *You can either visualize $B_2$ as the affine line minus the origin, or a hyperbola*
- *$B_3$ is a plane*

*Given $A \subseteq B_i$ (say $A$ is the affine coordinate ring of $X$, and $B_i$ is the affine coordinate ring of $Y_i$), there exists some map $Y_i \to X$ via precomposition. To see this, think of an element of $A$ as a map $X \to k$. Then given $\varphi : A \to B_i$, we get a map $Y_i \to X \to k$ i.e. an element of $B_i$.*

**Remark 6.** *If $k$ is an infinite field, different polynomial functions in $k[x]$ are different maps:*

*Let $f(x) = a_n x^n + \cdots + a_1 x + a_0 \in k[x]$. If $f$ is not the zero polynomial, then at least one $a_i \neq 0$; let $n$ be the highest index such that $a_n \neq 0$.*

*If $f(a) = 0$ for all $a \in k$, then every element of $k$ is a root of $f(x)$. Since $k$ is an infinite field, $f(x)$ has infinitely many roots. But a nonzero polynomial of degree $n$ over a field has at most $n$ roots, so $f$ must have at most $n$ roots in $k$. So $f$ must be the zero polynomial.*

*Then if $f(a) - g(a) = 0$ for all $a \in k$, then $f(a) - g(a)$ must be the zero polynomial, meaning $f(x) = g(x)$.*

**Example 7.** *Again let $A = k[x]$ and*

- $B_1 = A[y]/(y^2 - x)$
- $B_2 = A[x^{-1}]$
- $B_3 = A[y]$

*Say $A$ is the affine coordinate ring of $X$, and $B_i$ is the affine coordinate ring of $Y_i$.*
*Given $A \subseteq B_i$ there exists some map $Y_i \to X$ via precomposition. To see this, think of an element of $A$ as a map $X \to k$. Then given $\varphi : A \to B_i$, we get a map $Y_i \to X \to k$ i.e. an element of $B_i$.*

- *The map $Y_1 \to X$ is surjective and quasi-finite (i.e. only finitely many points map to a given point)*
- *The map $Y_2 \to X$ is not surjective but is quasi-finite*
- *The map $Y_3 \to X$ is surjective but is not quasi-finite*

**Recall:** integral and finite type (= finitely generated as an algebra) $\implies$ quasi-finite.
Finitely generated as an algebra is weaker than finitely generated as a module (e.g. $B_3/A$ is finitely generated as an algebra but not as a module, since you need all powers of $y$. $B_1/A$ is, however, finitely generated as a module.

**Example 8.** *Geometrically, the "Going up" (=Lying over) theorem says:*
*the map $V(B_1) = Y_1 \to V(A) = X$ (parabola to affine line) is surjective, since $B_1$ is integral over $A$.*

**Remark 9.** *If $A \subseteq B$ are domains and $B/A$ is integral, then (say $A = k[Y]$, $B = k[X]$) the induced map $X \to Y$ is surjective.*

**Recall:** $k[x]$ and $\mathbb{Z}$ are the key examples of Euclidean domains, and $\mathbb{Z} \subseteq \mathbb{Q}$ is the key example where an ideal in $\mathbb{Q}$ is maximal but its contraction is not.

## 1.2 Irreducible algebraic sets

**Definition 1.** *A **closed algebraic subset of** $k^n$ is a set consisting of all roots of a finite collection of polynomial equations, i.e.*

$$\{(x_1, \ldots, x_n) \mid f_1(x_1, \ldots, x_n) = \cdots = f_m(x_1, \ldots, x_n) = 0\}.$$

The above set depends only on the ideal $A = (f_1, \ldots, f_m)$ in $k[X_1, \ldots, X_n]$. So for any ideal $A \trianglelefteq k[X_1, \ldots, X_n]$, define

$$V(A) := \{x \in k^n \mid f(x) = 0 \text{ for all } f \in A\}.$$

The subsets of $k^n$ of the form $V(A)$ are exactly the closed algebraic sets.

If $\Sigma$ is a closed algebraic set, we define

$$I(\Sigma) := \{f \in k[X_1, \ldots, X_n] \mid f(x) = 0 \text{ for all } x \in \Sigma\}.$$

$I(\Sigma)$ is an ideal, and

$$\Sigma = V(I(\Sigma)).$$

**Theorem 2** (Hilbert's Nullstellensatz)**.**

$$I(V(A)) = \sqrt{A}.$$

**Corollary 3.** *$V$ and $I$ set up a bijection between the set of closed algebraic subsets of $k^n$ and the set of ideals $A \subseteq k[X_1, \ldots, X_n]$ such that $A = \sqrt{A}$.*

**Remark 4.** *This really is a generalization of Weak Nullstellensatz, since maximal ideals are radical, and points are minimal closed subsets.*

**Example 5.** *An example of the correspondence: in $k[x, y]$, the chain*

$$(0) \subseteq (x) \subseteq (x, y)$$

*corresponds to the chain in $k^2$ looking like*

$$\textit{Plane} \supseteq \textit{(Vertical) line} \supseteq \textit{Point}$$

.

**Example 6.** *In $k[x, y]$, the ideal $(xy)$ corresponds to the two coordinate axes. This ideal is not prime but is still radical, hence the geometric image is not irreducible but is still closed, since the correspondence is*

$$\textit{Closed subsets} \leftrightarrow \textit{Radical ideals}$$

*and*

$$\textit{Irreducible closed subsets} \leftrightarrow \textit{Prime ideals}$$

*and*

$$\textit{Minimal irreducible closed subsets (=points)} \leftrightarrow \textit{Maximal ideals}$$

**Definition 7.** *A closed algebraic set is* ***irreducible*** *if it is not the union of two strictly smaller closed algebraic sets. We will omit "closed" when referring to these sets.*

Primary Decomposition Theorem says any radical ideal can be written in exactly one way as an intersection of a finite set of prime ideals, none of which contain any other. And a prime ideal is not the intersection of any two strictly bigger ideals. Then:

**Proposition 8.** *In the bijection in our corollary, the irreducible algebraic sets correspond exactly to the prime ideals of* $k[X_1, \ldots, X_n]$.
*Every closed algebraic set* $\Sigma$ *can be written in exactly one way as*

$$\Sigma = \Sigma_1 \cup \cdots \cup \Sigma_k$$

*where the* $\Sigma_i$ *are irreducible sets and* $\Sigma_i \not\subseteq \Sigma_j$ *if* $i \neq j$.

**Definition 9.** *The* $\Sigma_i$ *of the proposition are called the* ***components of*** $\Sigma$.

Now that we have affine algebraic sets, we can also define projective algebraic sets:

$\mathbb{P}_n(k)$ is the set of $(n+1)$-tuples $(x_0, \ldots, x_n) \in k^{n+1}$ such that some $x_i \neq 0$, modulo the equivalence relation

$$(x_0, \ldots, x_n) \sim (\alpha x_0, \ldots, \alpha x_n), \quad \alpha \in k^\times.$$

An $(n+1)$-tuple $(x_0 : \cdots : x_n)$ is called a set of **homogeneous coordinates** for the point associated to it.

$\mathbb{P}_n(k)$ can be covered by $n+1$ subsets $U_0, \ldots, U_n$ where

$$U_i = \{\text{points represented by homogeneous coordinates } (x_0 : \cdots : x_n) \text{ with } x_i \neq 0\}.$$

And each $U_i$ is naturally isomorphic to $k^n$ under the map

$$U_i \to k^n, \quad (x_0 : \cdots : x_n) \mapsto \left(\frac{x_0}{x_i}, \ldots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \ldots, \frac{x_n}{x_i}\right).$$

$\mathbb{P}_n(k)$ adds to the affine space $\mathbb{A}^n(k) \cong U_0$ the extra points at infinity $\mathbb{P}_n(k) \setminus U_0$.

- The **linear subspaces** $L$ of $\mathbb{P}_n(k)$ are the sets of points $P \in \mathbb{P}_n(k)$ with homogeneous coordinates in $W$ for some vector subspace $W \subseteq k^{n+1}$.
- If $W$ has codimension 1, we get a hyperplane.
- The "points at infinity" with respect to the affine piece $U_i$ form the hyperplane associated to the vector subspace $x_i = 0$.
- And the linear subspace $L$ associated to $W$ is isomorphic to $\mathbb{P}_r(k)$ where $r = \dim W - 1$.

Linear subspaces are examples of projective algebraic sets.

**Definition 10.** *A* ***closed algebraic set*** *in* $\mathbb{P}_n(k)$ *is a set consisting of all roots of a finite collection of homogeneous polynomials* $f_i \in k[X_0, \ldots, X_n], 1 \leq i \leq m$.

This makes sense because if $f$ is homogeneous (i.e. all terms have the same degree) and $(x_0 : \cdots : x_n) = (\alpha x_0 : \cdots : \alpha x_n)$, then $f(x_0 : \cdots : x_n) = 0 \iff f(\alpha x_0 : \cdots : \alpha x_n) = 0$.

**Homogeneous ideals** $A \subseteq k[X_0, \ldots, X_n]$ refer to ideals that, when they contain a polynomial $f$, also contain the homogeneous components of $f$. Equivalently, homogeneous ideals are the ideals generated by a finite set of homogeneous polynomials.

If $A$ is a homogeneous ideal, define

$$V(A) := \{P \in \mathbb{P}_n(k) \mid \text{ If } (x_0 : \cdots : x_n) \text{ are coordinates of } P, \text{ then } f(x) = 0 \forall f \in A\}.$$

If $\Sigma \subseteq \mathbb{P}_n(k)$ is a closed algebraic set, define

$$I(\Sigma) := \langle F \in k[X_0, \ldots, X_n] \text{ homogeneous} \mid F|_\Sigma = 0 \rangle.$$

**Theorem 11.** *$V$ and $I$ set up a bijection between the closed algebraic subsets of $\mathbb{P}_n(k)$ and the homogeneous radical ideals $A \subseteq k[X_0, \ldots, X_n]$ satisfying*

$$A : (X_0, \ldots, X_n) = A.$$

*Equivalently, one must identify homogeneous ideals that define the same projective zero set; the condition above chooses the largest such ideal.*

Define irreducible algebraic sets analogously to the affine case.

**Proposition 12.** *In the bijection of the theorem, the nonempty irreducible algebraic sets correspond exactly to the homogeneous prime ideals other than $(X_0, \ldots, X_n)$. And every closed algebraic set $\Sigma$ in $\mathbb{P}_n(k)$ can be written in exactly one way as*

$$\Sigma = \Sigma_1 \cup \cdots \cup \Sigma_k$$

*where the $\Sigma_i$ are irreducible algebraic sets and $\Sigma_i \not\subseteq \Sigma_j$ if $i \neq j$.*

**Problem 1.1.** ***Let** $\Sigma \subseteq \mathbb{P}_n(k)$ **be a closed algebraic set, and let** $H$ **be the hyperplane** $X_0 = 0$. **Identify** $\mathbb{P}_n \setminus H$ **with** $\mathbb{A}^n(k)$ **in the usual way.***
***Prove that** $\Sigma \cap (\mathbb{P}_n \setminus H)$ **is a closed algebraic subset of** $k^n$ **and show that the ideal of** $\Sigma \cap (\mathbb{P}_n \setminus H)$ **is derived from the ideal of** $\Sigma$ **in a very natural way.***

*Identify $\mathbb{P}_n \setminus H$ with $k^n$ via*

$$\varphi : \mathbb{P}_n \setminus H \to k^n, \quad (x_0 : \cdots : x_n) = (1 : x_1/x_0 : \cdots : x_n/x_0) \mapsto (x_1/x_0, \ldots, x_n/x_0)$$

*Observe:* $(1 : y_1 : \cdots : y_n) \in \Sigma \iff F(1, y_1, \ldots, y_n) = 0$ *for all homogeneous* $F \in I(\Sigma)$. *Thus,*

$$\varphi(\Sigma \cap (\mathbb{P}_n \setminus H)) =$$

$$\{(x_1/x_0, \ldots, x_n/x_0) \in k^n \mid F(1, x_1/x_0, \ldots, x_n/x_0) = 0 \text{ for all homogeneous } F \in I(\Sigma)\}$$

$$= \{x \in k^n \mid F(1, x) = 0 \text{ for all homogeneous } F \in I(\Sigma)\}$$

*Define $f(x_1, \ldots, x_n) := F(1, x_1, \ldots, x_n)$ for each homogeneous $F$. Then $f \in k[x_1, \ldots, x_n]$, so*

$$\varphi(\Sigma \cap (\mathbb{P}_n \setminus H)) = \{x \in k^n \mid f(x) = F(1, x) = 0 \text{ for all homogeneous } F \in I(\Sigma)\}$$

*is closed.*

*For the ideal $I(\varphi(\Sigma \cap (\mathbb{P}_n \setminus H)))$, we should consider*

$$J := \{f(x) = F(1, x) \mid F \in I(\Sigma) \text{ homogeneous}\}.$$

*This is an ideal in $k[x_1, \ldots, x_n]$:*

- *If $F(1, x) \in J$ and $h \in k[x_1, \ldots, x_n]$, homogenize $h$ to a homogeneous polynomial $H$ of sufficiently large degree. Then $HF \in I(\Sigma)$, so $h(x)F(1, x) = (HF)(1, x) \in J$.*
- *If $F(1, x), G(1, x) \in J$, first multiply $F$ and $G$ by suitable powers of $X_0$ so they have the same degree. Their sum is then homogeneous and lies in $I(\Sigma)$, so $F(1, x) + G(1, x) \in J$.*

*We know (by construction) that $V(J) = \varphi(\Sigma \cap (\mathbb{P}_n \setminus H))$.*
*By Nullstellensatz, $I(\varphi(\Sigma \cap (\mathbb{P}_n \setminus H))) = \sqrt{J}$.*
*And*

$$\sqrt{J} = \left\{f \in k[x_1, \ldots, x_n] \mid f^\ell \in J \text{ for some } \ell \in \mathbb{Z}_{>0}\right\}$$

*The natural construction $F \mapsto f$ is called* ***dehomogenization:***
*Given $F(x_0, \ldots, x_n)$ homogeneous polynomial, its dehomogenization with respect to $x_0$ is*

$$F^\#(x_1, \ldots, x_n) := F(1, x_1, \ldots, x_n).$$

*So the affine ideal is*

$$I(\Sigma)^\# = \{F^\# \mid F \in I(\Sigma)\}$$

*Thus, the ideal of the affine chart is obtained by setting $X_0 = 1$.*

*To summarize:*

1. *Let $I(\Sigma) \subseteq k[x_0, \ldots, x_n]$ be the homogeneous ideal defining $\Sigma$. Under the identification $\mathbb{P}^n \setminus H \cong k^n$, $\quad (1 : x_1 : \cdots : x_n) \leftrightarrow (x_1, \ldots, x_n)$ we have*

$$\Sigma \cap (\mathbb{P}^n \setminus H) = \{x \in k^n \mid F(1, x) = 0 \forall F \in I(\Sigma)\} \implies$$

$$\Sigma \cap (\mathbb{P}^n \setminus H) = V(\{F(1, x_1, \ldots, x_n) \mid F \in I(\Sigma)\})$$

*so it's a closed algebraic subset of $k^n$.*

*2. The dehomogenization of $I(\Sigma)$ is:*

$$I(\Sigma)^{\#} = \{F(1, x_1, \ldots, x_n) \mid F \in I(\Sigma)\} \implies$$

$$I(\Sigma \cap (\mathbb{P}^n \setminus H)) = \sqrt{I(\Sigma)^{\#}}.$$

**Example 13.** *(Hypersurfaces)*
*Let $f(X_0, \ldots, X_n)$ be an irreducible homogeneous polynomial. Then $(f)$ is prime, so $f = 0$ defines an irreducible algebraic set in $\mathbb{P}_n(k)$ called a hypersurface.*

**Example 14.** *(Twisted cubic in $\mathbb{P}_3(k)$)*
*Start with the ideal*

$$A_0 = (xz - y^2, yw - z^2) \subseteq k[x, y, z, w].$$

*$V(A_0)$ is the intersection in $\mathbb{P}_3(k)$ of the 2 quadrics $xz = y^2$ and $yw = z^2$. Look in the affine space with coordinates*

$$X = x/w, \quad Y = y/w, \quad Z = z/w$$

*(the complement of $w = 0$).*

*In here $V(A_0)$ is the intersection of $XZ = Y^2$ and $Y = Z^2$. This intersection falls into $Y = Z = 0$ and the twisted cubic itself, meaning $A_0$ is the intersection of the ideal of the line and of the twisted cubic:*

$$A_0 = (y, z) \cap (xz - y^2, yw - z^2, xw - yz).$$

*So we let $A := (xz - y^2, yw - z^2, xw - yz)$. We define the **twisted cubic** to be $V(A)$.*

*$A$ is prime by being the kernel of the homomorphism $\varphi : k[x, y, z, w] \to k[s, t]$ sending:*

$$x \mapsto s^3, \quad y \mapsto s^2 t, \quad z \mapsto st^2, \quad w \mapsto t^3.$$

## 1.3 Definition of a Morphism

**Example 1.** *If $V \subseteq k^n$ is an irreducible algebraic set and if $f \in k[X_1, \ldots, X_n]$, then*

$$V^* := \{(x_1, \ldots, x_n, f(x_1, \ldots, x_n)) \mid (x_1, \ldots, x_n) \in V\} \subseteq k^{n+1}$$

*is an irreducible algebraic set, and the projection*

$$(x_1, \ldots, x_{n+1}) \mapsto (x_1, \ldots, x_n)$$

*should be an isomorphism $V \to V^*$.*

*So in particular e.g. the affine line $k$ and the parabola $y = x^2$ in $k^2$ should be isomorphic.*

**Example 2.** *It turns out that an irreducible conic $C \subseteq \mathbb{P}_2(k)$ will be isomorphic to $\mathbb{P}_1(k)$.*

*One sees this by identifying $\mathbb{P}_1(k)$ with the set of all lines through some point $P_0$ on $C$ and then defining the map $\alpha : \mathbb{P}_1(k) \to C$ by taking a line $\ell$ through $P_0$ and letting $\alpha(\ell)$ be its second intersection point with $C$.*

**Example 3.** *It turns out that $\mathbb{P}_1(k)$ and the twisted cubic in $\mathbb{P}_3(k)$ are isomorphic.*

$\mathbb{P}_1(k) = \{(s,t) \in k^2 \setminus \{0\}\}/\sim$ *where* $(s,t) \sim (\lambda s, \lambda t)$, *and the isomorphism is given by*

$$\alpha : \mathbb{P}_1(k) \to \mathbb{P}_3(k), \qquad (s:t) \mapsto (s^3 : s^2t : st^2 : t^3).$$

**Example 4.** *If $C$ is a smooth cubic curve in $\mathbb{P}_2(k)$ and $P_0 \in C$, then for any $P \in C$, if $\ell$ is the line through $P$ and $P_0$, let $\alpha(P)$ be the third point in which $\ell$ meets $C$ (counting intersection multiplicity). Then $\alpha$ is an automorphism of $C$ of order 2.*

**Definition 5.** *Let $\Sigma_1 \subseteq k^{n_1}$ and $\Sigma_2 \subseteq k^{n_2}$ be two irreducible algebraic sets. A map*

$$\alpha : \Sigma_1 \to \Sigma_2$$

*will be called a* ***morphism*** *if there exist $n_2$ polynomials $f_1, \ldots, f_{n_2}$ in the variables $X_1, \ldots, X_{n_1}$ such that*

$$\alpha(x) = (f_1(x_1, \ldots, x_{n_1}), \ldots, f_{n_2}(x_1, \ldots, x_{n_1}))$$

*for all points $x = (x_1, \ldots, x_{n_1}) \in \Sigma_1$.*

**Remark 6.** *Note that this definition implies that every morphism $\alpha$ from $\Sigma_1$ to $\Sigma_2$ is the restriction of a morphism $\alpha'$ from $k^{n_1}$ to $k^{n_2}$.*

**Definition 7.** *Let $\Sigma \subseteq k^n$ be an irreducible algebraic set. Then the* ***affine ring*** *$\Gamma(\Sigma)$ is the ring of $k$-valued functions on $\Sigma$ given by the polynomials in the coordinates, i.e.*

$$k[X_1, \ldots, X_n]/I(\Sigma).$$

**Proposition 8.** *If $\Sigma_1, \Sigma_2$ are two irreducible algebraic sets, then the set of morphisms from $\Sigma_1$ to $\Sigma_2$ and the set of $k$-homomorphisms from $\Gamma(\Sigma_2)$ to $\Gamma(\Sigma_1)$ are canonically isomorphic, i.e.*

$$\mathrm{Hom}(\Sigma_1, \Sigma_2) \cong \mathrm{Hom}_k(\Gamma(\Sigma_2), \Gamma(\Sigma_1)).$$

**Corollary 9.** *If $\Sigma$ is an irreducible algebraic set, then $\Gamma(\Sigma)$ is canonically isomorphic to the set of morphisms from $\Sigma$ to $k$.*

For any ideal $A \subseteq \Gamma(\Sigma)$, define

$$V(A) := \{x \in \Sigma \mid f(x) = 0 \text{ for all } f \in A\} \subseteq \Sigma$$

Nullstellensatz implies

$$\{f \in \Gamma(\Sigma) \mid f(x) = 0 \text{ for all } x \in V(A)\} = \sqrt{A}.$$

**Proposition 10.** *The assignment $\Sigma \mapsto \Gamma(\Sigma)$ extends to a contravariant functor $\Gamma$:*

$$\{\textit{Category of irreducible algebraic sets + morphisms}\} \to$$

$$\{\textit{Category of finitely generated integral domains over } k + k\textit{−homomorphisms}\}$$

*which is an equivalence of categories.*

**Example 11.** *Consider $\mathbb{P}^2$. Its ring of global regular functions is $\Gamma(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2})$. Note: $\mathbb{P}^2 = \mathbb{A}^2 \cup H$ where $H$ is the line at $\infty$ (= a copy of $\mathbb{P}^1$).*

*A polynomial $f \in k[x, y]$ on the affine chart $\mathbb{A}^2$ extends regularly across the line at infinity only if it is constant. Hence*

$$\Gamma(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}) = k.$$

*Thus $\mathbb{P}^2$ and a point have the same ring of global regular functions. This is why global functions alone do not determine a non-affine variety.*

**Remark 12.** *In general,*

$$\mathbb{P}^n = U_0 \cup H = \mathbb{A}^n \cup \mathbb{P}^{n-1}.$$

**Definition 13.** *A **closed set** in $k^n$ is to be a closed algebraic set $V(A)$. These define a topology in $k^n$, called the **Zariski topology.** An irreducible algebraic set $\Sigma \subseteq k^n$ is given the induced topology, also called the Zariski topology. The closed sets of $\Sigma$ are exactly the sets $V(A)$ where $A$ is an ideal in $\Gamma(\Sigma)$.*

**Remark 14.**

1. *All morphisms are continuous in the Zariski topology. In fact, this is the weakest topology making morphisms continuous.*
2. *A basis for the open sets in the Zariski topology on $\Sigma$ is given by the open sets $\Sigma_f := \{x \in \Sigma \mid f(x) \neq 0\}$ for elements $f \in \Gamma(\Sigma)$.*
3. *On $k$ itself (for $k$ infinite), the Zariski topology is the cofinite topology: the proper closed sets are finite.*
4. *Not all continuous maps are morphisms.*
5. *Zariski topology on the product $\neq$ Zariski product topology*

**Definition 15.** *A topological space $X$ is **noetherian** if its closed sets satisfy the descending chain condition. It is equivalent to require that every open subset be quasi-compact.*

Since ideals in $k[X_1, \ldots, X_n]$ satisfy ascending chain condition, the Zariski topology is noetherian.

**Definition 16.** *A **closed set in** $\mathbb{P}_n(k)$ is to be a closed algebraic set $V(A)$. These define a topology in $\mathbb{P}_n(k)$, called the **Zariski topology**. Irreducible algebraic sets are again given the induced topology. As in the affine case, a basis for the open sets is given by*

$$[\mathbb{P}_n(k)]_f = \{x \in \mathbb{P}_n(k) \mid f(x) \neq 0\}$$

*where $f$ is a **homogeneous** polynomial.*

**Remark 17.** *The Zariski topology on $\mathbb{P}_n(k)$ is noetherian.*

**Problem 1.2.** ***Check that*** $\mathbb{P}_n(k)_{X_0}$ ***is homeomorphic to*** $k^n$ ***under the usual map***

$$(x_0 : \cdots : x_n) \mapsto (\frac{x_1}{x_0}, \ldots, \frac{x_n}{x_0})$$

*Call this map* $\varphi : \mathbb{P}_n(k)_{X_0} \to k^n$.

*Well-defined: If* $(x_0 : \cdots : x_n) = (\lambda x_0 : \cdots : \lambda x_n)$, *then*

$$\varphi(x_0 : \cdots : x_n) = (x_1/x_0, \ldots, x_n/x_0) = (\lambda x_1/\lambda x_0, \ldots, \lambda x_n/\lambda x_0) = \varphi(\lambda x_0 : \cdots : \lambda x_n).$$

*Injective: If* $(x_0 : \cdots : x_n) \neq (y_0 : \cdots : y_n)$, *then for all* $\lambda \neq 0$, *there exists some* $i \in \{0, \ldots, n\}$ *such that* $x_i \neq \lambda y_i$. *If we had* $\varphi(x_0 : \cdots : x_n) = \varphi(y_0 : \cdots : y_n)$, *then*

$$(x_1/x_0, \ldots, x_n/x_0) = (y_1/y_0, \ldots, y_n/y_0) \implies x_i/x_0 = y_i/y_0 \text{ for all } i = 1, \ldots, n$$

$$\implies x_i = \left(\frac{x_0}{y_0}\right) y_i \text{ for all } i = 1, \ldots, n,$$

*so* $(x_0 : \cdots : x_n) = (y_0 : \cdots : y_n)$, *contradicting our assumption that the two projective points were distinct. Surjective: If* $(x_1, \ldots, x_n) \in k^n$ *then* $(1 : x_1 : \cdots : x_n) \in \mathbb{P}_n(k)_{X_0}$ *and*
$\varphi(1 : x_1 : \cdots : x_n) = (x_1, \ldots, x_n)$.

*Defining an inverse* $\psi$ :
*Let*

$$\psi : k^n \to \mathbb{P}_n(k)_{X_0}, \quad (x_1, \ldots, x_n) \mapsto (1 : x_1 : \cdots : x_n)$$

*Then*

$$\psi(\varphi(x_0 : \cdots : x_n)) = \psi(x_1/x_0, \ldots, x_n/x_0) = (1 : x_1/x_0 : \cdots : x_n/x_0) = (x_0 : \cdots : x_n)$$

*and*

$$\varphi(\psi(x_1, \ldots, x_n)) = \varphi(1 : x_1 : \cdots : x_n) = (x_1/1, \ldots, x_n/1) = (x_1, \ldots, x_n).$$

$\varphi$ *is continuous: Suppose* $f \in k[x_1, \ldots, x_n]$ *and* $V(f) \subseteq k^n$. *If* $(x_0 : \cdots : x_n) \in \mathbb{P}_n(k)_{X_0}$ *then:*

$$\varphi(x_0 : \cdots : x_n) \in V(f) \iff f(x_1/x_0, \ldots, x_n/x_0) = 0 \iff$$
$$x_0^{\deg f} f(x_1/x_0, \ldots, x_n/x_0) = 0 \iff F(x_0, \ldots, x_n) = 0$$

*where* $F(x_0, \ldots, x_n) := x_0^{\deg f} f(x_1/x_0, \ldots, x_n/x_0) = 0$ *is homogeneous.*

*Thus* $\varphi^{-1}(V(f)) = V(F) \cap \mathbb{P}_n(k)_{X_0}$ *by Problem 1. But* $V(F) \cap \mathbb{P}_n(k)_{X_0}$ *is closed in the subspace topology* $\implies \varphi^{-1}(V(f))$ *is closed. So* $\varphi$ *is continuous.*

*$\psi$ is continuous: Let $(F) \cap \mathbb{P}_n(k)_{X_0} \subseteq \mathbb{P}_n(k)_{X_0}$ be closed ($F$ is a homogeneous polynomial). Then $(x_1, \ldots, x_n) \in k^n$ has $\psi(x_1, \ldots, x_n) \in V(F) \cap \mathbb{P}_n(k)_{X_0} \iff F(1, x_1, \ldots, x_n) = 0$. Thus $\psi^{-1}(V(F) \cap \mathbb{P}_n(k)_{X_0}) = V(F(1, x_1, \ldots, x_n))$, which is closed in $k^n$. So $\psi$ is continuous.*

*Conclude: the map*

$$\mathbb{P}_n(k)_{X_0} \to k^n, \quad (x_0 : \cdots : x_n) \mapsto (x_1/x_0, \ldots, x_n/x_0)$$

*is bijective, continuous, and has a continuous inverse*

$$(a_1, \ldots, a_n) \mapsto (1 : a_1 : \cdots : a_n)$$

*and thus $\mathbb{P}_n(k)_{X_0} \cong k^n$ as topological spaces.*

## 1.4 Sheaves and affine varieties

**Definition 1.** *Let $X$ be a topological space. A **presheaf** $F$ on $X$ consists of*

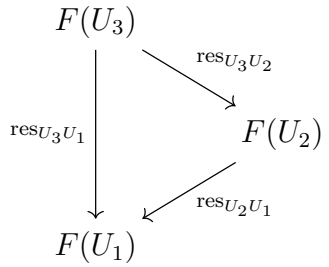
1. *for all open $U \subseteq X$, a set $F(U)$*
2. *for all pairs of open sets $U_1 \subseteq U_2$, a restriction map*

$$\mathrm{res}_{U_2 U_1} : F(U_2) \to F(U_1)$$

*such that*

*(a) $\mathrm{res}_{U,U} = \mathrm{id}_{F(U)}$ for all $U$*

*(b) If $U_1 \subseteq U_2 \subseteq U_3$, then the following diagram commutes:*




**Definition 2.** *If $F_1, F_2$ are presheaves on $X$, a **map of presheaves** $\varphi : F_1 \to F_2$ is a collection of maps $\varphi(U) : F_1(U) \to F_2(U)$ for each open $U$ such that if $U \subseteq V$, the following diagram commutes:*

$$\begin{array}{ccc} F_1(V) & \xrightarrow{\varphi(V)} & F_2(V) \\ \downarrow{\scriptstyle \mathrm{res}_{V,U}} & & \downarrow{\scriptstyle \mathrm{res}_{V,U}} \\ F_1(U) & \xrightarrow{\varphi(U)} & F_2(U) \end{array}$$

**Definition 3.** *A presheaf $F$ is a **sheaf** if for every collection $\{U_i\}$ of open sets in $X$ with $U = \bigcup_i U_i$, the diagram*

$$F(U) \longrightarrow \prod F(U_i) \underset{g}{\overset{f}{\rightrightarrows}} \prod_{i,j} F(U_i \cap U_j)$$

*is exact, i.e. the map*

$$\mathrm{res}_{U,U_i} : F(U) \to \prod F(U_i)$$

*is injective, and its image is the set on which*

$$\mathrm{res}_{U_i, U_i \cap U_j} : \prod_i F(U_i) \to \prod_{i,j} F(U_i \cap U_j)$$

*and*

$$\mathrm{res}_{U_j, U_i \cap U_j} : \prod_j F(U_j) \to \prod_{i,j} F(U_i \cap U_j)$$

*agree.*

What this really means:

1. If $x_1, x_2 \in F(U)$, and for all $i$, $\mathrm{res}_{U,U_i} x_1 = \mathrm{res}_{U,U_i} x_2$, then $x_1 = x_2$.
   (i.e. **elements are uniquely determined by local data**)

2. If we have a collection of elements $x_i \in F(U_i)$ such that $\mathrm{res}_{U_i, U_i \cap U_j} x_i = \mathrm{res}_{U_j, U_i \cap U_j} x_j$ for all $i, j$, then there exists an $x \in F(U)$ such that $\mathrm{res}_{U,U_i} x = x_i$ for all $i$.
   (i.e. **if we have local data which are compatible, then they patch together to form something in** $F(U)$)

**Example 4.** *Let $X, Y$ be topological spaces. For all open sets $U$, let*

$$F(U) := \{\textit{continuous maps } U \to Y\}$$

*This is a presheaf with restriction maps being usual restriction.*
*This is a sheaf, since a function is continuous on $U_i \iff$ its restrictions to each $U_i$ are continuous.*

**Example 5.** *Let $X, Y$ be differentiable manifolds. For all open sets $U$, let*

$$G(U) := \{\textit{continuous functions } U \to Y \textit{ which have relatively compact image}\}$$

*This is a subpresheaf of the first example, but need not be a sheaf (we can glue together bounded functions to get an unbounded function). Thus, this fails on gluing, but passes on locality.*

Continuity is local, but boundedness is not local.

**Example 6.** *Let $X$ be a topological space. For any open set $U$, let*

$$F(U) := \{\text{locally constant real-valued functions on } U\}/\{\text{constant functions on } U\}$$

*This is a presheaf.*
*This is not a sheaf: Every $s \in F(U)$ goes to zero in $\prod F(U_i)$ for some covering $\{U_i\}$. But if $U$ is not connected, $F(U) \neq (0)$.*

**Definition 7.** *Let $F$ be a sheaf on $X$, $x \in X$. The collection of $F(U)$, $U$ open containing $x$, is a direct system under restriction toward smaller neighborhoods, and we can form*

$$F_x := \varinjlim_{x \in U} F(U)$$

*called the **stalk of** $F$ **at** $x$.*

**Example 8.** *Let*

$$F(U) = \{\text{continuous functions } U \to \mathbb{R}\}$$

*Then $F_x$ is called the set of **germs of continuous functions at** $x$, and*

$$F_x = \bigcup_{x \in U} F(U)/\sim$$

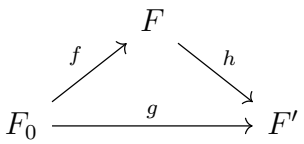
*where $f_1 \sim f_2$ if $f_1$ and $f_2$ agree in a neighborhood of $x$.*

**Example 9.** *If $\mathcal{F}$ = sheaf of holomorphic functions on $\mathbb{C}$, then the stalk at $0$ consists of germs represented by power series with positive radius of convergence at $0$, i.e.*

$$\mathcal{O}_{\mathbb{C},0} \cong \mathbb{C}\{z\},$$

*where $\mathbb{C}\{z\}$ denotes the ring of convergent power series. Question: What about the sheaf of $C^\infty$ functions $\mathbb{R}^2 \to \mathbb{R}$ at the origin?*
*Two functions can agree on an open set containing the origin and differ on another.*
*An element of this stalk is a germ of $C^\infty$ functions at the origin.*

**Sheafification of a presheaf:**
Let $F_0$ be a presheaf on $X$. Then there is a sheaf $F$ and a map $f : F_0 \to F$ such that if $g : F_0 \to F'$ is any map with $F'$ a sheaf, then there is a unique map $h : F \to F'$ such that the following diagram commutes:




To get $F$ from $F_0$, you basically just: first identify things which have the same restrictions, then add in all the things which can be patched together.

**Example 10.** *Revisiting* $G(U) = \{$*continuous functions* $U \to Y$ *which have relatively compact image*$\}$ *(which we said was a presheaf but not a sheaf), if* $X$ *is locally compact, the sheafification of this presheaf is the sheaf of all continuous functions on* $X$.

**Example 11.** *Revisiting* $F(U) = \{$*locally constant real-valued functions on* $U\}/\{$*constant functions on* $U\}$ *the sheafification of this presheaf is* $(0)$.

**Notation:** We write $\Gamma(U, F)$ for $F(U)$ (this is called the set of **sections of** $F$ **over** $U$)
$\Gamma(X, F)$ is the set of **global sections of** $F$.
In other contexts, we may denote $F(X)$ by $H^0(X, F)$ and call it the **zeroth cohomology group**.

If for all $U$, $F(U)$ is a group (ring, etc.) and the restriction maps are group (ring, etc.) homomorphisms, then $F$ is called a **sheaf of groups (rings, etc.)**. In this case, $F_x$ is a group (ring, etc.) and so on.

**Example 12.** *Let* $X$ *be a topological space. Let*

$$F_{cont,X}(U) := \{\textit{continuous functions } U \to \mathbb{R}\}.$$

*Then* $F_{cont,X}(U)$ *is a sheaf of rings.*

**Definition 13.** *If* $g : X \to Y$ *is a continuous function, the operation* $f \mapsto f \circ g$ *gives us the following maps: for every open* $U \subseteq Y$*, a map* $F_{cont,Y}(U) \to F_{cont,X}(g^{-1}U)$ *such that the following diagram commutes for all open sets* $V \subseteq U$*:*

$$\begin{array}{ccc} F_Y(U) & \longrightarrow & F_X(g^{-1}U) \\ \big\downarrow \text{res} & & \big\downarrow \text{res} \\ F_Y(V) & \longrightarrow & F_X(g^{-1}V) \end{array}$$

*This set-up is called a* ***morphism of the pair*** $(X, F_X)$ ***to the pair*** $(Y, F_Y)$***.***

**Example 14.** *Let* $X, Y$ *be differentiable manifolds. Let* $F_{diff,X}$ *and* $F_{diff,Y}$ *be subsheaves of* $F_{cont,X}$ *and* $F_{cont,Y}$ *of differentiable functions.*
*Let* $g : X \to Y$ *be a continuous map. Then* $g$ *is differentiable* $\iff$ *for all open sets* $U \subseteq Y$*,* $f \in F_{diff,Y}(U) \implies f \circ g \in F_{diff,X}(g^{-1}U)$.

**Example 15.** *Say* $X, Y$ *are complex analytic manifolds, Let* $F_{\text{an},X}$ *and* $F_{\text{an},Y}$ *be the sheaves of holomorphic functions.*

*A continuous map* $g : X \to Y$ *is holomorphic* $\iff$ *for all open sets* $U \subseteq Y$*,* $f \in F_{\text{an},Y}(U) \implies f \circ g \in F_{\text{an},X}(g^{-1}U)$.

**Definition 16.** *Let* $X \subseteq k^n$ *be an irreducible algebraic set. Let* $R$ *be its affine coordinate ring.*
$X$ *irreducible* $\implies$ $I(X)$ *is prime and* $R$ *is a domain. Let* $K = \text{Frac}(R)$.

*Recall: $R$ has been identified with a ring of functions on $X$.*
*For $x \in X$, let*

$$m_x := \{f \in R \mid f(x) = 0\}.$$

*This is a maximal ideal ($\ker(R \to k, f \mapsto f(x))$. Let*

$$\mathcal{O}_{X,x} := R_{m_x} = \{\frac{f}{g} \mid f, g \in R, g(x) \neq 0\} \subseteq K.$$

*Now for $U$ open in $X$, let*

$$\mathcal{O}_X(U) := \bigcap_{x \in U} \mathcal{O}_{X,x}.$$

- *Each $\mathcal{O}_X(U)$ is a subring of $K$.*
- *If $V \subseteq U$, then $\mathcal{O}_X(U) \subseteq \mathcal{O}_X(V)$.*

*Taking inclusion as the restriction map, this defines a sheaf $\mathcal{O}_X$. This is called the* ***structure sheaf of*** *$X$.*

**Remark 17.** *The elements of $\mathcal{O}_X(U)$ can be viewed as functions on $U$:*

*If $F \in \mathcal{O}_X(U)$ and $x \in U$, then $F \in \mathcal{O}_{X,x}$ so $F = \frac{f}{g}$ where $g(x) \neq 0$.*
*Define $F(x) = \frac{f(x)}{g(x)}$.*
*If $F(x) = 0$ for all $x \in U$, then $F = 0$, so we can identify $\mathcal{O}_X(U)$ with the associated ring of functions on $U$.*

*Here $\mathcal{O}(U)$ = polynomial functions on $U$ = regular functions on $U$. This is a sheaf because being regular is local.*

**Example 18.** $\mathcal{O}(\mathbb{A}^1) = k[X]$
*Let $U_1 : x \neq 1$. Then*

$$\mathcal{O}(U_1) = k[x, \frac{1}{x-1}] \cong k[x, y]/((x-1)y - 1)$$

*and letting $U_2 : x \neq 2$,*

$$\mathcal{O}(U_2) = k[x, \frac{1}{x-2}].$$

*It turns out: $\mathcal{O}_p = k[x]_{(p)}$.*

**Example 19.** *Say $\pi : Y \to X$. For every $U \subseteq X$, consider $\mathcal{F}(U) := \{$sections of $\pi|_U\}$. This is a sheaf. It is this example that motivates the name "sheaf."*

**Proposition 20.** *Let $X$ be an irreducible algebraic set. Let $R = \Gamma(X)$. Let $f \in R$, and $X_f = \{x \in X \mid f(x) \neq 0\}$. Then*

$$\mathcal{O}_X(X_f) = R_f.$$

**Corollary 21.** $\Gamma(X, \mathcal{O}_X) = R$.

So:
The set of global functions of $\mathcal{O}_X$ is

$$\mathcal{O}_X(X) = \bigcap_{x \in X} \mathcal{O}_{X,x} = \bigcap_{x \in X} \left\{ \frac{f}{g} \mid f, g, \in \Gamma(X), g(x) \neq 0 \right\}$$

and indeed equals $\Gamma(X)$.

**Remark 22.** *1. If $f \in \mathcal{O}_X(U)$ and $f$ vanishes nowhere on $U$, then $\frac{1}{f} \in \mathcal{O}_X(U)$, since: $f(x) \neq 0 \implies \frac{1}{f} \in \mathcal{O}_{X,x}$.*

*2. The stalk of $\mathcal{O}_X$ at $x$ is $\mathcal{O}_{X,x}$, since:*
*The sets $X_f$ are a basis of the Zariski topology of $X$, so*

$$\varinjlim_{x \in U} \mathcal{O}_X(U) = \varinjlim_{x \in X_f} \mathcal{O}_X(X_f) = \varinjlim_{f(x) \neq 0} R_f.$$

*Since all restriction maps in our sheaf are injective, this equals*

$$\bigcup_{f(x) \neq 0} R_f = \mathcal{O}_{X,x}.$$

*3. The field $K(= \mathrm{Frac}(\Gamma(X)))$ can be recovered from the sheaf $\mathcal{O}_X$:*
*$X$ is irreducible $\implies$ the intersection of any two nonempty open sets is nonempty $\implies$ we have an inverse system containing all open sets $\implies$ we can define a **generic stalk** of any sheaf $F$ on $X$.*
*$K$ is the generic stalk of the structure sheaf $\mathcal{O}_X$.*

*4. If $h \in \mathcal{O}_X(U)$ for some open $U \subseteq X$, then it's not necessarily true that $h = f/g$ with $f, g \in R$ and $g$ vanishing nowhere on $U$.*
*For example:*
*Let $X = V(xw - yz) \subseteq k^4$.*
*Let $U = X_y \cup X_w$.*
*Define $h \in \mathcal{O}_X(U)$ so that $h = x/y$ on $X_y$ and $h = z/w$ on $X_w$*
*One can show (Mumford page 21) that $h$ cannot equal $f/g$, $g \neq 0$ in $U$.*
*But if $U$ has the form $X_g$, then this is necessarily the case (by the Proposition).*

**Remark 23.** *$V(f)$ is called a principal hypersurface. $X_f = \{f \neq 0\}$ is called a distinguished open / basic open, and is the complement of $V(f)$.*

**Proposition 24.** *Let $X \subseteq k^n, Y \subseteq k^m$ be irreducible algebraic sets. Let $f : X \to Y$ be continuous. TFAE:*

1. *$f$ is a morphism.*
2. *For every open $U \subseteq Y$ and every $g \in \Gamma(U, \mathcal{O}_Y)$, one has $g \circ f \in \Gamma(f^{-1}(U), \mathcal{O}_X)$.*
3. *For every $x \in X$ and every regular function $g$ defined on a neighborhood of $f(x)$, the germ of $g \circ f$ at $x$ belongs to $\mathcal{O}_{X,x}$.*

4. *For every $x \in X$, pullback induces a local homomorphism $\mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$.*

**Definition 25.** *An **affine variety** is a topological space $X$ plus a sheaf of $k$-valued functions $\mathcal{O}_X$ on $X$, which is isomorphic to an irreducible algebraic subset of $k^n$, plus the structure sheaf of $X$ (as in Definition 16).*

**Remark 26.** *Many books do not require irreducible.*

**Definition 27.** *The affine variety $(k^n, \mathcal{O}_{k^n})$ is $\mathbb{A}^n$, called **affine $n$-space**.*

*Bijective morphisms need not be isomorphisms*:

- A bijective continuous map from a compact space to a Hausdorff space is a homeomorphism; analogous statements require additional hypotheses in other categories.
- For differentiable manifolds, a bijective smooth map need not be a diffeomorphism. Consider $f : \mathbb{R} \to \mathbb{R}, x \mapsto x^3$; its inverse is not differentiable at 0.
- This is also not true for affine varieties. Some examples:

  **Example 28.** *If $\mathrm{char}(k) = p \neq 0$, $f : \mathbb{A}^1 \to \mathbb{A}^1, t \mapsto t^p$ is bijective and on the ring level corresponds to $k[X] \hookleftarrow k[X^p]$ which are not equal.*

  **Example 29.** *Let $k$ be an algebraically closed field, $f : \mathbb{A}^1 \to \mathbb{A}^2, t \mapsto (t^2, t^3)$ has image the irreducible closed curve $C : X^3 = Y^2$. $f : \mathbb{A}^1 \to C$ is a bijection and corresponds to $k[T] \hookleftarrow k[T^2, T^3]$, which are not equal.*

  **Example 30.** *Let $f : \mathbb{A}^1 \to \mathbb{A}^2, X = t^2 - 1, Y = t(t^2 - 1)$. The image of this morphism is the curve $D : Y^2 = X^2(X + 1)$.*
  *$f$ is bijective between $\mathbb{A}^1$ and $D$ except that $t = 1$ and $t = -1$ both map to the origin.*
  *If $X_1 = \mathbb{A}^1 \setminus \{1\}$, then $f$ restricts to a bijection $f' : X_1 \to D$, but this morphism corresponds to $k[T, (T-1)^{-1}] \hookleftarrow k[T^2 - 1, T(T^2 - 1)]$, which are not equal.*

**Induced affine variety structure on open and closed subsets of affine varieties:**
Let $Y$ be an irreducible closed subset of an affine variety $(X, \mathcal{O}_x)$.
Define an induced sheaf $\mathcal{O}_X$ of functions on $Y$ as follows:
If $V$ is open in $Y$,

$$\mathcal{O}_Y(V) = \{f : V \to k \mid \forall x \in V, \exists \text{ a neighborhood } U \text{ of } x \text{ in } X \text{ and } F \in \mathcal{O}_X(U) \text{ so that } f = F|_{U \cap V}\}$$

**Proposition 31.** *$(Y, \mathcal{O}_Y)$ is an affine variety.*

**Proposition 32.** *Let $(X, \mathcal{O}_X)$ be an affine variety. Let $f \in \Gamma(X, \mathcal{O}_X)$. Then $(X_f, \mathcal{O}_X|_{X_f})$ is an affine variety.*

Here we pushed the zeroes of $f$ out to infinity. For example:
Suppose $X = \mathbb{A}^1$, $f$ is the coordinate $X_1$. Then $B = (1 - X_1X_2) \implies$ the projection of the $B$ down to the axis is an isomorphism with $X_f$.

**Example 33.** *Not all open subsets of affine varieties are affine varieties. You cannot push the origin in $\mathbb{A}^2$ out to infinity.*

*$X = \mathbb{A}^2 \setminus \{(0,0\}$ is not an affine variety: if we had some embedding $\mathbb{A}^2 \setminus \{(0,0\} \hookrightarrow k^n$, then since no rational function is well-defined on $\mathbb{A}^2 \setminus \{(0,0\}$ but not at $(0,0)$, the coordinate functions giving this embedding would have to extend to all of $\mathbb{A}^2$. So the restriction map $\Gamma(\mathbb{A}^2, \mathcal{O}_{\mathbb{A}^2}) = k[x,y] \to \Gamma(X, \mathcal{O}_X)$ is bijective.*

*Then if $X$ were affine, it would mean that since $X$ and $\mathbb{A}^2$ have the same coordinate rings, that they must be isomorphic as varieties, which is not true.*

*This is similar to the idea that a holomorphic function on $\mathbb{C} \times \mathbb{C} \setminus \{(0,0\}$ is necessarily holomorphic at the origin.*

**Remark 34.** *We call an open subset of an affine (which is not itself affine) "quasi-affine."*

## 1.5 Prevarieties and morphisms

**Definition 1.** *A topological space $X$ plus a sheaf $\mathcal{O}_X$ of $k$-valued functions on $X$ is a **prevariety** if*

1. *$X$ is connected, and*
2. *there is a finite open covering $\{U_i\}$ of $X$ such that for all $i$, $(U_i, \mathcal{O}_X|_{U_i})$ is an affine variety.*

**Definition 2.** *An open subset $U$ of $X$ is called an **open affine set** if $(U, \mathcal{O}_X|_U)$ is an affine variety.*

The open affine sets are a basis of the topology.

**Definition 3.** *A topological space is **irreducible** if it is not the union of two proper closed subsets (equivalently, the intersection of ay two nonempty sets is nonempty).*

**Proposition 4.** *Every prevariety $X$ is an irreducible topological space.*

So every open set is dense.

**Remark 5.** *If you remove the irreducible requirement, opens need not be dense. Consider the union $X$ of the $x$ and $y$ axes. In here, the $x$-axis minus the origin is open but not dense, since its closure does not contain the $x$-axis.*

**Proposition 6.** *If $X$ is a prevariety, then the closed sets of $X$ satisfy DCC, i.e. $X$ is noetherian.*

So every variety is quasi-compact.

**Proposition 7.** *Let $X$ be a noetherian topological space. Then every closed set $Z$ in $X$ can be written uniquely as an irredundant union of finitely many irreducible closed sets, called the components of $Z$.*

**Remark 8.** *The stalk at a point is all functions regular at the point. They are actually all regular in some neighborhood of the point. We often specify that we are talking about a set of functions all regular at "some point," and in these situations, we call the point the* ***generic point.***

**Example 9.**

$$k(\mathbb{P}^1) = \varinjlim_{\text{nonempty opens in } \mathbb{P}^1} \mathcal{O}_{\mathbb{P}^1}(U)$$

- *Why is $k(X)$ (in general from the definition) a field?*

  *Elements of $k(X)$ are equivalence classes of pairs $(U, f)$. If $f \in k(X)$ is a nonzero element represented by $(U, f)$, then because it is nonzero in the limit, $f$ does not identically vanish on any non-empty open subset. Thus $V(f) \subseteq U$ is a proper closed subset.*

  *Since $X$ is irreducible, $U_f = U \setminus V(f)$ is a nonempty, dense open set. By definition, $f(x) \neq 0$ for all $x \in U_f$. So $1/f$ is a well-defined regular function on $U_f$, meaning $(U_f, 1/f) \in k(X)$.*

  *Intuitively, $U$ irreducible $\implies$ $R = \mathcal{O}_X(U)$ is a domain $\implies$ $k(X) \cong \mathrm{Frac}(R)$, a field.*

  *More concretely, given $f/g$ with $g$ nonzero, pick $U = \{g \neq 0\}$. $U$ is a dense open. So $f/g \in k(\mathbb{A}^1)$. And conversely, everything in $k(\mathbb{A}^1)$ is in some $\mathcal{O}_X(U)$, so looks like some $f/g$ where $f, g$ are polynomials.*

- *Why is $k(\mathbb{P}^1) = k(t)$?*

  *Because $\mathbb{P}^1$ is irreducible, every non-empty open set is dense. The collection of open subsets contained within the standard affine chart $U_0 = \{[x_0 : x_1] \mid x_0 \neq 0\} \cong \mathbb{A}^1$ forms a cofinal system in the poset of all non-empty open sets. Therefore, the direct limit restricts isomorphically to $U_0$:*

$$\varinjlim_{U \subset \mathbb{P}^1} \mathcal{O}_{\mathbb{P}^1}(U) \cong \varinjlim_{V \subset U_0} \mathcal{O}_{U_0}(V) = k(U_0)$$

  *The open subset $U_0$ is isomorphic to the affine line $\mathbb{A}^1$, meaning its coordinate ring is the polynomial ring in one variable. Choosing the coordinate function $t = x_1/x_0$, we have:*

$$\mathcal{O}_{\mathbb{P}^1}(U_0) \cong k[t]$$

  *For an affine variety, the direct limit of the structure sheaf over all non-empty open sets is the fraction field of its coordinate ring. Thus:*

$$k(\mathbb{P}^1) \cong k(U_0) \cong \mathit{Frac}(\mathcal{O}_{\mathbb{P}^1}(U_0)) \cong \mathit{Frac}(k[t]) = k(t)$$

**Remark 10.** *This generalizes:*

$$k(\mathbb{P}^n) = k(x_1, \cdots, x_n).$$

**Example 11.** *Let $C = \{P(x,y) = 0\}$ be a curve in $\mathbb{A}^2$ (where $P$ is an irreducible polynomial). Then*

$$\mathcal{O}(C) = k[x,y]/(P)$$

*and*

$$k(C) = \mathrm{Frac}(k[x,y]/(P)) = \{f/g \mid f, g \in k[x,y], g \neq 0\} = k(x)[y]/(P).$$

**Definition 12.** *If $X$ is a prevariety, the **function field** $k(X)$ is the generic stalk of $\mathcal{O}_X$, i.e.*

$$k(X) = \varinjlim_{\text{non-empty open } U} \mathcal{O}_X(U).$$

- $k(X)$ equals the function field of each open affine set $U$ in $X$, because of density
- The elements of $k(X)$ are called **rational functions on** $X$

**Definition 13.** *Let $Y \subseteq X$ be an irreducible closed subset. Define*

$$\mathcal{O}_{Y,X} := \varinjlim_{\text{open sets } U \text{ s.t. } U \cap Y \neq \varnothing} \mathcal{O}_X(U).$$

Fix an open affine $U \subseteq X$ which meets $Y$. Let $R$ be the coordinate ring of $U$. Let $P = I(Y \cap U)$. Then

$$\mathcal{O}_{Y,X} = \varinjlim_{\text{open sets } U_f, f \in R, f \notin P} \mathcal{O}_X(U_f) = R_P.$$

So $\mathcal{O}_{Y,X}$ is a local ring with fraction field $k(X)$ and residue field $k(Y)$.

**Example 14.** *If $X = \mathbb{A}^2$ and $Y$ is a line in the plane, then taking $\mathcal{O}_{Y,X}$, we get all functions defined in a neighborhood of some point in $Y$ (i.e. on most points in $Y$), i.e. at a generic point of $Y$, or at **the generic point**.*

*Call this "the local ring at the generic point of $Y$."*

**Example 15.** *Let $X = \mathbb{A}^2, Y = x$-axis.*

*If $Z =$ origin, then $\mathcal{O}_{Z,X} = \mathcal{O}_Z = k[x,y]_{(x,y)}$ (invert everything that does not vanish at the origin)*

*$\mathcal{O}_{Y,X} =$ functions defined in a neighborhood of $y = 0 = k[x,y]_{(y)}$*

**Proposition 16.** *An open subset of a prevariety is a prevariety.*

Note that this was not true about affine varieties, since $\mathbb{A}^2 \setminus \{(0,0\}$ is not an affine variety. But since $\mathbb{A}^2$ is a prevariety, $\mathbb{A}^2 \setminus \{(0,0\}$ is a prevariety; we can write is as $\mathbb{A}^2 \setminus \{(y = 0\} \cup \mathbb{A}^2 \setminus \{(x = 0\}$.

Let $Y \subseteq X$ is a closed irreducible subset and $X$ is a prevariety. If $V$ is open in $Y$, define

$$\mathcal{O}_Y(V) := \{f : V \to k \mid \forall x \in V \exists \text{ a neighborhood } U \text{ of } x \text{ in } X, F \in \mathcal{O}_X(U) \text{ so that } f = F|_{U \cap V}\}$$

**Proposition 17.** *$(Y, \mathcal{O}_Y)$ is a prevariety.*

The last two propositions allow us to give a prevariety structure to any locally closed subset of a previariety $X$. Call the set of prevarieties obtained in this way **sub-prevarieties** of $X$.

**Example 18.** *We can obtain $\mathbb{P}^1$ by gluing two copies $U, V$ of $\mathbb{A}^1$:*

*If $u, v$ are the coordinates on $U, V$, let $U_0 = \{u \neq 0\} \subseteq U$ and $V_0 = \{v \neq 0\} \subseteq V$.*
*Then $\Gamma(U\mathcal{O}_U) = k[u]$, so $\Gamma(U\mathcal{O}_{U_0}) = k[u, u^{-1}]$ and similarly for $V$, $V_0$.*
*Define*

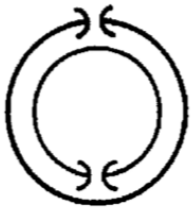
$$\varphi : U_0 \to V_0, u = a \mapsto v = \frac{1}{a}.$$

*This induces*

$$\varphi^* : k[v, v^{-1}] \to k[u, u^{-1}], v \mapsto u^{-1}, v^{-1} \mapsto u.$$

*$\varphi^*$ is an isomorphism of rings, so $\varphi$ is an isomorphism of varieties.*

*Patch $U$ and $V$ together via $\varphi$, i.e. form $U \cup V$ with $U_0$ and $V_0$ identified via $\varphi$. This space is homeomorphic to $\mathbb{P}^1$. We call this the variety $\mathbb{P}^1$:*



*If we had patched via $v \mapsto u, v^{-1} \mapsto u^{-1}$, we would get $\mathbb{A}^1$ with a point doubled, which is a prevariety but not a variety. It is not "separated" (this idea is the analog of Hausdorff in algebraic geometry).*

Improve the following exposition:
**Definition of Projective Varieties:**
Let $P \subseteq k[X_0, \ldots, X_n]$ be a homogeneous prime ideal. Let $X = V(P) \subseteq \mathbb{P}_n(k)$.

The elements of $k[X_0, \ldots, X_n]$ are not well-defined (because we can rescale anything, so

there are not unique coordinates) but the ratio of any two having the same degree is a function (rescaling in the numerator and denominator cancels out).

$P$ homogeneous $\implies$ $R = k[X_0, \ldots, X_m]/P$ is a graded ring $\bigoplus_{n=0}^{\infty} R_n$ and a domain. Let $k(X)$ be the zero-th piece of the localization of $R$ with respect to homogeneous elements, i.e.

$$k(X) = \left\{ \frac{f}{g} \mid f, g \in R_n \text{ for the same } n \right\}.$$

If $x \in X$ and $g \in R_n$, saying $g(x) \neq 0$ still makes sense. Define a ring in $k(X)$:

$$\mathcal{O}_{X,x} := \left\{ \frac{f}{g} \in k(X) \mid g(x) \neq 0 \right\}$$

and the maximal ideal

$$m_x := \left\{ \frac{f}{g} \in k(X) \mid g(x) \neq 0, f(x) = 0 \right\} \subseteq \mathcal{O}_{X,x}.$$

$\mathcal{O}_{X,x}$ is a local ring. Now define a sheaf: for all open $U \subseteq X$,

$$\mathcal{O}_X(U) := \bigcap_{x \in U} \mathcal{O}_{X,x}$$

We can identify $\mathcal{O}_X$ with a sheaf of $k$-valued functions:
If $x \in U$ and $a \in \Gamma(U, \mathcal{O}_X)$, then $a = f/g$ where $f, g \in R_n$, $g(x) \neq 0$.
Define $a(x) := f(x)/g(x)$ (i.e. lift $f, g$ to $F, G$ homogeneous polynomials of degree $n$ in $k[X_0, \ldots, X_n]$, let $\tilde{x}$ be a set of homogeneous coordinates of $x$, and take $F(\tilde{x})/G(\tilde{x})$)
Check: if $a \in \Gamma(U, \mathcal{O}_X)$ and $a(x) = 0$ for all $x \in U$, then $a = 0$. This will come out of the next step.

For all $i$ , $0 \leq i \leq n$,

$$(X \cap \mathbb{P}_n(k))_{X_i}, \text{ restriction of } \mathcal{O}_X)$$

is an affine variety (so in particular, $(X, \mathcal{O}_X)$ is locally isomorphic to an affine variety. Suffices to check this only for $X_0$:

For every homogeneous polynomial $F \in P$, $F/X_0^{\deg F}$ can be written as a polynomial $F'$ in the variables $\frac{X_1}{X_0}, \ldots, \frac{X_n}{X_0}$.
Let $P' \subseteq k[Y_1, \ldots, Y_n]$ be the ideal generated by all these $F'$.
Map $k[Y_1, \ldots, Y_n]$ into a subring of $k(X)$ via $Y_i \mapsto X_i/X_0$. The kernel of this map is $P'$ so $P'$ is prime.
We get an isomorphism

$$\varphi : (X \cap \mathbb{P}_n(k))_{X_0} \to X' = V(P') \subseteq k^n \text{ via } x \mapsto \left( \frac{x_1}{x_0}, \ldots, \frac{x_n}{x_0} \right).$$

For $x \in (X \cap \mathbb{P}_n(k))_{X_0}$, $\mathcal{O}_{X,x} = \{f/g \in k(X) \mid f, g, \text{ in same } R_m, g(x) \neq 0\}$.
$X'$ has affine coordinate ring $R' = k[Y_1, \ldots, Y_n]/P'$.

$\mathcal{O}_{X',\varphi(x)} = \{F/G \in k(X') \mid F, G \in R', G(\varphi(x)) \neq 0\}$.
$\varphi$ extends to an isomorphism $k(X') \to k(X)$.
Claim: the isomorphism takes $\mathcal{O}_{X',\varphi(x)}$ precisely onto $\mathcal{O}_{X,x}$. Proof:
It clearly (?) maps $\mathcal{O}_{X',\varphi(x)}$ into $\mathcal{O}_{X,x}$. And if $f, g \in R_m, g(x) \neq 0$, then

$$\frac{f}{g} = \frac{f/X_0^m}{g/X_0^m}$$

in $k(X)$ and $f/X_0^m, g/X_0^m$ come from $F, G$ in $R'$ with $G(\varphi(x)) \neq 0$.
So the local rings correspond, and since the sheaves are defined by intersecting local rings, they correspond.

**Definition 19.** *Let $X, Y$ be prevarieties. A map $f : X \to Y$ is a **morphism** if $f$ is continuous and for all open sets $V$ in $Y$,*

$$g \in \Gamma(V, \mathcal{O}_Y) \implies g \circ f \in \Gamma(f^{-1}(V), \mathcal{O}_X).$$

**Proposition 20.** *Let $f : X \to Y$ be any map. Let $\{V_i\}$ be a collection of open affine subsets covering $Y$. Suppose that $\{U_i\}$ is an open covering of $X$ such that 1) $f(U_i) \subseteq V_i$, and 2) $f^*$ maps $\Gamma(V_i, \mathcal{O}_Y)$ into $\Gamma(U_i, \mathcal{O}_X)$. Then $f$ is a morphism.*

**Example 21.** *Let $C$ be the plane cubic curve defined in homogeneous coordinates by*

$$zy^2 = x(x^2 - z^2).$$

*In $C \cap \mathbb{P}_2(k)_x$, with affine coordinates $X = x/z, Y = y/z$, the equation of $C$ becomes $Y^2 = X(X^2 - 1)$. For all lines $\ell$ through the origin, interchange the two points in $\ell \cap C$ (other than the origin).*

*When we write this out (Mumford page 31), we do not get polynomials. The problems are $P_1 = (0 : 1 : 0)$ (point at $\infty$), $P_2 = (0 : 0 : 1)$ (origin), $Q_1 = (1 : 0 : 1)$ and $Q_2 = (-1 : 0 : 1)$ (other points on $X$-axis).*

*But we can restrict to different subsets of $C$ and get morphisms there, and then, using the Proposition, $f$ will be a morphism (see Mumford page 32).*

***Note:** On a cubic curve, there are 9 inflection points. Using the operation we defined above, we get a group, and the 9 inflection points become 9 3-torsion points.*

**Problem 1.3.** ***Generalize the result by which we covered $\mathbb{P}_n$ by open affine sets as follows: for all homogeneous polynomials $H \in k[X_0, \ldots, X_n]$ of positive degree, show that $\mathbb{P}_n(k)_H$ is an affine variety.***

*Let $H \in k[x_0, \ldots, x_n]$ be a homogeneous polynomial of positive degree $d > 0$. $\mathbb{P}_n(k)_H = \{x \in \mathbb{P}_n(k) \mid H(x) \neq 0\}$. Want to show: there exists an isomorphism from $\mathbb{P}_n(k)_H$ to some closed algebraic set inside some affine space $\mathbb{A}^N(k)$.*

*Regular functions on $\mathbb{P}_n(k)_H$ can be represented by fractions $G/H^m$ where $G$ is a ho-*

*mogeneous polynomial of degree $m \cdot d$ ($d = \deg H$).*

*Let $M_1, \ldots, M_N$ be all monomials of degree $d$ in $x_0, \ldots, x_n$. (so $N = \binom{n+d}{d}$)*
*$H$ homogeneous polynomial of degree $d \implies H = \sum_{i=1}^{N} c_i M_i$ some $c_i \in k$. Define*

$$\psi : \mathbb{P}_n(k)_H \to \mathbb{A}^N(k), \quad (x_0 : \cdots : x_n) \mapsto \left( \frac{M_1(x)}{H(x)}, \ldots, \frac{M_N(x)}{H(x)} \right)$$

*This is well-defined: since $M_i$ and $H$ both have degree $d$, replacing $x_i$ with $\lambda x_i$ causes the $\lambda^d$ terms to cancel.*
*This is regular: $H(X)$ is nonzero on $\mathbb{P}_n(k)_H$.*

*The image of $\psi$ is a closed affine variety:*

- *Let $(Y_1, \ldots, Y_N)$ be the coordinates of $\mathbb{A}^N$.*
- *Notice that $\sum_{i=1}^{N} c_i \frac{M_i}{H} = \frac{\sum c_i M_i}{H} = H/H = 1$, so the image of $\psi$ must lie entirely on the hyperplane defined by $\sum_{i=1}^{N} c_i Y_i = 1$ (*)*
- *Also, any algebraic relations that hold between $M_i$ will translate into polynomial equations among the $Y_i$.*
- *The vanishing set of all these polynomial relations (including (*)) defines a closed algebraic set $V \subseteq \mathbb{A}^N$.*

*So $\mathrm{Im}\psi \subseteq V$.*

*Now, let $\varphi : V \to \mathbb{P}_n(k)_H$ be defined as:*
*Let the coordinates of $\mathbb{A}^N$ be labeled by monomials of degree $d$, so we have $Y_0, \ldots, Y_n$ corresponding to $M_1 = x_0^d, \ldots, M_n = x_n^d$, and $y_{j,i}$ corresponding to $x_j x_i^{d-1}$ for all $j \neq i$.*

*On $Y_i \neq 0$, $\varphi(Y) := (Y_{0,i} : \cdots : Y_{n,i})$.*
*Recall that*

$$Y_{j,i} = \frac{x_j x_i^{d-1}}{H(x)}, \quad Y_i = \frac{x_i^d}{H(x)}$$

*so*

$$(\varphi \circ \psi)(x_0 : \cdots : x_n) = \varphi(Y) = \varphi\left( \left( \frac{x_0 x_i^{d-1}}{H(x)} : \cdots : \frac{x_n x_i^{d-1}}{H(x)} \right) \right) = (x_0 : \cdots : x_n).$$

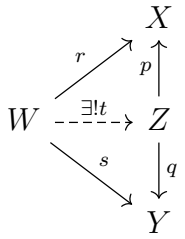
*Thus $\varphi$ is an inverse morphism (given by polynomials) for $\psi$. Thus $\psi$ is an isomorphism onto $V$, so $\mathbb{P}_n(k)_H$ is isomorphic to the affine variety $V$.*

*We call the map $\psi$ the **d-uple / Veronese embedding (see Hartshorne exercise 2.12)**.*

## 1.6 Products and the Hausdorff Axiom

Product of Zariski topologies in $\mathbb{A}^n$ and $\mathbb{A}^m$ is too weak to give the Zariski topology in $\mathbb{A}^{n+m}$. For example, in $\mathbb{A}^1 \times \mathbb{A}^1$, the only closed sets in the product topology are finite unions of horizontal and vertical lines.

**Definition 1.** *Let $\mathcal{C}$ be a category. Let $X, Y$ be objects in $\mathcal{C}$. An object $Z$ plus two morphisms $p : Z \to X$ and $q : Z \to Y$ is a **product** if it has the following UMP: for all objects $W$ and morphisms $r : W \to X, s : W \to Y$, there is a unique morphism $t : W \to Z$ such that $r = p \circ t$, $s = q \circ t$, i.e. such that the following diagram commutes:*




*The induced morphism $t$ will be denoted $(r, s)$.*
*We call $p, q$ the **projections** of the product onto its factors.*
*A product, if it exists, is unique up to a unique isomorphism commuting with the projections.*

The requirement of the definition can be rephrased as:

$$\mathrm{Hom}(W, Z) \cong \mathrm{Hom}(W, X) \times \mathrm{Hom}(W, Y)$$

We know: The underlying set must be the set-theoretic product $X \times Y$.
To see this, plug $W = \{\mathrm{pt}\}$ into $\mathrm{Hom}(W, Z) \cong \mathrm{Hom}(W, X) \times \mathrm{Hom}(W, Y)$. Get:

$$Z = \mathrm{Hom}(\{\mathrm{pt}\}, Z) \cong \mathrm{Hom}(\{\mathrm{pt}\}, X) \times \mathrm{Hom}(\{\mathrm{pt}\}, Y) = X \times Y$$

**Proposition 2.** *If $X, Y$ are affine varieties with coordinate rings $R, S$:*

1. *There is a product prevariety $X \times Y$.*
2. *$X \times Y$ is affine with coordinate ring $R \otimes_k S$.*
3. *A basis of the topology is given by the open sets defined by $\sum f_i(x) g_i(y) \neq 0$, $f_i \in R, g_i \in S$.*
4. *$\mathcal{O}_{(x,y)}$ is the localization of $\mathcal{O}_x \otimes_k \mathcal{O}_y$ at $m_x \mathcal{O}_y + \mathcal{O}_x m_y$.*

Recall (relevant): for irreducible algebraic sets, the sheaf of functions equals the ring of functions, i.e. if $X$ is an irreducible algebraic set and $R$ is the coordinate ring $\Gamma(X)$ of $X$, then for $f \in R$, $\mathcal{O}_X(X_f) = R_f$. So the structure sheaf $\mathcal{O}_X$ is completely determined by the ring $R$, and its global sections recover $R$.

**Theorem 3.** *For any prevarieties $X$ and $Y$ over $k$, the product $X \times Y$ exists.*

**Remark 4.** *1. If $U \subset X$ is an open subprevariety, then $U \times Y$ is an open subprevariety of $X \times Y$.*

*2. If $Z \subset X$ is a closed subprevariety, $Z \times Y$ is a closed subprevariety of $X \times Y$.*

Terminology: a **ringed space** is a pair $(X, \mathcal{O}_X)$ where $X$ is any topological space and $\mathcal{O}_X$ is any sheaf of rings.

**Theorem 5.** *The product of two projective varieties is a projective variety.*

- Proven by embedding $\mathbb{P}_n \times \mathbb{P}_m$ into $\mathbb{P}_{nm+n+m}$ using the Segre embedding.
- Map: $(x_0 : \cdots : x_n) \times (y_0 : \cdots : y_m) \mapsto (U_{ij} = x_i y_j)$.
- The image is a closed subvariety (intersection of quadrics).

**Example 6.** *Consider the embedding*

$$I : \mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3,$$

$$(u : v) \times (s : t) \mapsto (x = us : \quad y = vt : \quad z = ut : \quad w = vs)$$

- *The image of $I$ is $xy = zw$ (a quadric)*
- *Our quadric is isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$*
- *The images $\{(u_0 : v_0)\} \times \mathbb{P}^1$ and $\mathbb{P}^1 \times \{(s_0 : t_0)\}$ give two rulings of lines of our quadric.*
- *Over an algebraically closed field, all nondegenerate quadrics (those corresponding to nondegenerate quadric forms) are isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$.*

**Definition 7** (Hausdorff axiom)**.** *A prevariety $X$ is a **variety** if for all prevarieties $Y$ and morphisms $f, g : Y \to X$, the set $\{y \in Y | f(y) = g(y)\}$ is closed in $Y$.*

**Example 8.** *Let $X$ be a variety, $Y = X \times X$, $f = p_1, g = p_2$. The diagonal morphism is*

$$\Delta : X \to X \times X, \quad x \mapsto (x, x)$$

*So*

$$\Delta(X) = \{z \in X \times X \mid p_1(z) = p_2(z)\}$$

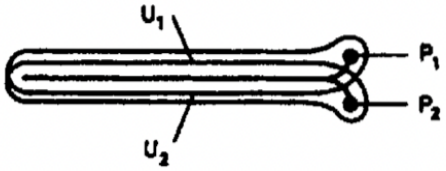
*is closed by the Hausdorff axiom.*

The converse also holds:

**Proposition 9.** *Let $X$ be a prevariety: $X$ is a variety $\iff$ $\Delta(X) \subset X \times X$ is closed.*

**Remark 10.** *People typically use the above as the definition for variety. Also, people would typically say $X$ "is separated", rather than saying $X$ "satisfies the Hausdorff axiom."*

*It is more common to refer to Mumford's prevarieties as varieties and refer to Mumford's varieties as separated varieties.*




**Example 11.** *The "line with two origins" $X$ (patching two copies of $\mathbb{A}^1$) is not a variety, because if $i_1, i_2 : \mathbb{A}^1 \to X$, then $\{y \in \mathbb{A}^1 \mid i_1(y) = i_2(y)\} = \mathbb{A}^1 \setminus \{0\}$ is not closed in $\mathbb{A}^1$ (closed sets in $\mathbb{A}^1$ are either all of $\mathbb{A}^1$ or finite sets of points)*

**Remark 12.** *1. Subprevarieties of a variety are varieties.*

2. *A product of 2 varieties is a variety.*
3. *Affine varieties are varieties.*
4. *If $f, g : X \to Y$ are morphisms of prevarieties, $\{y \in Y \mid f(y) = g(y)\}$ is always locally closed.*
5. *If $f : X \to Y$ is a morphism of prevarieties and $Y$ is a variety, then the graph $\Gamma_f$ of $f$ is closed. And it is a closed subprevariety of $X \times Y$ isomorphic to $X$.*

**Proposition 13.** *Let $X$ be a prevariety, Assume that for all $x, y \in X$, there is an open affine $U$ containing both $x$ and $y$. Then $X$ is a variety.*

**Corollary 14.** *Every projective variety $X$ is a variety.*

To see this: for every $x, y \in \mathbb{P}_n(k)$ there is a hyperplane $H$ not containing $x$ or $y$, so take $\mathbb{P}_n(k)_H$ as your open set.

**Proposition 15.** *Let $X$ be a variety, and let $U, V$ be affine open subsets with coordinate rings $R, S$. Then $U \cap V$ is an affine open subset with coordinate ring $R \cdot S \subseteq k(X)$.*

**Remark 16.** *As discussed in class, the ring compositum $\neq$ the field compositum:*

$$k(x) \cdot k(y) = k(x, y)$$

*is the field compositum, but*

$$k(x) \otimes_k k(y)$$

*is the ring compositum, and these are not equal because e.g. $\frac{1}{x+y}$ lies in $k(x, y)$ but not $k(x) \otimes_k k(y)$.*
*Observe:*

$$\mathrm{Frac}k[x] \otimes \mathrm{Frac}k[y] = k(x) \otimes_k k(y) \neq k(x, y) = \mathrm{Frac}k[x, y]$$

*The function field of $\mathbb{A}^1 \times \mathbb{A}^1$ is $k(x, y)$, and this does not equal the tensor product of the function fields $k(x)$, $k(y)$ of each copy of $\mathbb{A}^1$.*

**Example 17.** *Let $X = \mathbb{A}^2$, $U = \{x \neq 0\}$, $V = \{y \neq 0\}$. Then if $R$ is the coordinate ring of $U$, $R = k[x, y, x^{-1}]$, and if $S$ is the coordinate ring of $V$, $S = k[x, y, y^{-1}]$.*

*Now the proposition says that the coordinate ring of $U \cap V$ is $R \cdot S = k[x, y, x^{-1}, y^{-1}] \subsetneq k(x, y) = k(\mathbb{A}^2)$ (so here it was important that we took the ring compositum, not the field compositum).*

**Problem 1.4.** ***Prove the following converse: let $X$ be a prevariety, $\{U_i\}$ an affine open covering of $X$. Let $R_i$ be the coordinate ring of $U_i$. Then if $U_i \cap U_j$ is an affine subset of $X$ with coordinate ring $R_i \cdot R_j$, $X$ is a variety.***

*Want to show: $\Delta \subseteq X \times X$ is closed.*
*$\{U_i\}$ affine open cover of $X$ $\implies$*

$$X \times X = \bigcup_{i,j} (U_i \times U_j)$$

*A subset of a space is closed $\iff$ its intersection with every member of an open cover is closed.*

*Suffices to show: $\Delta \cap (U_i \times U_j)$ is closed in $U_i \times U_j$ for all $i, j$. So fix $i, j$:*
*By hypothesis, $U_i \cap U_j$ is affine with coordinate ring $R_i \cdot R_j$. Let $\delta : U_i \cap U_j \to U_i \times U_j, \quad x \mapsto (x, x)$. Every function in $R_i$ and every function in $R_j$ restricts to a regular function on $U_i \cap U_j$, so there is a natural homomorphism (actually induced by $\delta$)*

$$R_i \otimes R_j \to R_{ij}, \quad f \otimes g \mapsto fg.$$

*Since $R_{ij} = R_i \cdot R_j$, this map is surjective. Thus $R_{ij} \cong (R_i \otimes R_j)/I$ for some ideal $I$. By Chapter 1, $(R_i \otimes R_j)/I$ is the coordinate ring of the closed subvariety $V(i) \subseteq U_i \times U_j$.*

*$\delta$ induces an isomorphism*

$$\Gamma(V(I)) \cong \Gamma(U_i \cap U_j)$$

*so $U_i \cap U_j \cong V(I)$ as affine varieties. Thus $\delta(U_i \cap U_j)$ is closed in $U_i \times U_j$. Then $\Delta$ itself is closed in $X \times X$, since $\{U_i \times U_j\}$ form an open cover. So $X$ is a variety.*

**Definition 18.** *Let $\mathcal{O} \subseteq \mathcal{O}'$ be local rings. We say $\mathcal{O}'$ **dominates** $\mathcal{O}$ ($\mathcal{O}' > \mathcal{O}$) if $m' \supset m$ ($\iff m' \cap \mathcal{O} = m$) where $m, m'$ are the maximal ideals of $\mathcal{O}, \mathcal{O}'$ respectively.*

**Example 19.** *Let $\mathcal{O} = k[x, y]_{(x,y)}$. This embeds into $\mathcal{O}' = k[x, z]_{(x,z)}$ via $x \mapsto x, y \mapsto xz$.*

*$m = (x, y) = (x, xz)$ and $m' = (x, z)$, so $m \subseteq m'$.*

*$\mathcal{O}'$ dominates $\mathcal{O}$.*

**Local Criterion:** Let $X$ be a prevariety. Then $X$ is a variety $\iff$ for all $x, y \in X$ such that $x \neq y$, there is no local ring $\mathcal{O} \subseteq k(X)$ such that $\mathcal{O} > \mathcal{O}_x$ and $\mathcal{O} > \mathcal{O}_y$.

The fact that this holds for varieties will be proven later.

(Chevalley-Nagata) A variety is a finitely generated field extension $K$ of $k$ plus a collection of local subrings of $K$ satisfying various conditions.

**Problem 1.5.** ***Let*** $X \subseteq \mathbb{P}^n$ ***be defined by a homogeneous ideal*** $P \subseteq k[X_0, \ldots, X_n]$***. Let*** $P$ ***define the affine variety*** $X^* \subseteq \mathbb{A}^{n+1}$***. Show that for all*** $i$***,***

$$\{\textbf{\textit{Subset of}}\ X^*\ \textbf{\textit{where}}\ X_i \neq 0\} \cong (\mathbb{A}^1 \setminus \{0\}) \times \{\textbf{\textit{Subset of}}\ X\ \textbf{\textit{where}}\ X_i \neq 0\}$$

*Let* $W_i := \{a \in X^* \mid a_i \neq 0\}$.
*Let* $U_i := \{(a_0, \ldots, a_n) \in \mathbb{P}^n \mid a_i \neq 0\} \cong \mathbb{A}^n$.
*Let* $X_i := X \cap U_i$.
*Want to show:* $W_i \cong (\mathbb{A}^1 \setminus \{0\}) \times X_i$.
*Define*

$$\Phi : W_i \to (\mathbb{A}^1 \setminus \{0\}) \times X_i, \quad (a_0, \ldots, a_n) \mapsto (a_i, (a_0 : \cdots : a_n))$$

*This lands in* $(\mathbb{A}^1 \setminus \{0\}) \times X_i$*:*
$(a_0, \ldots, a_n) \in W_i \implies a_i \neq 0 \implies (a_0 : \cdots : a_n)$ *is a valid projective point*
$a \in X^* \implies$ *every homogeneous polynomial in* $P$ *vanishes at* $a \implies (a_0 : \cdots : a_n) \in X$ *(and* $a_i \neq 0$ *so* $(a_0 : \cdots : a_n) \in X_i$*)*

*Constructing an inverse:*
*Let* $\psi : (\mathbb{A}^1 \setminus \{0\}) \times X_i \to W_i$ *be the map so that if* $(t, p) \in (\mathbb{A}^1 \setminus \{0\}) \times X_i$*, then* $p \in X_i \implies p = (b_0 : \cdots : b_n)$ *with* $b_i \neq 0$*. And* $p \in U_i \implies \exists$ *unique representative satisfying* $b_i = 1$*:* $p = (b_0 : \cdots : b_{i-1} : 1 : b_{i+1} : \cdots : b_n)$*. Send:*

$$(t, p) \mapsto (tb_0, \ldots, tb_{i-1}, t, tb_{i+1}, \ldots, tb_n)$$

*This lands in* $W_i$*:*
*If* $f \in P$ *is homogeneous of degree* $d$*,* $p \in X \implies f(b_0, \ldots, 1, \ldots, b_n) = 0$*. And* $f$ *homogeneous* $\implies f(tb_0, \ldots, t, \ldots, tb_n) = t^d f(b_0, \ldots, 1, \ldots, b_n) = 0$*. So every polynomial of* $P$ *vanishes on* $\psi(t, p)$*, meaning* $\psi(t, p) \in X^*$*. If* $t \neq 0$*, its* $i$*-th coordinate equals* $t$*, so it lies in* $W_i$*.*

*This is an inverse:*
$a \in W_i \implies \Phi(a) = (a_i, (a_0 : \cdots : a_n))$*, and* $(a_0 : \cdots : a_n) = (a_0/a_i : \cdots : 1 : \cdots : a_n/a_i)$*, so*

$$\psi(\Phi(a)) = \left(a_i \frac{a_0}{a_i}, \ldots, a_i, \ldots, a_i \frac{a_n}{a_i}\right) = (a_0, \ldots, a_n) \implies \psi \circ \Phi = \mathrm{id}.$$

*Conversely, if* $(t,p) \in k^\times \times X_i$, *write* $p = (b_0 : \cdots : 1 : \cdots : b_n)$.

$$\psi(t,p) = (tb_0, \ldots, t, \ldots, tb_n) \implies$$

$$\Phi(\psi(t,p)) = (t, (tb_0 : \cdots : t : \cdots : tb_n)) = (t, (b_0 : \cdots : 1 : \cdots : b_n)) = (t,p)$$

*so* $\Phi \circ \psi = \mathrm{id}$.

*These maps are regular:*
*We can identify* $X_i$ *with affine variety having coordinates* $y_j = x_j/x_i, j \neq i$ *(this is from the* $\mathbb{P}_n(k)_{X_i} \cong \mathbb{A}^n$ *isomorphism).*
*So under these coordinates,* $\Phi(a) = (a_i, a_0/a_i, \ldots, a_{i-1}/a_i, a_{i+1}/a_i, \ldots, a_n/a_i)$ *and* $a_i \neq 0 \implies$ *these are regular on* $W_i$.
*Similarly,* $\psi(t, y_0, \ldots, y_{i-1}, y_{i+1}, \ldots, y_n) = (ty_0, \ldots, t, \ldots, ty_n)$, *which is polynomial.*
*Thus, they define an isomorphism of varieties.*

## 1.7 Dimension

**Definition 1.** *Let* $X$ *be a variety.* $\dim X := \mathrm{trdeg}_k k(X)$.

If $U$ is an open nonempty set in $X$, then $\dim U = \dim X$ since $k(U) = k(X)$.

Since $k$ is algebraically closed, TFAE:

1. $\dim X = 0$
2. $k(X) = k$
3. $X$ is a point.

**Example 2.** *If* $k = \mathbb{R}$, *and* $X = V(x^2+1) = \varnothing$ *in* $\mathbb{A}^1_{\mathbb{R}}$, *then* $\mathbb{R}[x]/(x^2+1) \cong \mathbb{C}$ *is a field, so* $\dim X = 0$ *but* $X = \varnothing$. *Thus when* $k$ *is not algebraically closed,* $\dim X = 0 \not\implies X$ *is a point.*

**Example 3.**
1. $k(\mathbb{A}^n) = k(x_1, \ldots, x_n) \implies \dim \mathbb{A}^n = n$.
2. $\dim(\mathbb{A}^m \times \mathbb{A}^n) = m + n$
3. $\mathbb{P}^1 \times \mathbb{P}^1 \hookrightarrow \mathbb{P}^3$, *image* $xy = zw$
   $\dim(\mathbb{P}^1 \times \mathbb{P}^1) = 1 + 1 = 2$ *so quadric surfaces have dimension 2.*
4. $C = V(y^2 - x^3) \subseteq \mathbb{A}^2 \implies k(C) \cong k(t)$ *via* $x = t^2, y = t^3 \implies \dim C = 1$.
5. $k(\mathbb{P}^n) = k(x_1/x_0, \ldots, x_n/x_0) \implies \dim \mathbb{P}^n = n$.

**Remark 4.**
- *Dimension of a variety is a birational invariant. Recall:*
  - *Rational map = equivalence class of morphisms that agree on a dense open subset of a domain*
  - *Birational map = rational map with rational inverse*

- *Removing a proper closed subset does not change dimension.*

**Proposition 5.** *Let $Y$ be a proper closed subvariety of $X$. Then* $\dim Y < \dim X$.

The proof uses this lemma:

**Lemma 6.** *Let $R$ be an integral domain over $k$, $P \subseteq R$ a prime. Then* $\operatorname{trdeg}_k R \geq \operatorname{trdeg}_k R/P$, *with equality only if $P = 0$ or both sides are $\infty$.*

In the situation of the proposition, call $\dim X - \dim Y$ the **codimension** of $Y$ in $X$.

**Theorem 7** (Geometric Version of Krull's Principal Ideal Theorem)**.** *Let $X$ be a variety, $U \subseteq X$ open, $g \in \Gamma(U, \mathcal{O}_X)$, $Z$ an irreducible component of $\{x \in U \mid g(x) = 0\}$. Then if $g \neq 0$,* $\dim Z = \dim X - 1$.

**Theorem 8** (Algebraic Version of Krull's Principal Ideal Theorem)**.** *Let $R$ be a finitely generated integral domain over $k$, $f \in R$, $P$ an isolated prime ideal of $(f)$ (i.e. minimal among the prime ideals containing it). Then if $f \neq 0$,* $\operatorname{trdeg}_k R/P = \operatorname{trdeg}_k R - 1$.

**Example 9.** *$V_{\mathbb{R}}(x^2 + y^2) = (0, 0)$ so has codimension 2 in $\mathbb{A}^2_{\mathbb{R}}$. This does not contradict Theorem 7 because $\mathbb{R}$ is not algebraically closed.*
*$V_{\mathbb{C}}(x^2 + y^2) = \{x = \pm yi\} = \{x = yi\} \cup \{x = -yi\}$ has codimension 1 in $\mathbb{A}^2_{\mathbb{C}}$.*
*$x^2 + y^2$ is irreducible over $\mathbb{R}$, reducible over $\mathbb{C}$.*

*$V_{\mathbb{R}}(x^2 + y^2)$ is irreducible but not geometrically irreducible.*

**Example 10.** *$V(xy) = V(x) \cup V(y)$ is reducible and pure codimension 1.*

**Example 11.** *Claim: There is no surjective morphism*

$$f : \mathbb{P}^2 \to \mathbb{P}^1.$$

*Recall (Bézout's Theorem):*
*If two projective plane curves of positive degree have no common irreducible component, then they intersect.*

*Suppose that*

$$f : \mathbb{P}^2 \to \mathbb{P}^1$$

*is a surjective morphism. Then*

$$f = (F : G),$$

*where $F$ and $G$ are homogeneous polynomials of the same degree with no common zero. Choose two distinct points*

$$(a : b), (c : d) \in \mathbb{P}^1.$$

*Their inverse images are the curves*

$$f^{-1}(a : b) = V(bF - aG), \qquad f^{-1}(c : d) = V(dF - cG).$$

*Since $f$ is surjective, both curves are nonempty.*

*If these curves met at a point $p$, then*

$$(F(p) : G(p)) = (a : b) = (c : d),$$

*contradicting the fact that $(a : b) \neq (c : d)$. Hence the two curves are disjoint.*

*They also cannot have a common irreducible component, since this would force them to intersect. Therefore, by Bézout's theorem, they must intersect, a contradiction.*

**Definition 12.** *Let $f : X \to Y$ be a morphism of affine varieties. Let $R$ and $S$ be the coordinate rings of $X$ and $Y$, and let $f^* : S \to R$ be the induced homomorphism. Then $f$ is* ***finite*** *if $R$ is integral over $f^*(S)$.*

**Remark 13.** *The restriction of a finite morphism from $X$ to $Y$ to a closed subvariety $Z$ of $X$ is also finite.*

**Example 14.**

1. *If* $\mathrm{char}k = p \neq 0$*, the map $f : \mathbb{A}^1 \to \mathbb{A}^1, t \mapsto t^p$ is finite. This induces $k[X^p] \hookrightarrow k[X]$.*
2. *If $k$ is algebraically closed, $f : \mathbb{A}^1 \to \mathbb{A}^2, t \mapsto (t^2, t^3)$ is finite. This induces $k[T^2, T^3] \hookrightarrow k[T]$.*
3. *$f : \mathbb{A}^1 \to \mathbb{A}^2$ defined by $X = t^2 - 1, \quad Y = t(t^2 - 1)$ is NOT finite. This induces $k[T^2 - 1, T(T^2 - 1)] \hookrightarrow k[T, (T - 1)^{-1}]$.*

**Proposition 15.** *Let $f : X \to Y$ be a finite morphism of affine varieties.*

1. *Then $f$ is a closed map.*
2. *For all $y \in Y, f^{-1}(y)$ is a finite set.*
3. *$f$ is surjective $\iff$ the corresponding map $f^*$ of coordinate rings is injective.*

**Geometric form of Noether's normalization lemma.**
Let $X$ be an affine variety of dimension $n$. Then there exists a finite surjective morphism $\pi : X \to \mathbb{A}^n$.

**Definition 16.** *Let $X$ be a variety, and let $Z \subseteq X$ be a closed subset. Then $Z$ has* ***pure dimension*** *$r$ if each of its components has dimension $r$ (similarly for* ***pure codimension*** *$r$).*

The conclusion of Theorem 7 may be stated as: $V(g)$ has pure codimension 1, for any nonzero $g \in \Gamma(X, \mathcal{O}_X)$.

The converse holds: if $Z$ is an irreducible closed subset of a variety $X$ of codimension 1, then for all open sets $U$ with $Z \cap U \neq \varnothing$ and for all nonzero functions $f$ on $U$ vanishing on $Z$, $Z \cap U$ is a component of $f = 0$.

**Corollary 17.** *Let $X$ be a variety and $Z$ a maximal closed irreducible subset, smaller than $X$ itself. Then* $\dim Z = \dim X - 1$.

**Corollary 18** (Topological characterization of dimension)**.** *Suppose*

$$\varnothing \neq Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_r = X$$

*is any maximal chain of closed irreducible subsets of $X$. Then* $\dim X = r$.

**Corollary 19.** *Let $X$ be a variety and let $Z$ be a component of $V(f_1, \ldots, f_r)$, where $f_1, \ldots, f_r \in \Gamma(X, \mathcal{O}_X)$. Then* $\mathrm{codim} Z \leq r$.

Equality need not hold: e.g. take $f_1 = \cdots = f_r, r > 1$.

**Corollary 20.** *Let $U$ be an affine variety, $Z$ a closed irreducible subset. Let $r = \mathrm{codim} Z$. Then there exist $f_1, \ldots, f_r$ in $R = \Gamma(U, \mathcal{O}_U)$ such that $Z$ is a component of $V(f_1, \ldots, f_r)$.*

Recall: every noetherian local ring $\mathcal{O}$ has a Krull dimension, which is defined as:

1. the length $r$ is the longest chain of prime ideals $P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_r = M$ where $M$ is the maximal ideal of $\mathcal{O}$, or
2. the least integer $n$ such that there exist elements $f_1, \ldots, f_n \in M$ for which $M = \sqrt{(f_1, \ldots, f_n)}$

Also recall that every irreducible closed subset $Z$ of every variety $X$ has a local ring $\mathcal{O}_{Z,X}$.

**Corollary 21.** *The Krull dimension of $\mathcal{O}_{Z,X}$ is* $\dim X - \dim Z$.

Suppose $Z \subseteq X$ is irreducible and of codimension 1. Question: is it true that for all $y \in Z$, there is some neighborhood $U$ of $y$ in $X$ and some function $f \in \Gamma(U, \mathcal{O}_X)$ such that $Z \cap U$ is not just a component of $f = 0$ but equal to it.

More generally, if $Z \subseteq X$ is a closed subset of pure codimension $r$, is it true that for all $y \in Z$, there is a neighborhood $U$ of $y$ and functions $f_1, \ldots, f_r \in \Gamma(U, \mathcal{O}_X)$ such that $Z \cap U = V(f_1, \ldots, f_r)$?

Even if $Z$ is irreducible of codimension 1, this is not always true. A closed set $Z$ with this property is referred to as a **local set-theoretic complete intersection**.

**Proposition 22.** *Let $X$ be an affine variety with coordinate ring $R$. Assume $R$ is a UFD. Then every closed subset $Z \subseteq X$ of pure codimension 1 equals $V(f)$ for some $f \in R$.*

**Proposition 23.** $\dim X \times Y = \dim X + \dim Y$.

We also have projective versions of all of these:
Let $X \subseteq \mathbb{P}_n(k)$ be a projective variety and let $I(X) \subseteq k[X_0, \ldots, X_n]$ be its ideal.

**Theorem 24.** *If $f \in k[X_0, \ldots, X_n]$ is homogeneous and not a constant and $f \notin I(X)$, then $X \cap V(f)$ is non-empty and of pure codimension 1 in $X$, unless* $\dim X = 0$.

**Corollary 25.** *If $f_1, \ldots, f_r \in k[X_0, \ldots, X_n]$ are homogeneous and not constants, then all components of $X \cap V(f_1, \ldots, f_r)$ have codimension at most $r$ in $X$. And if $\dim X \geq r$, then $X \cap V(f_1, \ldots, f_r)$ is non-empty.*

**Corollary 26.** *If $Y \subseteq X$ is a closed subvariety (resp. $Y$ is the empty subset), then there exist homogeneous non-constant elements $f_1, \ldots, f_r \in k[X_0, \ldots, X_n]$ where $r =$ codimension of $Y$ (resp. $r = \dim X + 1$) such that $Y$ is a component of $X \cap V(f_1, \ldots, f_r)$ (resp. $X \cap V(f_1, \ldots, f_r) = \varnothing$).*

This implies that every space curve (i.e. one-dimensional subvariety of $\mathbb{P}_3(k)$) is a component of an intersection $H_1 \cap H_2$ of 2 surfaces.

**Proposition 27.** *Every closed subset of $\mathbb{P}_n(k)$ of pure codimension 1 is a hypersurface, i.e. equals $V(f)$ for some homogeneous element $f \in k[X_0, \ldots, X_n]$.*

**Proposition 28.** *Suppose $f : \mathbb{P}^n \to \mathbb{P}^m$ is a morphism. Assume $W = f(\mathbb{P}^n)$ is closed (this turns out to always be true). Then either $W$ is a single point, or $\dim W = n$.*

**Example 29.** *The projection map $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^1$ has dimension 1 image.*

*Hence $\mathbb{P}^1 \times \mathbb{P}^1$ is very different from $\mathbb{P}^2$. But $k(\mathbb{P}^2) = k(\mathbb{P}^1 \times \mathbb{P}^1) = k(x, y)$, and each contains $\mathbb{A}^2$ as a dense open.*

**Example 30.** *Compute the total ring of fractions of $\mathbb{Z}[t]/(t^2)$. Observe: e.g. $a+t$ (for $a \neq 0$) is inverted after inverting all nonzero integers, since*

$$(a + t)(a - t) = a^2 - t^2 = a^2$$

*so $\frac{a-t}{a^2}$ is the inverse. Conclude*

$$\mathrm{Tot}(\mathbb{Z}[t]/(t^2)) = \mathbb{Q}[t]/(t^2)$$

*and another example of total ring of fractions (this time for a disconnected space):*

$$\mathrm{Tot}(k[x] \times k[y, z]) = k(x) \times k(y, z) = \Gamma(\mathbb{A}^1) \times \Gamma(\mathbb{A}^2) = \Gamma(\mathbb{A}^1 \sqcup \mathbb{A}^2)$$

*$k[x] \times k[y, z]$ has idempotents $(1, 0)$ and $(0, 1)$. Idempotents $\implies$ disconnected. However, while disconnected $\implies$ not domain, not domain $\not\implies$ disconnected.*

*Similarly, the total ring of fractions of $k[x, t]/(t^2)$ is $k(x)[t]/(t^2)$. Its spectrum has one underlying point, but the ring is nonreduced because it contains the nilpotent class of $t$.*

## 1.8 The fibers of a morphism

Let $f : X \to Y$ be a morphism of varieties.

**Definition 1.** *A morphism $f : X \to Y$ is **dominating** if its image is dense in $Y$, i.e. $Y = \overline{f(X)}$.*

**Proposition 2.** *If $f : X \to Y$ is any morphism, let $Z = \overline{f(X)}$. Then $Z$ is irreducible, the restricted morphism $f' : X \to Z$ is dominating, and $f'$ induces an injection*

$$f'^* : k(Z) \hookrightarrow k(X).$$

So when we study fibers of an arbitrary morphism, we can focus on the case of dominating morphisms.
Our situation thus is: given $f : X \to Y$, we get $k(Y) \hookrightarrow k(X)$. But this is not well-defined unless $f$ is dominating (e.g. given a morphism from a point to a line, there is no induced homomorphism of function fields $k(Y) \to k(X)$ because the generic point of $Y$ is not in the image). For this reason, we focus on the dominating case.

If $f : X \to Y$ is a map between affine varieties, it induces a map between the coordinate rings of the varieties. But if we remove the "affine" conditions, we get a map between the function fields. Homomorphisms between fields are always injective. So we end up with a well-defined injective morphism.

Note that a finite morphism is dominating $\iff$ it is surjective (because finite morphisms are closed).

**Theorem 3.** *Let $f : X \to Y$ be a dominating morphism of varieties, and let $r = \dim X - \dim Y$. Let $W \subseteq Y$ be a closed irreducible subset, and let $Z$ be a component of $f^{-1}(W)$ that dominates $W$. Then*

$$\dim Z \geq \dim W + r$$

*or*

$$\operatorname{codim}(Z \text{ in } X) =: \operatorname{codim}_X Z \leq \operatorname{codim}_Y W.$$

**Corollary 4.** *If $Z$ is a component of $f^{-1}(y)$ for some $y \in Y$, then $\dim Z \geq r$.*

**Example 5.** *Given $f : \mathbb{A}^2 \to \mathbb{A}^2, (x, z) \mapsto (x, y)$ via $y = xz$. Here $r = 0$. And $f^{-1}((0,0))$ is a line, which has dimension 1. So the whole axis goes to a point. So here we get strict inequality $\dim Z > r$.*

*This is a morphism. It is an isomorphism on a dense open subset (=birational isomorphism). But not an isomorphism, since $f^{-1}((0,0)) =$ line.*

**Theorem 6.** *Let $f : X \to Y$ be a dominating morphism of varieties and let $r = \dim X - \dim Y$. Then there exists a nonempty open set $U \subseteq Y$ such that*

1. *$U \subseteq f(X)$*
2. *For all irreducible closed subsets $W \subseteq Y$ such that $W \cap U \neq \varnothing$, and for all components $Z$ of $f^{-1}(W)$ such that $Z \cap f^{-1}(U) \neq \varnothing$,*

$$\dim Z = \dim W + r$$

*or*

$$\operatorname{codim}_X Z = \operatorname{codim}_Y W.$$

**Corollary 7.** *Let $f$ be as above. Then there is a nonempty open set $U \subseteq Y$ such that for all $y \in U$, $f^{-1}(y)$ is a nonempty "pure" $r$-dimensional set, i.e. all its components have dimension $r$.*

**Definition 8.** *Let $X$ be a variety. A subset $A$ of $X$ is **constructible** if it is a finite union of locally-closed subsets of $X$.*

The constructible sets form an algebra; they are the smallest algebra containing all open sets.

**Example 9.** *Let $Z = \{\mathbb{A}^2 \setminus V(x)\} \cup \{(0,0)\}$.*

*Then $Z$ is a constructible set, but is not locally closed.*

*This is because $\mathbb{A}^2 \setminus V(x)$ is open, and $\{(0,0)\}$ is closed—so $Z$ is a finite union of locally closed sets.*

*But $Z$ is not locally closed. A set is locally closed $\iff$ it is open in its closure. The closure $\overline{Z}$ of $Z$ is $\mathbb{A}^2$, since the closure of the open part is $\mathbb{A}^2$. If $Z$ were open in $\mathbb{A}^2$, then $\mathbb{A}^2 \setminus Z$ would have to be closed. And $\mathbb{A}^2 \setminus Z = V(x) \cap (\mathbb{A}^2 \setminus \{(0,0)\})$ = the $y$-axis minus the origin. This set is not closed.*

**Corollary 10** (Chevalley)**.** *Let $f : X \to Y$ be a morphism. Then the image of $f$ is a constructible set in $Y$. More generally, $f$ maps constructible sets in $X$ to constructible sets in $Y$.*

**Corollary 11** (Upper semi-continuity of dimension)**.** *Let $f : X \to Y$ be any morphism. For all $x \in X$, define*

$$e(x) = \max\{\dim Z \mid Z \text{ is a component of } f^{-1}(f(x)) \text{ containing } x\}$$

*Then $e$ is upper semi-continuous, i.e. for all integers $n$,*

$$S_n(f) = \{x \in X \mid e(x) \geq n\}$$

*is closed.*

**Definition 12.** *A morphism $f : X \to Y$ is **birational** if it is dominating and the induced map $f^* : k(Y) \to k(X)$ is an isomorphism.*

**Theorem 13.** *If $f : X \to Y$ is a birational morphism, then there is a nonempty open set $U \subseteq Y$ such that $f$ restricts to an isomorphism from $f^{-1}(U)$ to $U$.*

**Example 14.** *Let*

$$f : \mathbb{A}^2 \to \mathbb{A}^2, \quad (x,y) \mapsto (xy, y).$$

1. *The image of $f$ is the union of $(0,0)$ and $(\mathbb{A}^2)_y = \mathbb{A}^2 \setminus \{y = 0\}$.*
   *This set is not locally closed (we showed this in a previous example).*

2. *$f$ is birational since $k(x,y) \cong k(yx,y)$, and if $U = \mathbb{A}^2_y$, then $f^{-1}(U) = \mathbb{A}^2_y$ and the restriction of $f$ to a map from $f^{-1}(U)$ to $U$ is an isomorphism.*

3. *$f^{-1}((0,0))$ is the whole line of points $(x,0)$.*

4. *$S_0(f) = \mathbb{A}^2, \quad S_1(f) = \{(x,0)\}, \quad S_2(f) = \varnothing$.*
*(where $e(x) = \max\{\dim Z \mid Z$ is a component of $f^{-1}(f(x))$ containing $x\}$ and $S_n(f) = \{x \in X \mid e(x) \geq n\}$)*

**Example 15** (Previous examples revisited)**.** *Given the finite birational morphism $f : \mathbb{A}^1 \to C$, where $C$ is the affine plane curve $X^3 = Y^2$, then if $U = C \setminus \{(0,0)\}$, then $f^{-1}(U) = \mathbb{A}^1 \setminus \{(0)\}$, and $f^{-1}(U) \cong U$.*
*On rings, $k[T^2, T^3] \subsetneq k[T]$, but $k[T, T^{-1}] = k[T^2, T^3, T^{-2}]$.*

*Given the finite birational morphism $f : \mathbb{A}^1 \to D$ where $D$ is the curve $Y^2 = X^2(X+1)$, its restriction $f' : \mathbb{A}^1 \setminus \{1\} \to D$ is a bijection, and if $U = D \setminus \{(0,0)\}$, then $f'^{-1}(U) = \mathbb{A}^1 \setminus \{\pm 1\}$ and $f^{-1}(U) \cong U$.*

## 1.9 Complete varieties

An affine variety can be embedded in a projective variety by a birational inclusion. Projective varieties cannot be embedded birationally in anything even bigger. This makes them *complete* varieties. This is our algebraic geometric analog of compact spaces in topology.

Note: Over $\mathbb{C}$, complete = compact.

**Example 1.** *$V = V(x^2 + y^2 - 1)$ is not complete because over $\mathbb{C}$, it's clearly not compact (a change of variables makes it look like a hyperbola).*

**Example 2.** *Look at $\mathbb{P}^1(k)$. $R = k[x]_{(x)} \subseteq k(x) = k(\mathbb{P}^1)$ is a valuation ring.*

*Assertion: $\mathbb{P}^1$ complete $\implies$ $R$ contains $\mathcal{O}_P$ for some $P$. This is indeed true: take $P$ to be the point where $x = 0$.*

**Example 3.** *$\mathbb{P}^1 \setminus P \cong \mathbb{A}^1$ has $k(\mathbb{P}^1 \setminus P) = k(x)$. $k(x)$ still contains $k[x]_{(x)}$. But $k[x]_{(x)}$ is not the local ring of a point now, since we threw out $P$.*

*Similarly, let $\mathbb{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$. Now $k[x]_{(x)}$ corresponds to 0. But $k(x) = k(x^{-1}) \supseteq k[x^{-1}]_{(x^{-1})}$ does not correspond to any point because it would have to be $\infty$.*

**Definition 4.** *A variety $X$ is **complete** if for all varieties $Y$, the projection morphism $p_2 : X \times Y \to Y$ is a closed map.*

Some properties:

1. If $f : X \to Y$ is a morphism with $X$ complete, then $f(X)$ is closed in $Y$ and is complete.

2. If $X$ and $Y$ are complete, then $X \times Y$ is complete.

3. If $X$ is complete and $Y \subseteq X$ is a closed subvariety, then $Y$ is complete.

4. An affine variety $X$ is complete only if $\dim X - 0$, i.e. $X$ consists of a single point.

Complete means universally closed, i.e. closed in anything you put it into, and for all $Y$, $X \times Y \to Y$ is closed.
Notice that this is similar to topology, where compact $\iff$ any time it's embedded into a Hausdorff space, it's closed there.

**Example 5.** *$\mathbb{A}^1$ is not complete, because $\mathbb{A}^1 \times \mathbb{P}^1 \to \mathbb{P}^1$ is not closed:*

*Consider $\Delta \subseteq \mathbb{A}^1 \times \mathbb{P}^1$ (thinking of $\mathbb{A}^1 \hookrightarrow \mathbb{P}^1$).*
*The image is $\mathbb{A}^1$, which is not closed in $\mathbb{P}^1$.*

**Remark 6.** *Given $f : X \to Z$ a morphism of varieties, we can ask: after every base change $Y \to Z$, is the induced map $X \times_Z Y \to Y$ closed? (in this case $f$ is universally closed or **proper**)*

*Say this is true: taking $Z = \{\text{pt}\}$, $X$ being complete means $X \to \{\text{pt}\}$ is proper.*

**Theorem 7.** *$\mathbb{P}^n$ is complete.*

Let $Y$ be a variety, let $U_i = (\mathbb{P}^n)_{X_i} \times Y$, and let $Z$ be a closed subset of $\mathbb{P}^n \times Y$. Let $R = \Gamma(Y, \mathcal{O}_Y)$, let $S = R[X_0, \ldots, X_n]$, let $S_m$ be the graded piece of $S$ of degree $m$, and let $A_m \subseteq S_m$ be the vector space of homogeneous polynomials $f(X_0, \ldots, X_n)$ of degree $m$ with coefficients in $R$ such that for all $i$,

$$f\left(\frac{X_0}{X_i}, \ldots, \frac{X_n}{X_i}\right) \in I(Z \cap U_i).$$

The proof of Theorem 7 uses the following lemma:

**Lemma 8.** *For all $i$ and $g \in I(Z \cap U_i)$ there is a polynomial $f \in A_m$ for some $m$ such that*

$$g = f\left(\frac{X_0}{X_i}, \ldots, \frac{X_n}{X_i}\right).$$

The proof also uses Nakayama:

**Lemma 9** (Nakayama's Lemma)**.** *Let $M$ be a finitely generated $R$-module, and let $A \subseteq R$ be an ideal such that*

$$M = A \cdot M.$$

*Then there is an element $f \in 1 + A$ which annihilates $M$.*

Putting the theorem and property (3) together, every projective variety is complete.
Not all complete varieties are projective, but every complete variety can be dominated by a projective variety with the same function field.

Another way to prove Theorem 7:
**Valuative Criterion.** A variety $X$ is complete $\iff$ for all valuation rings $R \subseteq k(X)$ containing $k$ and with quotient field $k(X)$, $R \supseteq \mathcal{O}_{X,x}$ for some $x \in X$.

**Remark 10.**
- *Projective varieties are complete.*
- *For curves and regular surfaces, complete implies projective (though this is not always true)*

## 1.10 Complex varieties

When $k$ is a topological field, there is a unique way to endow all varieties $X$ over $k$ with a new topology called the **strong topology**, such that the following properties hold:

1. the strong topology is stronger than the Zariski topology (i.e. a Zariski-closed subset $Z \subseteq X$ is always strongly closed, and an open subset $Z \subseteq X$ is always strongly open).
2. all morphisms are strongly continuous
3. if $Z \subseteq X$ is a locally closed subvariety, then the strong topology on $Z$ is the one induced by the strong topology on $X$
4. the strong topology on $X \times Y$ is the product of the strong topologies on $X$ and $Y$ (product topology)
5. the strong topology on $\mathbb{A}^1$ is exactly the given topology on $k$

Note: recall
Locally closed = open subset of closed set = intersection of open and closed set
(for example $\mathbb{A}^1 \setminus \{0\}$ is locally closed)
Constructible = finite union of locally closed. So locally closed $\implies$ constructible, but the converse does not hold:
(for example, $(\mathbb{A}^2 \setminus V(x)) \cup \{(0,0)\}$ is constructible but not locally closed)

**Remark 1.** *All varieties $X$ are Hausdorff spaces in their strong topology ($\Delta(X)$ is strongly closed)*
*Here by Hausdorff we mean "separated"*

Now suppose $k = \mathbb{C}$ with its usual topology. Then there is a unique collection of sheaves (in the strong topology) of strongly continuous $\mathbb{C}$-valued functions $\{\Omega_X\}$, one for each variety $X$, such that:

1. for each Zariski-open set $U \subseteq X$, $\mathcal{O}_X(U) \subseteq \Omega_X(U)$
2. all morphisms $f : X \to Y$ are holomorphic, i.e. $f^*$ takes sections of $\Omega_Y$ to sections of $\Omega_X$
3. if $Z \subseteq X$ is a locally closed subvariety, then $\Omega_Z$ is the sheaf of $\mathbb{C}$-valued functions on $Z$ induced by the sheaf $\Omega_X$ on $X$
4. if $X = \mathbb{A}^n =$ (as a set) $\mathbb{C}^n$, then $\Omega_X$ is the usual sheaf of holomorphic functions on $\mathbb{C}^n$

Note: $\Omega_X$ is not standard notation. Typically this denotes differential 1-forms. Instead people use $\mathcal{H}$ or $\mathcal{O}_{\text{an}}$.

**Theorem 2.** *Let $X$ be a variety and $U$ a nonempty open subvariety. Then $U$ is strongly dense in $X$.*

**Corollary 3.** *If $Z \subseteq X$ is a constructible subset of a variety, then the Zariski closure and the strong closure of $Z$ are the same.*

**Theorem 4.** *Let $X$ be a variety over $\mathbb{C}$. Then $X$ is complete $\iff$ $X$ is compact in its strong topology.*

**Example 5.** *Over $\mathbb{R}$,$x^2 + y^2 = 1$ looks compact, but over $\mathbb{C}$ its points can go off to infinity: for large $x$, one can take $y = i\sqrt{x^2 - 1}$. Thus the affine curve is not complete/proper; its projective closure adds the missing points at infinity.*

**Remark 6.** *In topology, a compact space $Y$ has the property that for every space $Z$, the projection*

$$Y \times Z \to Z$$

*is a closed map; this is the topological model for being universally closed. In algebraic geometry, a morphism $Y \to X$ is universally closed if after every base change $Z \to X$, the projection*

$$Y \times_X Z \to Z$$

*is closed. Taking $X = \{\text{pt}\} = \operatorname{Spec} k$, this is one of the conditions for $Y$ to be proper (or, in Mumford's terminology, complete).*

**Lemma 7** (Chow's Lemma)**.** *Let $X$ be a complete variety (over any algebraically closed field $k$). Then there exists a closed subvariety $Y$ of $\mathbb{P}_n(k)$ for some $n$ and a surjective birational morphism $\pi : Y \to X$.*

Chow's lemma says: complete is almost projective.
Note: A projective variety is complete, because closed in $\mathbb{P}^n \implies$ since $\mathbb{P}^n$ is complete, it's complete.
For curves, complete implies projective. For surfaces, smooth and complete implies projective. In all other cases, there are counterexamples.

**Lemma 8** (Sublemma)**.** *Let $S$ and $T$ be varieties with isomorphic open subsets $V_S \subseteq S$, $V_T \subseteq T$. For simplicity, identify $V_S$ with $V_T$ and look at the morphism $\Delta : V \to V \times V \subseteq S \times T$. If $\overline{V}$ is the closure of the image, then $\overline{V} \cap (S \times V) = \overline{V} \cap (V \times T) = \Delta(V)$.*

**Remark 9** (GAGA)**.** *We know that from the structure sheaf, given an open set $U \subseteq X$, we associate a ring $\mathcal{O}(U)$.*

*Suppose you are given a sheaf $\mathcal{M}$ such that $\mathcal{M}(U)$ is an $\mathcal{O}(U)$-module for every open set $U$. If the module structures are compatible with the restriction maps, then we call $\mathcal{M}$ a* ***sheaf of $\mathcal{O}$-modules****. In particular, $\mathcal{M}$ is also a sheaf of abelian groups.*

*Examples of coherent sheaves of $\mathcal{O}$-modules include:*

- *$\mathcal{M} = \Omega^1_X$, the sheaf of differential 1-forms;*
- *if $X$ is a curve and $P$ is a point, $\mathcal{O}(P)$, which associates to an open set $U$ the functions on $U$ allowed to have a pole at $P$ of order at most 1;*

- *more generally, if $D$ is a divisor (a linear $\mathbb{Z}$-combination of points on a curve), the sheaf $\mathcal{O}(D)$.*

*A sheaf $\mathcal{M}$ of $\mathcal{O}$-modules is called* ***quasicoherent*** *if, locally on affine open sets $U = \operatorname{Spec} A$, it comes from an $A$-module. In particular, if*

$$V = D(f) = \{f \neq 0\} \subseteq U,$$

*then*

$$\mathcal{M}(V) = \mathcal{M}(U)_f.$$

*It is called* ***coherent*** *if, locally on affine open sets, the corresponding modules are finitely generated (more precisely, finitely presented). For a Noetherian scheme, this is equivalent to being quasicoherent and locally finitely generated.*

*So given a space $X$ and a structure sheaf $\mathcal{O}$, we get a category of coherent sheaves.*

*Now suppose $X$ is a projective variety over $\mathbb{C}$. We can view $X$ either algebraically or as a complex analytic space $X^{an}$. GAGA gives an equivalence of categories*

$$\{\textit{coherent algebraic sheaves on } X\} \simeq \{\textit{coherent analytic sheaves on } X^{an}\}.$$

*Moreover, GAGA says that the corresponding cohomology groups are the same:*

$$H^i(X, \mathcal{F}) \cong H^i(X^{an}, \mathcal{F}^{an}).$$

# 2 Preschemes

## 2.1 $\operatorname{Spec}(R)$

For a commutative ring $R$, define $\operatorname{Spec}(R) :=$ the set of prime ideals $P \subsetneq R$.
Write $[P]$ for the element of $\operatorname{Spec}(R)$ given by the prime ideal $P$.

Define a topology on $\operatorname{Spec}(R)$ called the **Zariski topology**:
Closed sets = sets of the form $\{[P] \mid P \supseteq A\}$ for some ideal $A \subseteq R$. (Denote this set $V(A)$)
One can check that

1. $V\left(\sum_\alpha A_\alpha\right) = \cap_\alpha V(A_\alpha)$
2. $V(A \cap B) = V(A) \cup V(B)$

So the collection of closed subsets $\{V(A)\}$ does define a topology.

Now define $\operatorname{Spec}(R)_f = \{[P] \mid f \notin P\}$.
Since $\operatorname{Spec}(R)_f = \operatorname{Spec}(R) \setminus V(f)$, $\operatorname{Spec}(R)_f$ is an open subset of $\operatorname{Spec}(R)$. We refer to these open subsets as **distinguished open subsets**.
These form a basis of the open sets of $\operatorname{Spec}(R)$, because

$$\operatorname{Spec}(R) \setminus V(A) = \bigcup_{f \in A} \operatorname{Spec}(R)_f.$$

The closure of $[P]$ is exactly $V(P)$, i.e. $\{P' \mid P' \supseteq P\}$. Thus $[P]$ is a closed point $\iff$ $P$ is a maximal ideal.
When $R$ is a domain, $(0)$ is a prime ideal and $[(0)]$ is called the **generic point** of $\mathrm{Spec}(R)$ since its closure is all of $\mathrm{Spec}(R)$.

**Definition 1.** *Suppose $Z$ is an irreducible closed subset of $\mathrm{Spec}(R)$. Then a point $z \in Z$ is a **generic point** of $Z$ if $Z$ is the closure of $z$, i.e. every open subset $Z_0$ of $Z$ contains $z$.*

**Proposition 2.** *If $x \in \mathrm{Spec}(R)$, then the closure of $\{x\}$ is irreducible and $x$ is a generic point of this set. Conversely, every irreducible closed subset $X \subseteq \mathrm{Spec}(R)$ equals $V(P)$ for some prime ideal $P \subseteq R$, and $[P]$ is its unique generic point.*

**Proposition 3.** *Let $\{f_\alpha \mid \alpha \in S\}$ be a set of elements of $R$. The*

$$\mathrm{Spec}(R) = \bigcup_{\alpha \in S} \mathrm{Spec}(R)_{f_\alpha} \iff 1 \in (\dots, f_\alpha, \dots).$$

Notice: $1 \in (\dots, f_\alpha, \dots) \iff$ there is a finite set $f_{\alpha_1}, \dots, f_{\alpha_n}$ of the $f_\alpha$'s and elements $g_1, \dots, g_n$ of $R$ such that

$$1 = \sum_{i=1}^{n} g_i \cdot f_{\alpha_i}.$$

**Corollary 4.** $\mathrm{Spec}(R)$ *is quasi-compact.*

You can generalize this to show $\mathrm{Spec}(R)_f$ is also quasi-compact.
Unless $R$ is e.g. noetherian, there may be some open sets in $\mathrm{Spec}(R)$ which are not quasi-compact.

We can endow $\mathrm{Spec}(R)$ with a sheaf of rings $\mathcal{O}_{\mathrm{Spec}(R)}$, called its **structure sheaf**:
Let $X = \mathrm{Spec}(R)$.
Some properties of distinguished opens:

1. $X_f \cap X_g = X_{fg}$ for all $f, g \in R$.
2. $X_f \supseteq X_g \iff g \in \sqrt{(f)}$. (In fact, since $\sqrt{(f)} = \cap\{P \mid f \in P\}$, $g \notin \sqrt{(f)} \iff \exists P, f \in P, g \notin P \iff \exists P, [P] \notin X_f, [P] \in X_g$)

If $X_g \subseteq X_f$ then there is a canonical map $R_f \to R_g$ ($g^n = h \cdot f$ and we map $a/f^m \mapsto (a \cdot h^m)/g^{nm}$). Further, if $[P] \in X_f$, then $f \notin P$ and there is a natural map $R_f \to R_P$ since $\{f, f^2, \dots\} \subseteq R \setminus P$. One can check that

$$R_P = \varinjlim_{f \in R \setminus P} R_f = \varinjlim_{f \text{ such that } [P] \in X_f} R_f.$$

**Lemma 5.** *Suppose $X_f = \bigcup_{\alpha \in S} X_{f_\alpha}$. If an element $g \in R_f$ has image 0 in all the rings $R_{f_\alpha}$, then $g = 0$.*

**Lemma 6.** *Suppose $X_f = \bigcup_{\alpha \in S} X_{f_\alpha}$. Suppose we have elements $g_\alpha \in R_{f_\alpha}$ such that $g_\alpha$ and $g_\beta$ have the same image in $R_{f_\alpha f_\beta}$. Then there is an element $g \in R_f$ which has image $g_\alpha$ in $R_{f_\alpha}$ for all $\alpha$.*

For each open $U \subseteq X$, let

$$\Gamma(U, \mathcal{O}_X) := \left\{ \{s_P\} \in \prod_{[P] \in U} R_P \right\}$$

such that $\exists$ a covering of $U$ by distinguished open sets $X_{f_\alpha}$ together with $s_\alpha \in R_{f_\alpha}$ such that $s_P$ equals the image of $s_\alpha$ in $R_P$ whenever $[P] \in X_{f_\alpha}$.

This is a ring. If $V \subseteq U$, then the coordinate projection

$$\prod_{[P] \in U} R_P \to \prod_{[P] \in V} R_P$$

takes $\Gamma(U, \mathcal{O}_X)$ into $\Gamma(V, \mathcal{O}_X)$. Taking this as the restriction map, we get a presheaf $\mathcal{O}_X$. Some facts:

1. $\mathcal{O}_X$ is a sheaf.
2. $\Gamma(X_f, \mathcal{O}_X) = R_f$.
3. The stalk of $\mathcal{O}_X$ at $[P]$ is $R_P$.

**Example 7.** *Let $k$ be a field. Then* $\mathrm{Spec}(k)$ *has just one point* $[(0)]$*, and the structure sheaf is just $k$ sitting on that point.*

*More generally, let $R$ be any commutative ring with DCC. Then $R \cong \prod_{i=1}^n R_i$ is a finite product of Artinian local rings.*

$$\mathrm{Spec}(R) = \bigsqcup_{i=1}^{n} \mathrm{Spec}(R_i).$$

*where each* $\mathrm{Spec}(R_i)$ *is open in* $\mathrm{Spec}(R)$*. Each Artinian local ring $R_i$ has a unique prime ideal, namely its maximal ideal*

$$M_i = \{x \in R_i \mid x \text{ is nilpotent}\},$$

*and* $\mathrm{Spec}(R_i)$ *is one point. So* $\mathrm{Spec}(R)$ *itself consists of $n$ points with the discrete topology. The structure sheaf consists of $R_i$ sitting as the stalk on the $i$-th point.*

*Geometrically, the presence of nilpotents in $M_i$ means the $i$-th point is surrounded by some infinitesimal normal neighborhood.*

**Example 8.** *The* ***affine line over*** *$k$ is* $\mathrm{Spec}(k[X])$*. This is denoted* $\mathbb{A}^1_k$*.*
*$k[X]$ has two types of prime ideals:* $(0)$ *and* $(f(X))$ *for $f$ an irreducible polynomial. So* $\mathrm{Spec}(k[X])$ *has one closed point for each monic irreducible polynomial, and one generic point* $[(0)]$ *whose closure is all of* $\mathrm{Spec}(k[X])$*.*

*If $k$ is algebraically closed, then the closed points are all of the form $[(X-a)]$. We call this the point $X=a$. In that case, $\mathbb{A}^1_k$ is the ordinary $X$-line together with a generic point. And proper closed sets are finite unions of closed points.*

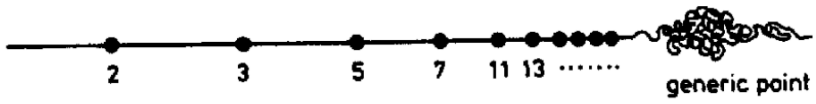
*The stalk of $\mathcal{O}_{\mathbb{A}^1}$ at $[(X-a)]$ is*

$$k[X]_{(X-a)} = \left\{ \frac{f(X)}{g(X)} \mid f, g \text{ polynomials}, \quad g(a) \neq 0 \right\}.$$

*The stalk at $[(0)]$ is $k(X)$, which we called before the function field.*

**Example 9.** *$\mathbb{Z}$ is a PID, like $k[X]$. $\mathrm{Spec}(\mathbb{Z})$ is usually visualized as a line.*

*There is one closed point for each prime number, plus a generic point.*




*The stalk at $[(p)]$ is $\mathbb{Z}_{(p)}$, and at $[(0)]$, it is $\mathbb{Q}$, so $\mathbb{Q}$ is the "function field" of $\mathrm{Spec}(\mathbb{Z})$.*

*The nonempty open sets of $\mathrm{Spec}(\mathbb{Z})$ are obtained by throwing away finitely many primes. If $m = \prod p_i$, then this is the distinguished open set $\mathrm{Spec}(\mathbb{Z})_m$, and*

$$\Gamma(\mathrm{Spec}(\mathbb{Z})_m, \mathcal{O}_{\mathrm{Spec}(\mathbb{Z})}) = \{a/m^k \mid a \in \mathbb{Z}, k \geq 0\}.$$

*The residue fields of the stalks $\mathcal{O}_x$ are $\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z}, \ldots, \mathbb{Q}$.*

**Example 10.** *Almost identical comments apply to $\mathrm{Spec}(R)$ for any Dedekind domain $R$. All prime ideals are maximal or $(0)$. So again there is a line of closed points plus a generic point.*

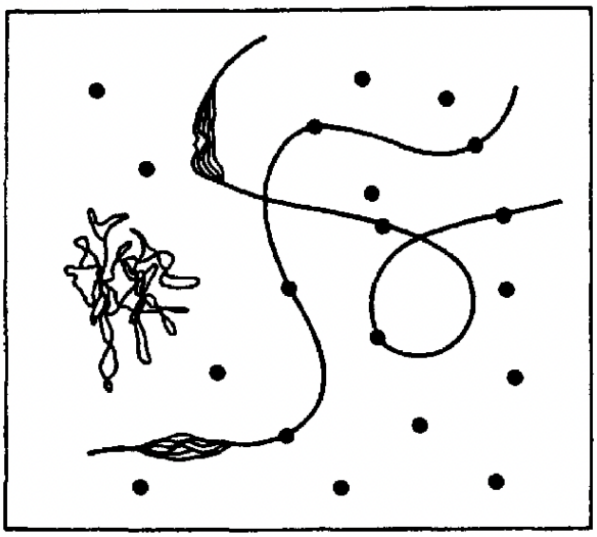
*If $R$ is not a PID, we cannot conclude as before that all non-empty open sets are distinguished; we only know that they are obtained by throwing out a finite set of closed points.*

*When $R$ is a discrete valuation ring (of rank 1), $R$ has a unique maximal ideal $M$ , hence $\mathrm{Spec}(R)$ has two points $[(0)]$ and $[M]$. Then $[(0)]$ is an open point and $[M]$ is closed.*

**Example 11.** *$\mathbb{A}^2_k = \mathrm{Spec}(k[X,Y])$, $k$ algebraically closed.*
*We get the maximal ideal $(X-a, Y-b)$, the principal primes $(f)$ for $f$ irreducible, and $(0)$. The set of maximal ideals gives a set of closed points isomorphic to the usual $X,Y$-plane. Then add one generic point, and for every irreducible curve, a point generic in that curve but not sticking out of it.*

*A proper closed set is obtained by taking a finite number of irreducible curves (including their generic points) plus a finite number of closed points.*



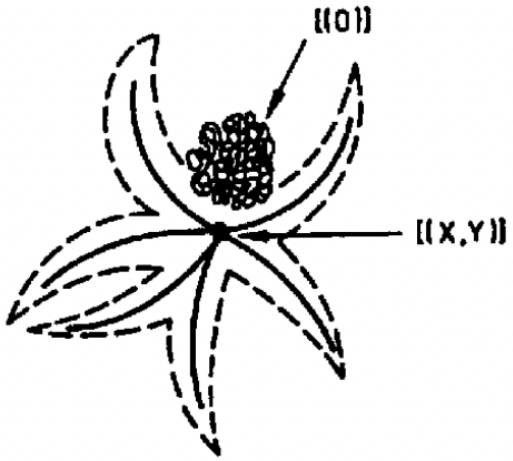
*Given a closed subset of the set of closed points in the old topology, there is a unique set of non-closed points to add to get a closed set in our new plane.*

**Example 12.** *Let*

$$\mathcal{O} = \left\{ \frac{f(X,Y)}{g(X,Y)} \mid f, g \in k[X,Y], g(0,0) \neq 0 \right\}.$$

*Then $\mathcal{O}$ is the stalk of $\mathcal{O}_X$ at $(0,0)$ if $X = \mathrm{Spec}(k[X,Y])$.*
*$\mathcal{O}$ has the maximal ideal $(X,Y)$, the principal prime ideals $(f)$ where $f$ is irreducible and $f(0,0) = 0$, and $(0)$.*
*Thus $\mathrm{Spec}(\mathcal{O})$ has only one closed point: If $K = k(X)$, this scheme is $\mathrm{Spec}(K[Y]_S)$ for some $S \subseteq K[Y]$, i.e. is part of the affine line over $K$.*
*But if you throw away the closed point and the $Y$-axis, you get the distinguished open set $X \neq 0$, which contains none of the original closed points.*

**Example 13** (Arithmetic surface)**.** *The prime ideals of $\mathbb{Z}[X]$ are $(0)$, the principal primes $(f)$ where $f$ is either a prime $p$ or a $\mathbb{Q}$-irreducible polynomial with coefficients having gcd 1, or the maximal ideals $(p, f)$ where $p$ is a prime and $f$ is a monic integral polynomial irreducible mod $p$.*




**Remark 14.** • *A point of* $\operatorname{Spec}\mathbb{Z}[X]$ *is a prime ideal of* $\mathbb{Z}[X]$*. There are three kinds visible in the picture:*

1. *The generic point* $(0)$*. Its closure is* $V(0) = \operatorname{Spec}\mathbb{Z}[X]$*.*
2. *Height-one primes* $(p)$ *and* $(f)$*.*
   *$(p)$ is prime since $\mathbb{Z}[X]/(p) \cong \mathbb{F}_p[X]$ is a domain.*
   *Likewise if $f \in \mathbb{Z}[X]$ is primitive and irreducible, then $(f)$ is prime.*
   *These are represented by the vertical and horizontal curves in the picture.*
3. *Closed points* $(p, g)$*. These are the maximal ideals.*

- *The columns are the fibers over the closed points of* $\operatorname{Spec}\mathbb{Z}$*. The inclusion* $\mathbb{Z} \hookrightarrow \mathbb{Z}[X]$ *gives* $\pi : \operatorname{Spec}\mathbb{Z}[X] \to \operatorname{Spec}\mathbb{Z}$*.*

  *Since* $\operatorname{Spec}\mathbb{Z} = \{(0), (2), (3), (5), \dots\}$*, the fiber over* $(p)$ *is*

  $$\pi^{-1}((p)) = \operatorname{Spec}(\mathbb{Z}[X] \otimes_{\mathbb{Z}} \mathbb{F}_p) = \operatorname{Spec}\mathbb{F}_p[X].$$

  *Equivalently, as a closed subset of* $\operatorname{Spec}\mathbb{Z}[X]$*,* $\pi^{-1}((p)) = V(p)$*. So for example,* $(2)$ *is the unique generic point of* $V(2)$*, and* $V(2)$ *is exactly the fiber over* $(2) \in \operatorname{Spec}\mathbb{Z}$*, i.e. the column labeled* $V((2))$ *is* $V(2) = \operatorname{Spec}\mathbb{F}_2[X]$*, its generic point is* $(2)$*, and its closed points correspond to irreducible polynomials in* $\mathbb{F}_2[X]$*:*

  $$(2, X), \quad (2, X + 1), \quad (2, X^2 + X + 1).$$

  *These are the vertical fibers.*

- *Look at the fiber over the generic point* $(0) \in \operatorname{Spec}\mathbb{Z}$*. It is* $\operatorname{Spec}\mathbb{Q}[X]$*. The closed points of* $\operatorname{Spec}\mathbb{Q}[X]$ *correspond to irreducible polynomials over* $\mathbb{Q}$*:*

  $$(X), \quad (X^2 + 1), \quad (X^2 - 2), \dots$$

  *Take one of those points and take the closure inside* $\operatorname{Spec}\mathbb{Z}[X]$*. For example,* $(X) \in \operatorname{Spec}\mathbb{Q}[X]$ *spreads out to* $V(X) \subseteq \operatorname{Spec}\mathbb{Z}[X]$*. These are the horizontal curves in the*

*drawing. So the horizontal row marked* $[(X^2+1)]$ *represents the irreducible closed subset* $V(X^2+1)$. *Its unique generic point is* $[(X^2+1)]$. *Note:* $V(X^2+1) = \operatorname{Spec}\mathbb{Z}[X]/(X^2+1) \cong \operatorname{Spec}\mathbb{Z}[i]$.

- *Since* $V(I) \cap V(J) = V(I+J)$, *the intersection between the p-column and the* $X^2+1$ *row is* $V(p) \cap V(X^2+1) = V(p, X^2+1)$. *So we determine the intersection simply by reducing* $X^2+1$ *mod p and factoring it in* $\mathbb{F}_p[X]$.

- *So e.g. reducing* $X^2+1$ *mod 3, the possible squares mod 3 are 0 and 1, so there is no solution to* $X^2 = -1 = 2$. *Thus* $X^2+1$ *is irreducible in* $\mathbb{F}_3[X]$. *Hence* $V(3) \cap V(X^2+1)$ *consists of one point:* $(3, X^2+1)$.

  *Its residue field is*

  $$\mathbb{Z}[X]/(3, X^2+1) \cong \mathbb{F}_3[X]/(X^2+1) \cong \mathbb{F}_9.$$

  *So the picture has one intersection over 3. The prime is inert, not ramified.*

- *If you reduce* $X^2+1$ *mod 5,* $2^2 = 4 = -1 \pmod 5$, *so* $X^2+1 = (X-2)(X+2)$ *in* $\mathbb{F}_5[X]$. *So* $X^2+1 = (X+3)(X+2)$ *and then we get two different intersection points:*

  $$(5, X+2), \quad (5, X+3).$$

  *This is why the picture splits into 2 branches near the 5-column. This means* $(5)$ *splits in* $\mathbb{Z}[i]$.

- *Modulo 2,* $X^2+1 = X^2-1 = (X-1)^2$. *So there is only one underlying point* $(2, X+1)$. *But the factor occurs with multiplicity 2. This is why there is a loop at* $p = 2$. *Scheme-theoretically,*

  $$V(2, X^2+1) = \operatorname{Spec}\mathbb{F}_2[X]/((X+1)^2)$$

  *and this ring is not reduced. Since*

  $$\mathbb{F}_2[X]/((X+1)^2) \cong \mathbb{F}_2[\varepsilon]/(\varepsilon^2),$$

  *there is one geometric point but it has multiplicity 2. So the prime in the* $X^2+1$ *row lying over 2 is ramified.*

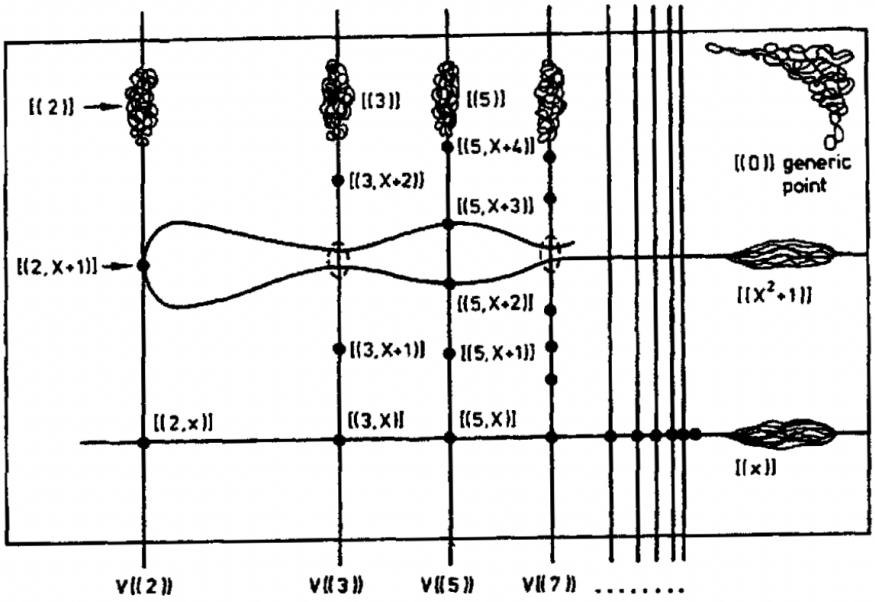
- $V(X^2+1)$ *intersects every vertical fiber* $V(p)$.
  *For an odd prime p,* $X^2+1$ *either splits or stays irreducible modulo p. The complete rule is: if* $p = 2$ *it ramifies. If* $p \equiv 1 \pmod 4$, *it splits. If* $p \equiv 3 \pmod 4$, *it is unramified/inert (one degree 2-point).*

- *The bottom row* $V(X)$ *is much simpler:* $V(X) = \operatorname{Spec}\mathbb{Z}[X]/(X) \cong \operatorname{Spec}\mathbb{Z}$.
  *at every prime p,* $V(X) \cap V(p) = V(p, X)$, *so there is exactly one point* $(p, X)$.

- *For any prime ideal P,* $\overline{\{P\}} = V(P)$ *so the point P is the generic point of* $V(P)$.

- *This is an arithmetic surface,* $\mathbb{A}^1_{\mathbb{Z}}$. *The vertical direction is arithmetic: 2, 3, 5, 7,... and the horizontal direction is polynomial/geometric:* $X, X^2+1, \ldots$

$\operatorname{Spec}(\mathbb{Z}[X])$ *looks like:*




**Problem 2.1.** ***What is*** $V((p)) \cap V((f))$ ***where*** $f$ ***is a*** $\mathbb{Q}$***-irreducible polynomial? What is*** $V((f)) \cap V((g))$ ***where*** $f, g$ ***are distinct*** $\mathbb{Q}$***-irreducible polynomials?***

*Recall:* $V(I) \cap V(J) = V(I + J)$.

*1)* $V((p)) \cap V((f)) = V((p) + (f)) = V(p, f)$.
*A point here is a prime ideal* $I \subseteq \mathbb{Z}[X]$ *containing both* $p$ *and* $f$. *So since it contains* $p$, *reduce mod* $p$: $\mathbb{Z}[X]/(p) \cong \mathbb{F}_p[X]$. *Let* $\overline{f}$ *denote* $f \pmod p$. *Write*

$$\overline{f} = h_1^{e_1} \cdots h_r^{e_r}$$

*where* $h_i$ *are distinct irreducible polynomials in* $\mathbb{F}_p[x]$.

*Then* $V((p)) \cap V((f)) = \{(p, h_1), \ldots, (p, h_r)\}$.
*Geometrically, the horizontal curve* $V(f)$ *meets the vertical fiber* $V(p)$ *at the points corresponding to the irreducible factors of* $f \pmod p$.

*2)* $V(f) \cap V(g) = V(f, g)$. *Since* $f, g$ *are distinct irreducible polynomials over* $\mathbb{Q}$, $\gcd(f, g) = 1$. *By Bezout, we can write*

$$A(x)f(x) + B(x)g(x) = 1, \quad A, B \in \mathbb{Q}[X].$$

*Clearing denominators, there exist* $A_0, B_0 \in \mathbb{Z}[X]$ *and a nonzero integer* $N$ *such that*

$$A_0 f + B_0 g = N.$$

*Thus $N \in (f, g)$.*

*Suppose $P \in V(f) \cap V(g)$. Then $(f, g) \subseteq P \implies N \in P \implies P$ contains some prime number $p$ dividing $N$. So every intersection point lies over one of finitely many rational primes $p$. For such a $p$, reduce $f \bmod p$:*
*A closed point $(p, h)$ lies on both $\iff h \mid \overline{f}$ and $h \mid \overline{g}$ in $\mathbb{F}_p[X]$.*

*Thus $V(f) \cap V(g) = \{(p, h) \mid h$ irreducible and $h \mid \gcd(\overline{f}, \overline{g})\}$.*
*Conclude:*

$$V(p) \cap V(f) = \{\text{irreducible factors of } f \bmod p.\}$$

$$V(f) \cap V(g) = \{\text{common irreducible factors of } f \bmod p, g \bmod p \text{ for finitely many } p\}.$$

*Each closed subset $V((p))$ is a copy of $\mathbb{A}^1_{\mathbb{Z}/p\mathbb{Z}}$.*
*All maximal chains of irreducible proper subsets have length 2. And if $\mathcal{O}$ is the local ring at a closed point $x$, then $\mathcal{O}$ has Krull dimension 2.*

The elements of $R$ can always be interpreted as functions on $\mathrm{Spec}(R)$. For all $x = [P] \in \mathrm{Spec}(R)$, let

$$\kappa(x) = R_P/(P \cdot R_P) = \text{ quotient field } (R/P) = \text{ residue field}(\mathcal{O}_x).$$

If $a \in R$, $a$ induces an element of $R_P$, and then in $\kappa(x)$. We call this the **value $a(x)$ of $a$ at $x$.** These values lie in different fields at different points.

For instance, if $m \in \mathbb{Z}$, its value at $[(p)]$ is "$m \pmod p$" in $\mathbb{Z}/p\mathbb{Z}$.

More generally, if $a \in \Gamma(U, \mathcal{O}_X)$, and $x \in U$, we can let $a(x)$ be the image of $a$ in $\kappa(x)$. For varieties we embedded $\mathcal{O}_X$ in the sheaf of $k$-valued functions on $X$, but this does not generalize: if $a \in R$, $a(x) = 0 \forall x \in \mathrm{Spec}(R) \iff a \in P$ for all prime ideals $P \subseteq R \iff a$ is nilpotent.

**Example 15.** *Let $Z = \mathrm{Spec}(k[X]/(X^2))$ for $k$ algebraically closed. Then $Z$ is a single point. $Z$ can also be embedded in $\mathbb{A}^1$ so that its unique point goes to the origin and so that its sheaf (which is just $k[X]/(X^2)$ can be obtained by taking all functions $f \in \mathcal{O}_{0,\mathbb{A}^1}$ modulo functions vanishing to second order at 0.*

*Then the function $X$ which just vanishes to first order is still nonzero as a function on $Z$, and if $f$ is any polynomial in $Z$, then one can compute from the restriction of $f$ to the subscheme $Z$ its value at 0 as well as its first derivative at 0.*

*More generally, $\mathrm{Spec}(k[X_1, \ldots, X_n]/A)$ can be interpreted as $V(A) \subseteq \mathbb{A}^n_k$ but with a sheaf of rings on it suitably fattened so as to include the first few terms of the Taylor expansion of polynomials in directions normal to $V(A)$.*

*If $R$ is any ring with nilpotents, one can visualize $\mathrm{Spec}(R)$ as containing some extra normal material.*

**Proposition 16.** *Let $R$ be a ring and let $f \in R$. Then the topological space $\mathrm{Spec}(R)_f$ together with the restriction of the sheaf of rings $\mathcal{O}_{\mathrm{Spec}(R)}$ to $\mathrm{Spec}(R)_f$ is isomorphic to $\mathrm{Spec}(R_f)$ together with the sheaf of rings $\mathcal{O}_{\mathrm{Spec}(R_f)}$.*

## 2.2 The category of preschemes

**Definition 1.** *A **prescheme** is a topological space $X$, plus a sheaf of rings $\mathcal{O}_X$ on $X$, provided that there exists an open covering $\{U_\alpha\}$ of $X$ such that each pair $(U_\alpha, \mathcal{O}_X|_{U_\alpha})$ is isomorphic to $(\mathrm{Spec}(R_\alpha), \mathcal{O}_{\mathrm{Spec}(R_\alpha)})$ for some commutative ring $R$.*

Note: modern language calls Mumford's preschemes "schemes" and his schemes "separated schemes."

**Definition 2.** *An **affine scheme** is a prescheme $(X, \mathcal{O}_X)$ isomorphic to $(\mathrm{Spec}(R), \mathcal{O}_{\mathrm{Spec}(R)})$ for some ring $R$.*

Note: An open subset of a prescheme is also a prescheme.

**Definition 3.** *If $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ are 2 preschemes, a **morphism** from $X$ to $Y$ is a continuous map $f : X \to Y$ plus a collection of homomorphisms*

$$f_V^* : \Gamma(V, \mathcal{O}_Y) \to \Gamma(f^{-1}(V), \mathcal{O}_X)$$

*one for each open set $V \subseteq Y$, such that*

1. *Whenever $V_1 \subseteq V_2$ are two open sets in $Y$, the diagram*

$$\begin{array}{ccc} \Gamma(V_2, \mathcal{O}_Y) & \xrightarrow{f_{V_2}^*} & \Gamma(f^{-1}(V_2), \mathcal{O}_X) \\ \downarrow \mathrm{res} & & \downarrow \mathrm{res} \\ \Gamma(V_1, \mathcal{O}_Y) & \xrightarrow{f_{V_1}^*} & \Gamma(f^{-1}(V_1), \mathcal{O}_X) \end{array}$$

   *commutes, and*

2. *If $V \subseteq Y$ is open and $x \in f^{-1}(V)$ and $a \in \Gamma(V, \mathcal{O}_Y)$, then $a(f(x)) = 0$ implies $f_V^*(a)(x) = 0$.*

Condition (2) may be expressed in other ways too. If $x \in X$ and $y = f(x)$, for all open neighborhoods $U$, $V$ of $x, y$ respectively (such that $f(U) \subseteq V$), we have a homomorphism

$$\Gamma(V, \mathcal{O}_Y) \underset{f_V^*}{\longrightarrow} \Gamma(f^{-1}(V), \mathcal{O}_X) \underset{\mathrm{res}}{\longrightarrow} \Gamma(U, \mathcal{O}_X)$$

Passing to the limit over such $U$ and $V$ we get a homomorphism between stalks $f_x^* : \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$. These stalks are always local rings. Thus condition (2) asserts that $f_x^*$ is a local homomorphism, i.e. $f_x^*(m_y) \subseteq m_x$. When this is the case $f_x^*$ also induces a homomorphism

$$\kappa(y) = \mathcal{O}_{Y,y}/m_y \xrightarrow{\overline{f_x^*}} \mathcal{O}_{X,x}/m_x = \kappa(x).$$

Call this induced residue-field map $\kappa(f(x)) \to \kappa(x)$. This induced map on residue fields has the property:
(2'): For all $a \in \Gamma(V, \mathcal{O}_Y)$, for some neighborhood $V$ of $y$, $f_V^*(a)(x)$ is the image of $a(y)$ under $\kappa(y) \to \kappa(x)$.

**Theorem 4.** *Let $X$ be a prescheme and let $R$ be a ring. To every morphism $f : X \to \mathrm{Spec}(R)$, associate the homomorphism:*

$$R \cong \Gamma(\mathrm{Spec}(R), \mathcal{O}_{\mathrm{Spec}(R)}) \xrightarrow{f^*} \Gamma(X, \mathcal{O}_X).$$

*Then this induces a bijection between* $\mathrm{Hom}(X, \mathrm{Spec}(R))$ *and* $\mathrm{Hom}(R, \Gamma(X, \mathcal{O}_X))$.

Recall: given a map on schemes, if we restrict to affines, we get a map of rings in the opposite direction. If we take the direct limit, we get a map of local rings (between stalks of corresponding points), and this is how you get $f_x^*$.
To get $f^*$, if $a \in \Gamma(V, \mathcal{O}_Y)$, then $a \in \mathcal{O}_Y(V) \implies a$ is a function on $V$ so that if $a$ vanishes at $f(x)$, $f^*a$ vanishes at $x$.

**Remark 5.**

- $\mathbb{P}_2, \mathbb{P}_1 \times \mathbb{P}_1$, *and* $\mathbb{P}_1$ *have the same ring of functions (=constants)*
- *Requiring the morphism to map into affines allows the correspondence, since affines have large enough rings of functions.*

**Corollary 6.** *The category of affine schemes is isomorphic to the category of commutative rings with unit, with arrows reversed.*

**Corollary 7.** $\mathrm{Spec}\,\mathbb{Z}$ *is the final object in the category of preschemes, i.e. for every prescheme $X$, there is a unique morphism $f : X \to \mathrm{Spec}\,\mathbb{Z}$.*

Every prescheme is a fibered object, with one fiber for each prime $p$, and one over the generic point of $\mathrm{Spec}\,\mathbb{Z}$. The fibering is given by $x \mapsto \mathrm{char}(\kappa(x))$.

**Proposition 8.** *Let $X$ be a prescheme, and $Z \subseteq X$ an irreducible closed subset. Then there is one and only one point $z \in Z$ such that $Z = \overline{\{z\}}$.*

The point $z$ will be called the **generic point** os $Z$. So there is a 1-1 correspondence between the points of $X$ and the irreducible closed subsets of $X$.

If $x \in X$, there is a 1-1 correspondence between:

1. irreducible closed subsets $Z \subseteq U$ such that $x \in Z$
2. points $z \in X$ such that $x \in \overline{\{z\}}$
3. prime ideals $P \subseteq \mathcal{O}_{X,x}$

**Example 9.** *Let $\mathcal{O}$ be a noetherian local ring with maximal ideal $M$. Thus* $\mathrm{Spec}(\mathcal{O})$ *has a unique closed point $x = [M]$.*
*Let $X$ be the open subscheme* $\mathrm{Spec}(\mathcal{O}) \setminus \{x\}$. *The closed points of $X$ correspond to prime ideals $P \subsetneq M$ such that there are no prime ideals between $P$ and $M$.*

*$X$ is rarely affine itself: to cover $X$, choose $f_1, \ldots, f_n \in M$ such that $M = \sqrt{(f_1, \ldots, f_n)}$. Then*

$$X = \bigcup_{i=1}^{n} \mathrm{Spec}(\mathcal{O}_{f_i})$$

*where* $\operatorname{Spec}(\mathcal{O}_{f_i})$ *and* $\operatorname{Spec}(\mathcal{O}_{f_j})$ *are patched along* $\operatorname{Spec}(\mathcal{O}_{f_i f_j})$.

*For instance if* $\mathcal{O} = \mathbb{C}[[X_1, \ldots, X_n]]$ *then* $X$ *turns out to have topological properties identical to the* $(2n-1)$*-sphere.*

**Example 10** ($\mathbb{P}^n_{\mathbb{Z}}$)**.** *Consider the union of* $n+1$ *copies* $U_0, \ldots, U_n$ *of* $\mathbb{A}^n_{\mathbb{Z}} = \operatorname{Spec} \mathbb{Z}[X_1, \ldots, X_n]$.

*Introduce* $X_{ij}, 0 \leq i, j \leq n$ *and* $i \neq j$ *and set*

$$U_i = \operatorname{Spec} \mathbb{Z}[X_{0i}, X_{1i}, \ldots, X_{i-1,i}, X_{i+1,i}, \ldots, X_{ni}].$$

*Then since*

$$(U_i)_{X_{ji}} = \operatorname{Spec} \mathbb{Z}[X_{0i}, \ldots, X_{ni}, X_{ji}^{-1}]$$
$$(U_j)_{X_{ij}} = \operatorname{Spec} \mathbb{Z}[X_{0j}, \ldots, X_{nj}, X_{ij}^{-1}]$$

*these two preschemes can be identified by*

$$X_{kj} = X_{ki}/X_{ji}, \quad 0 \leq k \leq n, k \neq i, j$$

$$X_{ij} = 1/X_{ji}$$

*with inverse*

$$X_{ki} = X_{kj}/X_{ij}, \quad 0 \leq k \leq n, k \neq i, j$$
$$X_{ji} = 1/X_{ij}.$$

*This is usually abbreviated by introducing* $X_0, \ldots, X_n$ *and replacing* $X_{ij}$ *by* $X_i/X_j$. *Then*

$$U_i = \operatorname{Spec} \mathbb{Z}[X_0/X_i, \ldots, X_n/X_i]$$

*and the identification of* $U_i$ *and* $U_j$ *is given by the identity map between* $\mathbb{Z}[X_0/X_i, \ldots, X_n/X_i, X_i/X_j]$ *and* $\mathbb{Z}[X_0/X_j, \ldots, X_n/X_j, X_j/X_i]$.

*And then if* $F(X_0, \ldots, X_n)$ *is a homogeneous integral polynomial, we can talk about the set* $F = 0$ *where by definition*

$$\{F = 0\} \cap U_i = V((F/X_i^d))$$

*where* $d = \deg F$.

Recall:

**Definition 11.** *Given morphisms* $r : X \to S$ *and* $s : Y \to S$, *a* ***fiber product*** *is a commutative diagram*

$$\begin{array}{ccc} X \times_S Y & \xrightarrow{p_2} & Y \\ \downarrow{\scriptstyle p_1} & & \downarrow{\scriptstyle s} \\ X & \xrightarrow{r} & S \end{array}$$

*with universal property so that given any commutative diagram*

$$\begin{array}{ccc} Z & \xrightarrow{q_2} & Y \\ \downarrow{\scriptstyle q_1} & & \downarrow{\scriptstyle s} \\ X & \xrightarrow{r} & S \end{array}$$

*there is a unique morphism* $t : Z \to X \times_S Y$ *such that* $q_1 = p_1 \circ t, \quad q_2 = p_2 \circ t.$

The fiber product is unique up to canonical isomorphism.

**Remark 12.**

- *The product is a special case of fiber product (take* $S = \{pt\}$*)*
- *Fiber product is also called pullback (reversal of arrows gives the pushback)*
- *Direct products are products, direct sums in familiar categories are pushouts, tensor products are pushouts*
- *The pullback of the fiber product in the category of affine schemes = pushout in category of rings*
- *When not specified,* $\times$ *just means fiber product over* $\operatorname{Spec}\mathbb{Z}$

t

**Theorem 13.** *If* $A$ *and* $B$ *are* $C$*-algebras, let the diagram of affine schemes*

$$\begin{array}{ccc} \operatorname{Spec}(A \otimes_C B) & \xrightarrow{p_2} & \operatorname{Spec}(B) \\ \downarrow{\scriptstyle p_1} & & \\ \operatorname{Spec}(A) & \xrightarrow{r} & \operatorname{Spec}(C) \end{array}$$

*be defined by the canonical homomorphisms* $C \to A, C \to B, A \to A \otimes_C B, B \to A \otimes_C B$*. This makes* $\operatorname{Spec}(A \otimes_C B)$ *into a fiber product.*

**Theorem 14.** *Given any morphisms* $r : X \to S$*,* $s : Y \to S$*, a fiber product exists.*

Note: for example, if $\mathbb{A}^n_{\mathbb{Z}} = \operatorname{Spec}(\mathbb{Z}[X_1, \ldots, X_n])$, then for all rings $R$,

$$\mathbb{A}^n_{\mathbb{Z}} \times \operatorname{Spec}(R) = \operatorname{Spec}(R[X_1, \ldots, X_n]).$$

**Definition 15.** $\mathbb{A}^n_R := \operatorname{Spec}(R[X_1, \ldots, X_n])$.

**Example 16.** *Since* $\mathbb{P}^n_{\mathbb{Z}}$ *is the union of the* $n+1$ *affines*

$$U_i = \operatorname{Spec}\mathbb{Z}[X_0/X_i, \ldots, X_n/X_i],$$

$\mathbb{P}^n_{\mathbb{Z}} \times \operatorname{Spec}(R)$ *is the union of the affines*

$$U_i' = \operatorname{Spec} R[X_0/X_i, \ldots, X_n/X_i]$$

*with the standard patching. All homogeneous polynomials* $F(X_0, \ldots, X_n)$ *over* $R$ *define closed subsets* $F = 0$ *of* $\mathbb{P}^n_{\mathbb{Z}} \times \operatorname{Spec}(R)$ *.*

**Definition 17.** $\mathbb{P}^n_R := \mathbb{P}^n_{\mathbb{Z}} \times \operatorname{Spec}(R)$ *.*

**Problem 2.2.** ***For all homogeneous*** $F \in R[X_0, \ldots, X_n]$***, show that*** $\{x \in \mathbb{P}^n_R \mid F(X) \neq 0\}$ ***is isomorphic to***

$$\operatorname{Spec} R[X_0^d/F, \ldots, (\prod X_i^{d_i})/F, \ldots, X_n^d/F]$$

***where*** $\sum d_i = d = \deg(F)$.

*Let*

$$S = R[X_0, \ldots, X_n],$$

*and let* $F \in S$ *be homogeneous of degree* $d$*. Set*

$$U = \{x \in \mathbf{P}^n_R \mid F(x) \neq 0\}.$$

*We will show that*

$$U \cong \operatorname{Spec} A,$$

*where*

$$A = R\left[\frac{X_0^{d_0} \cdots X_n^{d_n}}{F} \;\middle|\; d_0 + \cdots + d_n = d\right].$$

*For each* $i$*, let*

$$U_i = \{X_i \neq 0\} \subseteq \mathbf{P}^n_R.$$

*Recall that*

$$U_i \cong \operatorname{Spec} R\left[\frac{X_0}{X_i}, \ldots, \frac{\widehat{X_i}}{X_i}, \ldots, \frac{X_n}{X_i}\right].$$

*Since* $F$ *is homogeneous of degree* $d$*, the expression*

$$f_i = \frac{F}{X_i^d}$$

*is a polynomial in the affine coordinates* $X_j/X_i$*. Hence*

$$U \cap U_i = D(f_i),$$

*so*

$$U \cap U_i \cong \operatorname{Spec} R\left[\frac{X_0}{X_i}, \ldots, \frac{\widehat{X_i}}{X_i}, \ldots, \frac{X_n}{X_i}\right]_{f_i}.$$

*Now put*

$$t_i = \frac{X_i^d}{F} \in A.$$

*We claim that*

$$A_{t_i} = R\left[\frac{X_0}{X_i}, \ldots, \frac{\widehat{X_i}}{X_i}, \ldots, \frac{X_n}{X_i}\right]_{f_i}.$$

*Indeed, for every $j$ we have*

$$\frac{X_j}{X_i} = \frac{X_j X_i^{d-1}/F}{X_i^d/F} = \frac{X_j X_i^{d-1}/F}{t_i},$$

*and $X_j X_i^{d-1}/F$ is one of the generators of $A$. Thus*

$$\frac{X_j}{X_i} \in A_{t_i}.$$

*Moreover,*

$$\frac{1}{f_i} = \frac{X_i^d}{F} = t_i.$$

*Therefore*

$$R\left[\frac{X_0}{X_i}, \ldots, \frac{\widehat{X_i}}{X_i}, \ldots, \frac{X_n}{X_i}\right]_{f_i} \subseteq A_{t_i}.$$

*Conversely, consider a generator*

$$\frac{X_0^{d_0} \cdots X_n^{d_n}}{F}$$

*of $A$, where $\sum_j d_j = d$. We may write*

$$\frac{X_0^{d_0} \cdots X_n^{d_n}}{F} = \frac{(X_0/X_i)^{d_0} \cdots (X_n/X_i)^{d_n}}{F/X_i^d} = \frac{(X_0/X_i)^{d_0} \cdots (X_n/X_i)^{d_n}}{f_i}.$$

*Hence every generator of $A$ belongs to*

$$R\left[\frac{X_0}{X_i}, \ldots, \frac{\widehat{X_i}}{X_i}, \ldots, \frac{X_n}{X_i}\right]_{f_i}.$$

*This proves the claimed equality of rings, and therefore*

$$D(t_i) \cong U \cap U_i.$$

*It remains to check that the $D(t_i)$ cover* $\operatorname{Spec} A$. *Write*

$$F = \sum_{|\alpha|=d} c_\alpha X^\alpha.$$

*Dividing by $F$ gives*

$$1 = \sum_{|\alpha|=d} c_\alpha \frac{X^\alpha}{F}.$$

*Thus the elements $X^\alpha/F$, $|\alpha| = d$, generate the unit ideal of $A$.*

*Let*

$$J = (t_0, \dots, t_n) = \left( \frac{X_0^d}{F}, \dots, \frac{X_n^d}{F} \right).$$

*For any* $\alpha = (\alpha_0, \dots, \alpha_n)$ *with* $|\alpha| = d$,

$$\left( \frac{X^\alpha}{F} \right)^d = \prod_{i=0}^{n} \left( \frac{X_i^d}{F} \right)^{\alpha_i} \in J.$$

*Hence every* $X^\alpha/F$ *lies in* $\sqrt{J}$. *Since these elements generate the unit ideal, we obtain*

$$\sqrt{J} = A.$$

*Therefore*

$$D(t_0) \cup \cdots \cup D(t_n) = \operatorname{Spec} A.$$

*The isomorphisms*

$$D(t_i) \cong U \cap U_i$$

*agree on overlaps, since they are induced by the usual projective coordinate changes*

$$\frac{X_j}{X_i} = \frac{X_j/X_k}{X_i/X_k}.$$

*Consequently they glue to an isomorphism of preschemes*

$$\boxed{\{x \in \mathbf{P}_R^n \mid F(x) \neq 0\} \cong \operatorname{Spec} R \left[ \frac{X_0^{d_0} \cdots X_n^{d_n}}{F} \;\middle|\; d_0 + \cdots + d_n = d \right].}$$

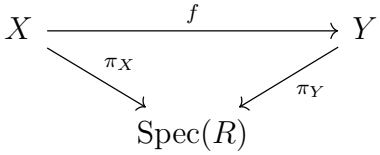
**Proposition 18.** *If* $r : X \to S$ *is surjective, then* $p_2 : X \times_S Y$ *is surjective.*

## 2.3 Varieties and preschemes

**Definition 1.** *Let* $R$ *be a ring. Then a* ***prescheme*** $X$ ***over*** $R$ *is a morphism* $\pi : X \to \operatorname{Spec}(R)$.

Note: This is the same as giving $\Gamma(X, \mathcal{O}_X)$ an $R$-algebra structure.

**Definition 2.** *Let* $X, Y$ *be preschemes over* $R$. *An* $R$***-morphism*** *from* $X$ *to* $Y$ *is a morphism* $f : X \to Y$ *such that*




*commutes.*

This just means that the maps $f_V^*$ ($V \subseteq Y$ open) are all $R$-algebra homomorphisms. This gives us the **category of preschemes over** $R$.

**Definition 3.** *A prescheme $X$ over $R$ is of* ***finite type over*** *$R$ if $X$ is quasicompact and for all open affine subsets $U \subseteq X$, $\Gamma(U, \mathcal{O}_X)$ is a finitely-generated $R$-algebra.*

**Example 4.** *$R[[X]]$ is not of finite type but is complete.*

Note: We typically mean f.g. as an algebra, not as a module. e.g. $R[1/f]$ is not a f.g. $R$-module.
Finite = f.g. as a module
Finite type = f.g. as an algebra

**Proposition 5.** *Let $X$ be a prescheme over $R$. If there exists a finite affine open covering $\{U_i\}$, $1 \leq i \leq n$ of $X$ such that $R_i = \Gamma(U_i, \mathcal{O}_X)$ is a finitely generated $R$-algebra, then $X$ is of finite type over $R$.*

**Corollary 6.** *If $X$ is an affine scheme and $U \subseteq X$ is an open affine subscheme, then $\Gamma(U, \mathcal{O}_X)$ is finitely generated over $\Gamma(X, \mathcal{O}_X)$.*

Warning: $X$ may be of finite type over $R$ but $\Gamma(X, \mathcal{O}_X)$ may not be a finitely generated $R$-algebra. This is the fact that Hilbert's 14th Problem was false.

**Definition 7.** *A prescheme $X$ is* ***reduced*** *if $\mathcal{O}_X$ contains no nilpotent sections, i.e. $\Gamma(U, \mathcal{O}_X)$ has no nilpotent elements for all open sets $U \subseteq X$.*

This holds if and only if all the stalks have no nilpotents. And that holds if there is a covering of $X$ by open affines $U_i$ so that $\Gamma(U_i, \mathcal{O}_X)$ has no nilpotents.

**Example 8.** *Consider $\mathbb{C}[x,t]/(t^2)$. This has $(t)$ as a prime. $[(t)]$ is a generic point ($(t)$ is the unique minimal prime; $t^2$ is in every prime). All other primes $(x-a,t)$ are maximal. Suppose the image of $t$ does not vanish. It vanishes at all $(x-a,t)$. So it vanishes at every point, but it is not $0$. Having nilpotents is what allows this.*

Recall: the value of something at a point (= maximal ideal = closed point) is its image in the ring modulo that maximal ideal. So for $\operatorname{Spec}\mathbb{Z}$, the value at $(p)$ is the value mod $p$.

**Theorem 9.** *Let $k$ be an algebraically closed field. Then there is an equivalence of categories between*

1. *The category of reduced, irreducible preschemes of finite type over $k$, and $k$-morphisms*
2. *The category of prevarieties over $k$ and morphisms of these (as in Ch. 1)*

**Remark 10.** *We have maximal ideals, and then we have the extra primes, which are the generic points of irreducible subvarieties. The minimal prime in a domain is unique. To go from a scheme to a prevariety, we forget the generic points.*

*And* $\operatorname{Spec} R$ *is reduced and irreducible (=integral)* $\iff$ *$R$ is a domain.*

**Lemma 11.** *Let $X$ be a prescheme of finite type over $k$, and let $x \in X$. Then $x$ is closed* $\iff$ *the composition*

$$k \to \mathcal{O}_{X,x} \to \kappa(x)$$

*is surjective.*

So for preschemes $X$ of finite type over $k$, a point $x \in X$ is closed $\iff$ it is closed in some open neighborhood of itself. This is false for general preschemes.

**Example 12.** *Consider $\mathbb{A}^1_{\mathbb{C}}$. $(x)$ is a closed point. Looking at the composition*

$$k \to \mathcal{O}_{X,x} \to \mathcal{O}_{X,x}/m_x$$

*we get*

$$\mathbb{C} \to \mathbb{C}[x]_{(x)} \to \mathbb{C}[x]_{(x)}/(x) \cong \mathbb{C}$$

*which turns out to be the identity. But if we take a non-closed point like $(0)$, we get*

$$\mathbb{C} \to \mathbb{C}(x) \to \mathbb{C}(x)/(0)$$

*which cannot be surjective.*

**Lemma 13.** *Let $X$ be a prescheme of finite type over $k$. Then the closed points of $X$ are dense in every closed subset of $X$.*

Note:

**Definition 14** (Redefinition)**.** *If $k$ is an algebraically closed field, a **prevariety over** $k$ is a reduced and irreducible prescheme of finite type over $k$.*

**Proposition 15.** *The product of two prevarieties $X$ and $Y$ over $k$, in the category of prevarieties over $k$, is the same as their fiber product $X \times_{\mathrm{Spec}(k)} Y$, in the category of all preschemes.*

An application of the generic point of a variety:
Suppose $X$ and $Y$ are two prevarieties over $k$. Suppose we're given $k$-isomorphisms $\alpha : k(X) \to K$, $\beta : k(Y) \to K$ of their function fields with a third field $K$. Then $\alpha$ and $\beta$ define morphisms $A : \operatorname{Spec} K \to X$, $B : \operatorname{Spec} K \to Y$ (by requiring that the images of $A, B$ be the generic points $x, y$ of $X$ and $Y$ and that $A^* : \mathcal{O}_{X,x} = k(X) \to K$ be $\alpha$ and that $B^* : \mathcal{O}_{Y,y} = k(Y) \to K$ be $\beta$.

By the proposition, we get a morphism $(A, B) : \operatorname{Spec} K \to X \times_k Y$.

Let $t$ be the image point and let $T = \overline{\{t\}}$, a closed subset of $X \times Y$. $T$ is irreducible, as it is the closure of a point, so $T$ is a closed subprevariety of $X \times Y$. And $k(T)$ is isomorphic via $(A, B)^*$ to $K$.

We describe this situation by saying that $\beta^{-1} \circ \alpha$ is a **birational correspondence** between $X$ and $Y$ and that $T$ is its **graph**.

**Example 16.** *Let $X = \mathbb{P}_2$ and let $x_0, x_1, x_2$ and $y_0, y_1, y_2$ be homogeneous coordinates in $X$ and $Y$.*

*Let $U_0 \subseteq X$ and $V_0 \subseteq Y$ be defined as the open sets $x_0 \cdot x_1 \cdot x_2 \neq 0$ and $y_0 \cdot y_1 \cdot y_2 \neq 0$. Define an isomorphism between $U_0$ and $V_0$ by the map:*

$$y_0 = 1/x_0, \quad y_1 = 1/x_1, \quad y_2 = 1/x_2.$$

*This is an extension of the isomorphism of function fields*

$$k(x_1/x_0, x_2/x_0) \to k(y_1/y_0, y_2/y_0), \quad x_1/x_0 \mapsto y_1/y_0, x_2/x_0 \mapsto y_2/y_0.$$

*Now we look at the closure $T \subseteq \mathbb{P}_2 \times \mathbb{P}_2$ of this isomorphism. Specifically, we'll just look at one piece of $T$. Let*

$$U = \operatorname{Spec} k[x_1/x_0, x_2/x_0] = X_{x_0}$$
$$U = \operatorname{Spec} k[y_0/y_1, y_2/y_1] = Y_{y_1}.$$
$$T \cap (U \times V) = \operatorname{Spec} k[x_1/x_0, x_2/x_0, y_0/y_1, y_2/y_1]/P = \operatorname{Spec} k[A, B, C, D]/P$$

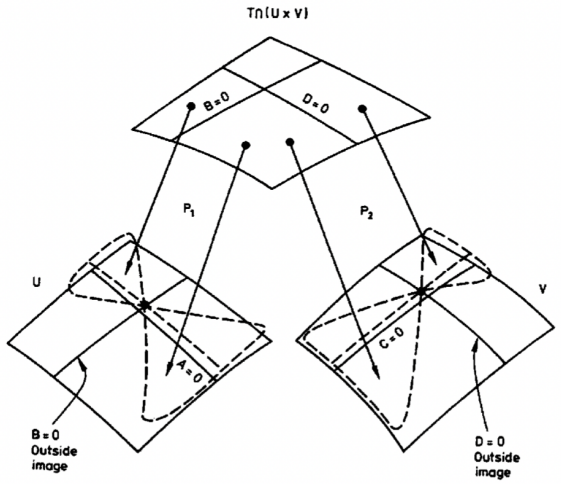
*where $P$ is the ideal of relations we get by putting all functions in the same function field (so $A - C \in P$ and $BD - C \in P$). Dividing out by $A - C$ and $BD - C$ gives $k[B, D]$ which is a domain of the correct dimension $\implies$ $P = (A - C, BD - C)$ and*

$$T \cap (U \times V) = \operatorname{Spec}(k[B, D]).$$

*And the projections $p_1 : T \cap (U \times V) \to U$, $p_2 : T \cap (U \times V) \to V$ give*

$$p_1^* : k[A, B] \to k[B, D], \quad p_2^* : k[C, D] \to k[B, D]$$

*where $p_1^*(A) = B \cdot D$ and $p_2^*(C) = B \cdot D$. So the line $B = 0$ in $T \cap (U \times V)$ collapses to the point $A = B = 0$ in $U$, and the line $D = 0$ in $T \cap (U \times V)$ collapses to the point $C = D = 0$ in $V$.*




## 2.4 Fields of definition

Let $k$ be an algebraically closed field and let $X$ be a closed subvariety of $\mathbb{A}^n$ defined by equations $f_i(X_1, \ldots, X_n) = 0$ for $1 \leq i \leq m$. Say $k_0$ is a subfield of $k$. Then if all coefficients of the $f_i$ lie in $k_0$, we say $X$ is **defined over** $k_0$. Let

$$A = (f_1, \ldots, f_m) = I(X).$$

Then if the coefficients of the $f_i$ lie in $k_0$, $A = k \cdot A_0$ where $A_0 = A \cap k_0[X_1, \ldots, X_n]$, and if $R_0 = k_0[X_1, \ldots, X_n]/A_0$, then the affine ring $R = k[X_1, \ldots, X_n]/A$ of $X$ is of the form $R_0 \otimes_{k_0} k$.

So when $X$ is defined over $k_0$, there is an affine scheme $X_0 = \mathrm{Spec}(R_0)$ of finite type over $k_0$ such that $X = X_0 \times_{\mathrm{Spec}(k_0)} \mathrm{Spec}(k)$.

Conversely, if $X$ is an affine variety over $k$ and $X_0$ is a prescheme of finite type over $k_0$ such that $X = X_0 \times_{\mathrm{Spec}(k_0)} \mathrm{Spec}(k)$, then let $R_0 = \Gamma(X_0, \mathcal{O}_{X_0})$ and write $R_0 = k_0[X_1, \ldots, X_n]/(f_1, \ldots, f_m)$. Then if $R = \Gamma(X, \mathcal{O}_X)$, then $R \cong k[X_1, \ldots, X_n]/(f_1, \ldots, f_m)$, hence $X$ is isomorphic to the closed subvariety of $\mathbb{A}^n$ defined by $f_i, 1 \leq i \leq m$. And since the coefficients of the $f_i$ are in $k_0$, $X$ is defined over $k_0$.

Often write $X = X_0 \times_{k_0} k$ for $X_0 \times_{\mathrm{Spec}(k_0)} \mathrm{Spec}(k)$.
If $X_0$ is the union of open affines $(U_0)_i = \mathrm{Spec}(R_i)$, then $X$ is the union of the affine schemes $U_i = \mathrm{Spec}(R_i \otimes_{k_0} k)$.

**Definition 1.** *Let $\sigma$ be an automorphism of $k$ over $k_0$. The conjugation map $\sigma_X : X \to X$ is the underlying map of the morphism*

$$X = X_0 \times_{\mathrm{Spec}(k_0)} \mathrm{Spec}(k) \xrightarrow{1_{X_0} \times \phi} X = X_0 \times_{\mathrm{Spec}(k_0)} \mathrm{Spec}(k)$$

*where $\phi : \mathrm{Spec}(k) \to \mathrm{Spec}(k)$ is defined by taking $\sigma^{-1}$ as the homomorphism $\phi^* : k \to k$.*

Check: $(\sigma \circ \tau)_X = \sigma_X \circ \tau_X$, i.e. the Galois group of $k/k_0$ is acting on the topological space $X$. So each $\sigma_X$ is a homeomorphism of $X$.

If we have:

- $X_0 = \mathrm{Spec}\, k_0[X_1, \ldots, X_n]/(f_1, \ldots, f_m)$
- $X = \mathrm{Spec}\, k[X_1, \ldots, X_n]/(f_1, \ldots, f_m)$
- $x \in X$ is the closed point $X_1 = \alpha_1, \ldots, X_n = \alpha_n$ where $\alpha_1, \ldots, \alpha_n \in k$ and $f_i(\alpha_1, \ldots, \alpha_n) = 0 \forall i$

then $\sigma_X(x)$ is the closed point $X_1 = \sigma(\alpha_1), \ldots, X_n = \sigma(\alpha_n)$.

**Theorem 2.** *Let $X_0$ be a prescheme over $k_0$, let $X = X_0 \times_{k_0} k$, and let $p : X \to X_0$ be the projection. Assume $k$ is an algebraic closure of $k_0$. Then*

1. *$p$ is surjective and both open and closed*
2. *For all $x, y \in X$, $p(x) = p(y) \iff x = \sigma_X(y)$ for some $\sigma \in \mathrm{Gal}(k/k_0)$, i.e. for all $x \in X_0, p^{-1}(x)$ is an orbit of $\mathrm{Gal}(k/k_0)$. Moreover, $p^{-1}(x)$ is a finite set.*

**Corollary 3.** *$X_0$, as a topological space, is the quotient of $X$ by the action of $\mathrm{Gal}(k/k_0)$.*

**Definition 4.** *The $k_0$-**topology** on $X$ is the set of $\mathrm{Gal}(k/k_0)$-invariant open sets, i.e. the set of open sets $\{p^{-1}(U) \mid U$ open in $X_0\}$.*

**Theorem 5.** *Let $X_0$ be a prescheme over $k_0$, let $X = X_0 \times_{k_0} k$ and let $p : X \to X_0$ be the projection. Assume $X_0$ is of finite type over $k_0$ (hence $X$ is of finite type over $k$).*

1. *For all $U \subseteq X_0$ open, the canonical map*

$$\Gamma(U, \mathcal{O}_{X_0}) \otimes_{k_0} k \to \Gamma(p^{-1}(U), \mathcal{O}_X)$$

*is bijective. Moreover, the functions $f \in \Gamma(U, \mathcal{O}_{X_0})$ satisfy*

$$f(\sigma_X(x)) = \sigma(f(x))$$

*for all closed points $x \in p^{-1}(U)$, all $\sigma \in \mathrm{Gal}(k/k_0)$.*

2. *If $k_0$ is perfect and $X$ is reduced, then $\Gamma(U, \mathcal{O}_{X_0})$ is exactly the subring of elements $f \in \Gamma(p^{-1}(U), \mathcal{O}_X)$ satisfying "$f(\sigma_X(x)) = \sigma(f(x))$ for all closed points $x \in p^{-1}(U)$, all $\sigma \in \mathrm{Gal}(k/k_0)$."*

So $X_0$ as a prescheme over $k_0$ can be reconstructed from:

1. the prescheme $X$ over $k$
2. the action of $\mathrm{Gal}(k/k_0)$ on $X$ via conjugation
3. the subsheaf of $\mathcal{O}_X$, defined only on the $k_0$-open sets, which is $\mathcal{O}_{X_0}$.

Call $\mathcal{O}_{X_0}$ (viewed as a sheaf in the $k_0$-topology on $X$) the **sheaf of $k_0$-rational functions**. Presenting $X$ as a fiber product $X_0 \times_{k_0} k$ is the same thing as endowing $X$ with a **$k_0$-structure** consisting of the conjugation action $\{\sigma_X\}$ and the subsheaf $\mathcal{O}_{X_0}$ of $k_0$-rational functions.

Note: if $x \in X$ is closed, then $p(x)$ is closed in $X_0$, and if $y \in X_0$ is closed, then $p^{-1}(y)$ is a finite set of conjugate closed points of $X$.

**Proposition 6.** *Let $y \in X_0$ be a closed point. Then $\kappa(y)$ is a finite algebraic extension of $k_0$, and there is a natural bijection between the set of points $x \in p^{-1}(y)$ and the set of $k_0$-embeddings of $\kappa(y)$ into $k$.*

**Example 7.** *Let $k_0 = \mathbb{R}$ and $k = \mathbb{C}$. Let $X_0 = \mathbb{A}^1_{\mathbb{R}}$ and $X = \mathbb{A}^1_{\mathbb{C}}$.*
*Take $y \in X_0$ closed (say $y = [(x^2+1)]$). Then $\kappa(y) = \mathbb{R}[x]/(x^2+1)$. The Proposition says: there are two embeddings of $\mathbb{R}[x]/(x^2+1)$ into $\mathbb{C}$. The embeddings are actually $x \mapsto i$ and $x \mapsto -i$.*

*But also let $p : \mathbb{A}^1_{\mathbb{C}} \to \mathbb{A}^1_{\mathbb{R}}$. Then $p^{-1}(y) = \{x - i, x + i\} = \{x = \pm i\}$, so the points are in bijection with the embeddings.*

**Example 8.** *Take a point where the residue field has $q^d$ elements. We get $d$ points upstairs. We get $d$ embeddings of the residue field into $\overline{\mathbb{F}_q}$. And $\mathrm{Gal}(\mathbb{F}_{q^d}/\mathbb{F}_q)$ has $d$ elements. There are $d$ embeddings $\mathbb{F}_{q^d} \hookrightarrow \overline{\mathbb{F}_q}$.*

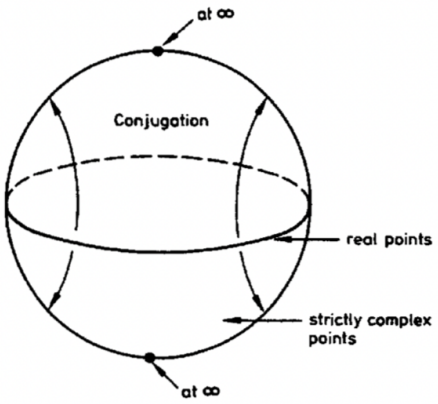
**Definition 9.** *A closed point $y \in X_0$ is **rational over** $k_0$ if $\kappa(y) \cong k_0$.*

**Corollary 10.** *If $k_0$ is perfect and $x \in X$, then $p(x)$ is rational over $k_0 \iff x$ is left fixed by all conjugations.*

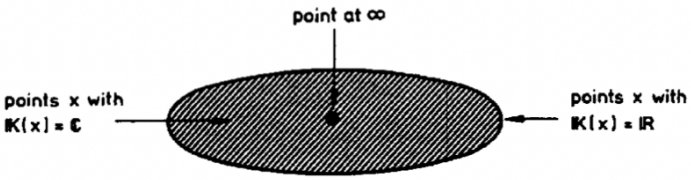
**Example 11.** *Take $k = \mathbb{C}$, $k_0 = \mathbb{R}$,*

$$X_0 = \mathrm{Spec}(\mathbb{R}[X, Y]/(X^2 + Y^2 - 1)).$$

*Then $X$ is a complex affine conic with two points at infinity:*




*And $X_0$ looks like:*




*The boundary of the disc is the circle of points rational over $\mathbb{R}$ corresponding to the maximal ideals*

$$(X - \alpha, Y - \beta) \quad \alpha, \beta \in \mathbb{R} \text{ and } \alpha^2 + \beta^2 = 1.$$

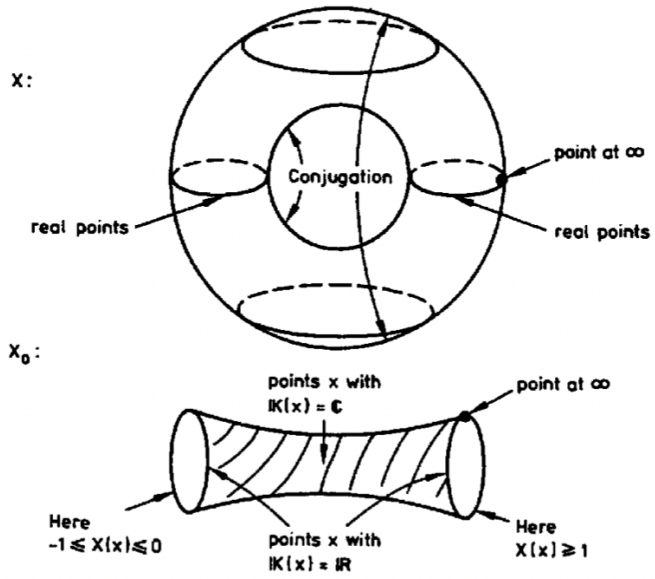
*The interior points $\alpha, \beta \in \mathbb{R}$ and $\alpha^2 + \beta^2 < 1$ correspond to the maximal ideals*

$$(X^2 + Y^2 - 1, \alpha X + \beta Y - 1)$$

*and are not rational over $\mathbb{R}$.*

*For example, $x = \sqrt{2}$ meets the circle (in $\mathbb{C}$) at $(\sqrt{2}, i)$ and $(\sqrt{2}, -i)$. These correspond to $(x - \sqrt{2}, y^2 + 1)$, one closed point with residue field $\mathbb{C}$.*

**Example 12.** *$k_0 = \mathbb{R}$, $k = \mathbb{C}$, $X_0$ the real plane curve $Y^2 = X(X^2 - 1)$:*




**Example 13.** $k_0 = \mathbb{R}$, $k = \mathbb{C}$, $X_0 = \mathrm{Spec}(\mathbb{R}[X,Y]/(X^2+Y^2))$. *Since* $\mathbb{R}[X,Y]/(X^2+Y^2)$ *is a domain,* $X_0$ *is reduced and irreducible. But*

$$X = \mathrm{Spec}(\mathbb{C}[X,Y]/(X+iY)(X-iY))$$

*is the union of two lines, so* $X$ *is reduced but not irreducible. Also* $X_0$ *has exactly one point rational over* $\mathbb{R}$ *(the intersection of two lines).*

$k_0(X_0) = \mathbb{R}(X_0) \cong \mathbb{C}(t)$. $\mathbb{R}$ *is not separably algebraically closed in* $\mathbb{C}(t)$.

*So* $X_0$ *is reduced and irreducible. The ring is not integrally closed, is a domain.* $X$ *is reduced, not irreducible.*

**Example 14.** *If* $k_0$ *is imperfect of characteristic* $p$ *and* $a \in k_0 \setminus k_0^p$, *then let* $\ell(X,Y)$ *be a linear polynomial and let*

$$X_0 = \mathrm{Spec}(k_0[X,Y]/(\ell(X,Y)^p - a))$$

$$X = \mathrm{Spec}(k[X,Y]/((\ell(X,Y) - a^{1/p})^p))$$

$X_0$ *is reduced and irreducible but* $X$ *is just a* $p$*-fold line (set-theoretically just* $\ell = a^{1/p}$, *but* $\ell - a^{1/p}$ *is a nonzero nilpotent function on* $X$*)*

$k_0(X_0) = \mathrm{Frac}(k_0[X,Y]/(\ell^p - a)) \cong k_0(a^{1/p})(t)$ *is not separable over* $k_0$.

**Proposition 15.** *Let* $X_0$ *be a reduced and irreducible prescheme over* $k_0$. *Let* $k$ *be an algebraically closed extension of* $k_0$. *Let* $x \in X_0$ *be its generic point and let* $k_0(X_0)$ *be the stalk of* $\mathcal{O}_{X_0}$ *at* $x$. *Then*

1. $X_0 \times_{k_0} k$ *is reduced* $\iff$ $k_0(X_0)$ *is separable over* $k_0$

*2. $X_0 \times_{k_0} k$ is irreducible $\iff$ $k_0$ is separably algebraically closed in $k_0(X_0)$.*

**Definition 16.** *Let $X_0$ be a prescheme of finite type over $k_0$. Then $X_0$ is a **prevariety over** $k_0$ if it is reduced and irreducible and if a) its function field $k_0(X_0)$ is separable over $k_0$, and b) $k_0$ is algebraically closed in $k_0(X_0)$.*

A prevariety over $k_0$ is a geometrically integral prescheme of finite type over $k_0$.

## 2.5 Closed subpreschemes

Let $\mathcal{F}, \mathcal{G}$ be sheaves of abelian groups on the topological space $X$, $\varphi : \mathcal{F} \to \mathcal{G}$ a morphism. For $U$ open, let $\mathcal{K}(U) = \ker(\mathcal{F}(U) \to \mathcal{G}(U))$. Can check: $\mathcal{K}$ is a sheaf, which we call the **kernel** of $\varphi$.

We say $\varphi$ is **injective** if $\mathcal{K} = 0$ (i.e. if $\mathcal{F}(U) \to \mathcal{G}(U)$ is injective for all $U$ ( $\iff$ $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ is injective for all $x$.) This *does not imply* that $\varphi_U$ is surjective for all $U$.

Let $\mathcal{L}^*$ be the presheaf given by $\mathcal{L}^*(U) = \mathcal{G}(U)/\varphi(\mathcal{F}(U))$. Let $\mathcal{L}$ be the associated sheaf and call it the **cokernel** of $\varphi$.

We say $\varphi$ is **surjective** if $\mathcal{L} = 0$ ( $\iff$ $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ is surjective for all $x$.) This *does not imply* that $\varphi_U$ is surjective for all $U$.

**Remark 1.** *The category of sheaves of abelian groups on $X$ is an abelian category.*

**Proposition 2.** *If $\varphi : \mathcal{F} \to \mathcal{G}$ is injective and surjective, then it is an isomorphism.*

**Remark 3.** *If $X$ is a topological space and $Y \subseteq X$ is a closed subset, then the sheaves of abelian groups on $Y$ correspond bijectively to the sheaves of abelian groups on $X$ such that $\mathcal{F}_x = \{0\}$ if $x \notin Y$:*
*If $\mathcal{F}_0$ is a sheaf on $Y$, **extend it by zero** by defining, for $U \subseteq X$ open,*

$$\mathcal{F}(U) = \begin{cases} \{0\}, & U \cap Y = \varnothing \\ \mathcal{F}_0(U \cap Y), & U \cap Y \neq \varnothing \end{cases}$$

*And the inverse to this is given by restricting a sheaf on $X$ to one on $Y$ (precise definition recalled in Mumford's footnote on p.104)*

Suppose $Y \subseteq X$ is a closed subset. If $X$ were just an ordinary variety with ordinary functions, you might think {functions on $X$} $\to$ {functions on $Y$} should be obtained simply by restricting functions. But for a prescheme, the underlying points do not capture all the information in the structure sheaf. For example: $\operatorname{Spec} k[t]/(t^2)$ has exactly one point, just like $\operatorname{Spec} k[t]/(t)$. But these are not the same prescheme. The first has a nonzero nilpotent $t$.

**Definition 4.** *A **closed subprescheme** of $X$ is*

*1. A closed set $Y \subseteq X$*

2. *A sheaf of rings $\mathcal{O}_Y$ on $Y$ such that $(Y, \mathcal{O}_Y)$ is a prescheme*

3. *A surjective homomorphism $\pi : \mathcal{O}_X \to \mathcal{O}_Y$ (extended by zero)*

Let $\mathcal{Q} = \ker \pi$. $\mathcal{Q}$ is a **sheaf of ideals** in $\mathcal{O}_X$,
i.e. $\mathcal{Q}$ is a subsheaf of $\mathcal{O}_X$ such that for all $U$, $\mathcal{Q}(U)$ is an ideal in $\mathcal{O}_X(U)$. We will call such an object a $\mathcal{O}_X$**-ideal.**

**Remark 5.** *Observe:*

- *We have a stalk sequence $0 \to \mathcal{Q}_x \to \mathcal{O}_{X,x} \to \mathcal{O}_{Y,x} \to 0$*
- *If $\mathcal{Q}_x = \mathcal{O}_{X,x}$, the map to $\mathcal{O}_{Y,x}$ is zero, so $x$ is not actually in the support of the prescheme.*
- *And if $\mathcal{Q}_x \neq \mathcal{O}_{X,x}$, then something survives at $x$.*

*Conclude that $Y = \{x \in X \mid \mathcal{Q}_x \neq \mathcal{O}_{X,x}\}$, and $\mathcal{O}_Y \cong \operatorname{coker}(\mathcal{Q} \to \mathcal{O}_X)$. Thus $\mathcal{Q} = \ker \pi$ determines the closed subpreschemes up to canonical isomorphism, because:*

1. *$\mathcal{O}_Y = \mathcal{O}_X/\mathcal{Q}$*
2. *$Y = \{x \in X \mid \mathcal{Q}_x \neq \mathcal{O}_{X,x}\}$*

*So from $\mathcal{Q}$ and $X$, we can recover $(Y, \mathcal{O}_Y)$.*

**Definition 6.** *A morphism of preschemes $f : Y \to X$ is a* ***closed immersion*** *if*

1. *$f$ is injective*
2. *$f$ is closed, i.e. $Z \subseteq Y$ closed $\implies$ $f(Z)$ closed*
3. *$f_y^* : \mathcal{O}_{X,f(y)} \to \mathcal{O}_{Y,y}$ is surjective for all $y \in Y$.*

Note: a closed immersion is exactly a morphism that identifies $Y$ with a closed subprescheme of $X$.

**Proposition 7.** *Let $R$ be a ring, $A \subseteq R$ an ideal. The canonical map $\pi : R \to R/A$ defines a morphism $f : \operatorname{Spec}(R/A) \to \operatorname{Spec}(R)$. Then $f$ is a closed immersion and the corresponding sheaf of ideals $\mathcal{Q}$ satisfies:*

1. *$\Gamma(\operatorname{Spec}(R)_f, \mathcal{Q}) = A \cdot R_f$*
2. *$\mathcal{Q}_x = A \cdot \mathcal{O}_{\operatorname{Spec}(R),x}$*

Thus an ideal produces a closed subprescheme, via

$$A \subseteq R \mapsto \operatorname{Spec}(R/A) \subseteq \operatorname{Spec} R$$

because $\operatorname{Spec}(R/A) \to \operatorname{Spec} R$ is a closed immersion. And for the defining ideal sheaf $\mathcal{Q}$ of $\operatorname{Spec}(R/A)$: On a principal affine open $D(f) = (\operatorname{Spec} R)_f$, $\mathcal{Q}(D(f)) = AR_f$, and at a point $x = [P]$, $\mathcal{Q}_x = AR_P$.

**Theorem 8.** *Let $X = \operatorname{Spec}(R)$ and let $Y \subseteq X$ be a closed subprescheme, $f : Y \to X$ the inclusion. Let $\mathcal{Q}$ be the $\mathcal{O}_X$-ideal defining $Y$, $A = \Gamma(X, \mathcal{Q})$. Then $Y$ is canonically isomorphic to $\operatorname{Spec}(R/A)$, i.e. there is a commutative diagram*

$$\begin{array}{ccc} Y & \xhookrightarrow{\quad f \quad} & \operatorname{Spec}(R) \\ & \searrow^{=} \quad \nearrow & \\ & \operatorname{Spec}(R/A) & \end{array}$$

Thus every affine closed subprescheme comes from an ideal. Given a closed subprescheme $Y \subseteq X = \operatorname{Spec} R$, $Y$ has its defining ideal sheaf $\mathcal{Q} = \ker(\mathcal{O}_X \to \mathcal{O}_Y)$. We can get an ideal via

$$(Y \subseteq X) \mapsto (A = \Gamma(X, \mathcal{Q}) \subseteq \Gamma(X, \mathcal{O}_X) = R).$$

Further, $Y \cong \operatorname{Spec}(R/A)$.

**Remark 9.** *Nullstellensatz gave us:*

$$\{\textit{radical ideals}\} \leftrightarrow \{\textit{closed subsets}\}$$

*Our results from II.5 give:*

$$\{\textit{all ideals}\} \leftrightarrow \{\textit{closed subpreschemes}\}$$

**Corollary 10.** *Let $f : Y \to X$ be a morphism of preschemes. Then TFAE:*

*(i) $f$ is a closed immersion;*

*(ii) for all affine open sets $U \subseteq X$, $f^{-1}(U)$ is affine and the map*

$$\Gamma(U, \mathcal{O}_X) \longrightarrow \Gamma(f^{-1}(U), \mathcal{O}_Y)$$

*is surjective;*

*(iii) there exists an affine open covering $\{U_i\}$ of $X$ such that $f^{-1}(U_i)$ is affine and the map*

$$\Gamma(U_i, \mathcal{O}_X) \longrightarrow \Gamma(f^{-1}(U_i), \mathcal{O}_Y)$$

*is surjective for all $i$.*

**Corollary 11.** *Let $X$ be a prescheme and let $\mathcal{Q}$ be an $\mathcal{O}_X$-ideal. Let*

$$Y = \{y \in X \mid 1 \notin \mathcal{Q}_y\},$$

*a closed subset of $X$. The cokernel of $0 \longrightarrow \mathcal{Q} \longrightarrow \mathcal{O}_X$ is $0$ outside of $Y$, so consider it as a sheaf of rings $\mathcal{O}_Y$ on $Y$. Then:*

$$(Y, \mathcal{O}_Y) \textit{ is a subprescheme of } (X, \mathcal{O}_X) \iff$$

*For all $y \in X, \exists$ a neighborhood $U$ of $y$ and sections $\{s_\alpha\}$ in $\Gamma(U, \mathcal{Q})$ such that* $\underbrace{\mathcal{Q}_x = \sum_\alpha \operatorname{res}_x(s_\alpha)\mathcal{O}_{X,x}}_{\textit{quasi-coherence condition}}$

**Example 12.** *(Closed subschemes of* $\operatorname{Spec}(k[t])$*)*
*Say* $k$ *is algebraically closed. Since* $k[t]$ *is a PID, all nonzero ideals are of the form*

$$A = \left( \prod_{i=1}^{n} (t - a_i)^{r_i} \right).$$

*The corresponding subscheme* $Y = \operatorname{Spec}(k[t]/A)$ *of* $\mathbb{A}^1 = \operatorname{Spec}(k[t])$ *has support* $\{a_1, \ldots, a_n\}$*, and at* $a_i$*, its structure sheaf is*

$$\mathcal{O}_{Y,a_i} = \mathcal{O}_{\mathbb{A}^1,a_i}/\mathfrak{m}_i^{r_i},$$

*where*

$$\mathfrak{m}_i = \mathfrak{m}_{\mathbb{A}^1,a_i} = (t - a_i).$$

*Focus on the schemes that are supported just by* $\{0\}$*. Since*

$$V((t^n)) = V(\sqrt{(t^n)}) = V((t)) = \{0\},$$

*these are the schemes*

$$Y_n = \operatorname{Spec}\left(k[t]/(t^n)\right).$$

*Then*

$$Y_1 \subset Y_2 \subset Y_3 \subset \cdots \subset Y_n \subset \cdots \subset \mathbb{A}^1.$$

*Suppose*

$$g(t) = a_0 + a_1 t + a_2 t^2 + \cdots.$$

*Modulo* $t^n$*, every term of degree* $n$ *or higher disappears:*

$$[g(t)] = a_0 + a_1 t + \cdots + a_{n-1} t^{n-1}.$$

*So* $g|_{Y_n}$ *remembers exactly the first* $n$ *coefficients of* $g$*. In general*

$$a_j = \frac{g^{(j)}(0)}{j!}.$$

*Thus*

$$g(t) \equiv g(0) + g'(0)t + \frac{g''(0)}{2!}t^2 + \cdots + \frac{g^{(n-1)}(0)}{(n-1)!}t^{n-1} \pmod{t^n}.$$

*So if you restrict a function* $g$ *on* $\mathbb{A}^1$ *to* $Y_n$*, it remembers the derivatives*

$$g(0), g'(0), \ldots, g^{(n-1)}(0).$$

**Example 13.** *(Closed subschemes of* $\operatorname{Spec}(k[x, y])$*)*
*Let* $k$ *be algebraically closed. Let*

$$X = \mathbb{A}^2_k = \operatorname{Spec} k[x, y], \qquad P = (0, 0), \qquad \mathfrak{m}_P = (x, y).$$

*For* $(\alpha, \beta) \neq (0, 0)$*, consider*

$$A_{\alpha,\beta} = (\alpha x + \beta y,\ x^2,\ xy,\ y^2) = (\alpha x + \beta y) + \mathfrak{m}_P^2.$$

*The corresponding closed subprescheme is*

$$Y_{\alpha,\beta} = \operatorname{Spec}\left(\frac{k[x,y]}{(\alpha x + \beta y, x^2, xy, y^2)}\right).$$

*Since* $\sqrt{A_{\alpha,\beta}} = (x,y)$*, the underlying closed set is just*

$$|Y_{\alpha,\beta}| = \{P\}.$$

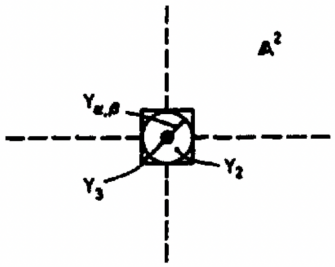
*But* $Y_{\alpha,\beta}$ *is not reduced: it retains one infinitesimal direction at* $P$*.*

*All of the following have underlying space* $\{P\}$*:*

$$Y_{\text{red}} = \operatorname{Spec}\frac{k[x,y]}{(x,y)}, \quad Y_{\alpha,\beta} = \operatorname{Spec}\frac{k[x,y]}{(\alpha x + \beta y, x^2, xy, y^2)},$$

$$Y_2 = \operatorname{Spec}\frac{k[x,y]}{(x^2, xy, y^2)} = \operatorname{Spec}\frac{k[x,y]}{\mathfrak{m}_P^2}, \quad \textit{and } Y_3 = \operatorname{Spec}\frac{k[x,y]}{(x^2, y^2)}.$$

- $Y_{\text{red}}$ *remembers only the value* $g(P)$*.*
- $Y_{\alpha,\beta}$ *remembers the value and one directional first-order variation.*
- $Y_2$ *remembers* $g(P), \frac{\partial g}{\partial x}(P), \frac{\partial g}{\partial y}(P)$*.*
- $Y_3$ *retains still more infinitesimal information, since* $xy \neq 0$ *in* $k[x,y]/(x^2,y^2)$*.*




*Now consider*

$$A = (x^2, xy) = x(x,y) \subseteq k[x,y].$$

*Then*

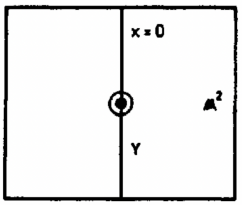
$$Y = \operatorname{Spec}\frac{k[x,y]}{(x^2,xy)}.$$

*Its underlying closed set is* $V(x^2,xy) = V(x)$ *because* $\sqrt{(x^2,xy)} = (x)$*. Thus the support is the line* $\{x = 0.\}$ *But the scheme has extra infinitesimal structure at the origin:* $x \neq 0, x^2 = 0,$ *and* $xy = 0$ *in its coordinate ring.*

**Proposition 14.** *Let* $X$ *be a prescheme and* $Z \subseteq X$ *a closed subset. Consider the various closed subschemes*

$$Z_1 = (Z, \mathcal{O}_X/\mathcal{I}_1)$$

*that can be defined with support* $Z$*.*




1. *Among all these, there is a unique reduced subscheme (call it $Z_0 = (Z, \mathcal{O}_X/\mathcal{I}_0)$).*
2. *Furthermore, if $Z_1 = (Z, \mathcal{O}_X/\mathcal{I}_1)$ is any other, then $\mathcal{I}_1 \subseteq \mathcal{I}_0$ (so $Z_0$ is a subscheme of $Z_1$) and $\mathcal{I}_0 = \sqrt{\mathcal{I}_1}$, i.e.,*
$$\Gamma(U, \mathcal{I}_0) = \sqrt{\Gamma(U, \mathcal{I}_1)}$$
*for $U$ affine.*

So if $Z \subseteq X$ is a closed subset, then among all the different closed subpreschemes having support $Z$, there is a unique reduced one.

**Example 15.** *(Fibers of a morphism)*
*Let $f : X \to Y$ be a morphism and let $y \in Y$ be a closed point.*
*Consider the fiber product $X_y = X \times_Y \operatorname{Spec}(\kappa(y))$ in the diagram*

$$\begin{array}{ccc} X_y & \xrightarrow{j} & X \\ \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle f} \\ \operatorname{Spec}(\kappa(y)) & \xrightarrow{i} & Y \end{array}$$

*where $i$ is the morphism with image $y$ and $i^* : \mathcal{O}_{Y,y} \to \kappa(y)$ is the canonical map to the residue field.*
*Since $y$ is closed,*
$$i : \operatorname{Spec} \kappa(y) \hookrightarrow Y$$
*is a closed immersion. Closed immersions are preserved under base change, so*

$$\boxed{j : X_y \hookrightarrow X \text{ is a closed immersion.}}$$

*Thus the fiber $X_y = X \times_Y \operatorname{Spec}(\kappa(y))$ is naturally a closed subprescheme of $X$ supported on the set-theoretic fiber $f^{-1}(y)$.*
*If*

- *$V = \operatorname{Spec}(S)$ is an affine open neighborhood of $y$,*
- *$\{U_i = \operatorname{Spec}(R_i)\}$ covers $f^{-1}(V)$,*
- *and $y = [M]$ where $M \subseteq S$ is maximal*

*then by our results about fiber products, $X_y = X \times_Y \operatorname{Spec}(\kappa(y))$ is covered by open affine pieces*
$$X_y^{(i)} = \operatorname{Spec}(R_i \otimes_S \kappa(y)),$$

*and since* $\kappa(y) \cong S/M$,

$$X_y^{(i)} \cong \operatorname{Spec}(R_i/M \cdot R_i).$$

*Thus, scheme-theoretic fibers can be nonreduced, because the ideal* $MR_i$ *need not be radical. For example, let*

$$X = \operatorname{Spec}(k[X_1, \ldots, X_n]/(f_1, \ldots, f_m))$$

*be an affine scheme of finite type over* $k$. *Define*

$$\varphi : \mathbb{A}_k^n \to \mathbb{A}_k^m$$

*such that* $\varphi^* : k[Y_1, \ldots, Y_m] \to k[X_1, \ldots, X_n]$ *maps* $Y_i \mapsto f_i$.

*Then* $X$ *can be realized as the fiber of* $\varphi$ *over the origin.*

*And if* $(f_1, \ldots, f_m)$ *is not radical then the fiber need not be reduced.*

*For example:* $X_0 = \operatorname{Spec}(k[t]/(t^2))$ *can be realized as the fiber of* $f : \mathbb{A}_k^1 \to \mathbb{A}_k^1, t \mapsto t^2$ *over* $0$, *and this ends up being a nonreduced double point.*

**Problem 2.3.** ***Show that the closed subpreschemes of*** $\mathbb{P}_k^n$ ***correspond bijectively with the homogeneous ideals***

$$A \subseteq k[X_0, \ldots, X_n]$$

***such that***

$$A : (X_0, \ldots, X_n) = A$$

***(i.e., if*** $f \in k[X_0, \ldots, X_n]$, $X_i \cdot f \in A$ ***for all*** $i$, ***then*** $f \in A$.***) This condition says that*** $A$ ***is biggest among those ideals that induce the same ideals in each affine piece of*** $\mathbb{P}^n$.***)***

*Let*

$$S = k[X_0, \ldots, X_n], \qquad \mathfrak{m} = (X_0, \ldots, X_n).$$

*We prove that closed subpreschemes of* $\mathbb{P}_k^n$ *are in bijection with homogeneous ideals* $A \subseteq S$ *satisfying*

$$A : \mathfrak{m} = A.$$

*Recall that* $\mathbb{P}_k^n$ *is covered by the standard affine open sets*

$$U_i = \{X_i \neq 0\} \cong \operatorname{Spec} k \left[ \frac{X_0}{X_i}, \ldots, \frac{\widehat{X_i}}{X_i}, \ldots, \frac{X_n}{X_i} \right].$$

*Step 1: A homogeneous ideal determines a closed subprescheme.*
*Let* $A \subseteq S$ *be homogeneous. On the affine chart* $U_i$, *define*

$$A_i = \left\{ \frac{f}{X_i^d} \;\middle|\; f \in A \text{ homogeneous of degree } d \right\}.$$

*Thus $A_i$ is an ideal in*

$$k\left[\frac{X_0}{X_i},\dots,\frac{\widehat{X_i}}{X_i},\dots,\frac{X_n}{X_i}\right].$$

*It therefore defines a closed subprescheme*

$$Y_i = V(A_i) \subseteq U_i.$$

*On the overlap $U_i \cap U_j$, the ideals $A_i$ and $A_j$ agree after localization, since both come from the same homogeneous ideal $A$. Hence the $Y_i$ agree on overlaps and glue to a closed subprescheme*

$$Y(A) \subseteq \mathbb{P}^n_k.$$

*Step 2: The condition $A : \mathfrak{m} = A$ makes $A$ the largest ideal giving the same affine ideals. Suppose $f \in S$ is homogeneous of degree $d$ and*

$$X_i f \in A \qquad \text{for every } i.$$

*Then on $U_i$,*

$$\frac{f}{X_i^d} = \frac{X_i f}{X_i^{d+1}} \in A_i.$$

*Thus adjoining $f$ to $A$ would not change the ideal induced on any of the affine charts $U_i$.*

*Therefore, if $A$ is to be the largest homogeneous ideal inducing the given ideals $A_i$, we must have*

$$X_i f \in A \text{ for all } i \quad \Longrightarrow \quad f \in A.$$

*This is exactly the condition*

$$A : \mathfrak{m} = A.$$

*Conversely, suppose $A : \mathfrak{m} = A$. Then, by induction,*

$$A : \mathfrak{m}^r = A$$

*for every $r \geq 1$.*
*Indeed, if*

$$\mathfrak{m}^r f \subseteq A,$$

*then for each $i$,*

$$\mathfrak{m}^{r-1}(X_i f) \subseteq A.$$

*Inductively $X_i f \in A$ for every $i$, and since $A : \mathfrak{m} = A$, it follows that $f \in A$.*

*Thus if some homogeneous polynomial can be added to $A$ without changing any of the ideals on the affine charts, it must already belong to $A$.*

*Step 3: Recover an ideal from a closed subprescheme.*
*Conversely, let*

$$Y \subseteq \mathbb{P}^n_k$$

*be a closed subprescheme.*

*For each $i$, the intersection*

$$Y \cap U_i$$

*is a closed subprescheme of the affine scheme $U_i$, so it is defined by a unique ideal*

$$I_i \subseteq k\left[\frac{X_0}{X_i}, \ldots, \widehat{\frac{X_i}{X_i}}, \ldots, \frac{X_n}{X_i}\right].$$

*Since the $I_i$ arise from the same closed subprescheme, they agree on the overlaps $U_i \cap U_j$.*

*Define a homogeneous ideal $A(Y) \subseteq S$ as follows. For a homogeneous polynomial $f$ of degree $d$, declare*

$$f \in A(Y)$$

*if and only if*

$$\frac{f}{X_i^d} \in I_i \qquad \text{for every } i.$$

*This defines a homogeneous ideal of $S$.*

*By construction, the ideal induced by $A(Y)$ on each $U_i$ is $I_i$. Hence*

$$Y(A(Y)) = Y.$$

*We now check that*

$$A(Y) : \mathfrak{m} = A(Y).$$

*Suppose $f$ is homogeneous of degree $d$ and*

$$X_i f \in A(Y) \qquad \text{for all } i.$$

*Fix $i$. Since $X_i f \in A(Y)$, we have*

$$\frac{X_i f}{X_i^{d+1}} = \frac{f}{X_i^d} \in I_i.$$

*This holds for every $i$, so by the definition of $A(Y)$,*

$$f \in A(Y).$$

*Therefore*

$$A(Y) : \mathfrak{m} = A(Y).$$

*Step 4: The two constructions are inverse.*
*We have already shown that*

$$Y(A(Y)) = Y.$$

*Now let $A$ be a homogeneous ideal satisfying*

$$A : \mathfrak{m} = A.$$

*Certainly*

$$A \subseteq A(Y(A)).$$

*For the reverse inclusion, let $f$ be homogeneous and suppose*

$$f \in A(Y(A)).$$

*Then on every chart $U_i$,*

$$\frac{f}{X_i^{\deg f}} \in A_i.$$

*By the definition of localization, for each $i$ there exists an integer $r_i \geq 0$ such that*

$$X_i^{r_i} f \in A.$$

*Choose $N$ large enough that every monomial of degree $N$ is divisible by at least one of*

$$X_0^{r_0}, \ldots, X_n^{r_n}.$$

*For example, one may take*

$$N = r_0 + \cdots + r_n.$$

*It follows that*

$$\mathfrak{m}^N f \subseteq A.$$

*But, as observed above,*

$$A : \mathfrak{m}^N = A,$$

*so*

$$f \in A.$$

*Hence*

$$A(Y(A)) = A.$$

*Thus the assignments*

$$A \longmapsto Y(A) \qquad \text{and} \qquad Y \longmapsto A(Y)$$

*are mutually inverse.*

*Therefore the closed subpreschemes of* $\mathbb{P}^n_k$ *correspond bijectively to the homogeneous ideals*

$$A \subseteq k[X_0, \ldots, X_n]$$

*such that*

$$\boxed{A : (X_0, \ldots, X_n) = A.}$$

*Equivalently, these are exactly the homogeneous ideals that are largest among all homogeneous ideals inducing the same ideals on each standard affine piece of* $\mathbb{P}^n_k$.

## 2.6 The functor of points of a prescheme

**Example 1.** *The point set of a fiber product* $X \times_S Y$ *does not necessarily map injectively into the set-theoretic product of* $X$ *and* $Y$.

*For example,*

$$\operatorname{Spec}(\mathbb{C}) \times_{\operatorname{Spec}(\mathbb{R})} \operatorname{Spec}(\mathbb{C}) \cong \operatorname{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}) .$$

*Now*

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}[X]/(X^2 + 1) \cong \mathbb{C}[X]/((X - i)(X + i)) \cong \mathbb{C} \times \mathbb{C}.$$

*Thus*

$$\operatorname{Spec}(\mathbb{C}) \times_{\operatorname{Spec}(\mathbb{R})} \operatorname{Spec}(\mathbb{C}) \cong \operatorname{Spec}(\mathbb{C} \times \mathbb{C}) \cong \operatorname{Spec}(\mathbb{C}) \sqcup \operatorname{Spec}(\mathbb{C}).$$

*Therefore the fiber product has two points, whereas the set-theoretic product of two one-point spaces has only one point.*

*In other words, the two points of*

$$\operatorname{Spec}(\mathbb{C}) \times_{\operatorname{Spec}(\mathbb{R})} \operatorname{Spec}(\mathbb{C})$$

*both map to the same pair of points in*

$$\operatorname{Spec}(\mathbb{C}) \times \operatorname{Spec}(\mathbb{C}).$$

*Hence the natural map from the point set of the fiber product to the set-theoretic product is not injective.*

**Example 2.** *Let* $C =$ *category of differentiable manifolds, let* $z$ *be the manifold with one point. Then for any manifold* $X$,

$$\operatorname{Hom}_C(z, X) \cong X$$

*as a point set.*

**Example 3.** *Let* $C =$ *category of groups, let* $z = \mathbb{Z}$. *Then for any group* $G$,

$$\operatorname{Hom}_C(z, G) \cong G$$

*as a point set.*

**Example 4.** *Let $C$ = category of rings with 1 (and homomorphisms $f$ such that $f(1) = 1$), let $z = \mathbb{Z}[X]$. Then for any ring $R$,*

$$\mathrm{Hom}_C(z, R) \cong R$$

*as a point set.*

So if $C$ is any category whose objects may not be point sets to begin with and $z$ is an object, we can think of $\mathrm{Hom}_C(z, X)$ as the underlying set of points of the object $X$.

$$X \mapsto \mathrm{Hom}_C(z, X)$$

extends to a function $C \to$ Sets. But we can only call $\mathrm{Hom}_C(z, X)$ the set of points of $X$ if this function is **faithful**, a morphism $f : X_1 \to X_2$ is determined by the map of sets $\tilde{f} : \mathrm{Hom}_C(z, X_1) \to \mathrm{Hom}_C(z, X_2)$.

**Example 5.** *Let* Hot *be the category of CW complexes, where* $\mathrm{Hom}(X, Y)$ *is the set of homotopy-classes of continuous maps from $X$ to $Y$. If $z =$ the one point complex, then*

$$\mathrm{Hom}_{\mathrm{Hot}}(z, X) = \pi_0(X) \quad \textit{(the set of components of } X\textit{)}$$

*and this does not give a faithful functor. The issue is that maps out of a point detect only connected components, so distinct homotopy classes of maps can induce the same map on $\pi_0$.*

**Example 6.** *Let $C$ = category of preschemes. Take for $z$ the final object of the category $C$: $z = \mathrm{Spec}(\mathbb{Z})$. Then* $\mathrm{Hom}_C(\mathrm{Spec}(\mathbb{Z}), X)$ *is really small and does not give a faithful functor.*

*The problem here is that no scheme plays the role of $\mathbb{Z}[X]$, so we must choose $\mathbb{Z}$, and then we get things like e.g.* $\mathrm{Hom}(\mathrm{Spec}\,\mathbb{Z}, \mathrm{Spec}\,\mathbb{F}_{p^2}) = \varnothing$ *(otherwise there would be a unital ring homomorphism $\mathbb{F}_{p^2} \to \mathbb{Z}$).*

To remedy this, one considers all test objects at once; informally, attach to $X$ the collection

$$\bigcup_z \mathrm{Hom}_C(z, X).$$

The correct formal version of this idea is the functor of points below. We have extra structure on the above set too:

1. the decomposition of $\bigcup_z \mathrm{Hom}_C(z, X)$ into subsets $S_z = \mathrm{Hom}_C(z, X)$
2. the natural maps from one $S_z$ to another $S_{z'}$, one for each morphism $g : z \to z'$ in the category

Formally: attach to each $X$ in $C$ the functor $h_X$ (contravariant from $C$ to Sets) via

(*) $h_X(z) = \mathrm{Hom}_C(z, X)$, $\quad z$ an object in $C$

(**) $h_X(g) = \{\text{induced map from } \mathrm{Hom}_C(z, X) \text{ to } \mathrm{Hom}_C(z', X)\}$, $\quad g : z' \to z$ a morphism in $C$.
$(h_X(f : Y \to Z) = (-) \circ f : \mathrm{Hom}_C(Z, X) \to \mathrm{Hom}_C(Y, X))$

And $h_X$ lives in the category $\mathrm{Funct}(C^{\mathrm{op}}, \mathrm{Sets})$. Also if $g : X_1 \to X_2$ is a morphism in $C$, then we get a morphism of functors $h_g : h_{X_1} \to h_{X_2}$. So we get one big functor

$$h : C \to \mathrm{Funct}(C^{\mathrm{op}}, \mathrm{Sets}).$$

**Proposition 7** (Yoneda Lemma)**.** *$h$ is fully faithful, i.e. if $X_1, X_2$ are objects of $C$, then under $h$,*

$$\mathrm{Hom}_C(X_1, X_2) \cong \mathrm{Hom}_{\mathrm{Funct}}(h_{X_1}, h_{X_2}).$$

Sketch of proof:

$$h_X(-) = \mathrm{Hom}_C(-, X).$$

Given a morphism $f : X_1 \to X_2$, composition with $f$ gives a natural transformation

$$h_{X_1} \to h_{X_2}, \qquad (g : T \to X_1) \mapsto f \circ g.$$

Conversely, given a natural transformation $\eta : h_{X_1} \to h_{X_2}$, evaluate it at $X_1$ and at the identity:

$$f := \eta_{X_1}(\mathrm{id}_{X_1}) \in \mathrm{Hom}_C(X_1, X_2).$$

Naturality implies that for every $g : T \to X_1$,

$$\eta_T(g) = f \circ g.$$

Thus $\eta$ is completely determined by $f$, and these two constructions are inverse. Hence

$$\mathrm{Hom}_C(X_1, X_2) \cong \mathrm{Hom}_{\mathrm{Funct}}(h_{X_1}, h_{X_2}).$$

**Definition 8.** *If $X$ and $K$ are preschemes, a $K$-**valued point of** $X$ is a morphism $f : K \to X$.*
*If $K = \mathrm{Spec}(R)$, we call this an $R$-**valued point of** $X$.*
*If $X$ and $K$ are preschemes over a third prescheme $S$ (i.e. we are given morphisms $p_X : X \to S, p_K : K \to S$), then $f$ is a $K$-**valued point of** $X/S$ if $p_X \circ f = p_K$.*
*If $K = \mathrm{Spec}(R)$, we call this an $R$-**valued point of** $X/S$.*
*The set of all $R$-valued points of a prescheme $X$, or of $X/S$, is denoted $X(R)$.*

So $X(K) =$ morphisms $K \to X$ of $S$-*schemes*.

If $K, X, Y$ are preschemes over $S$ then the set of $K$-valued points of $X \times_S Y/S$ is the set-theoretic product of the set of $K$-valued points of $X/S$ and the set of $K$-valued points of $Y/S$ (by def. of fiber product).

**Example 9.** *What are the $\mathbb{C}$-valued points of*

$$\mathrm{Spec}(\mathbb{C}) \times_{\mathrm{Spec}(\mathbb{R})} \mathrm{Spec}(\mathbb{C})?$$

*Recall that*

$$\mathrm{Spec}(\mathbb{C}) \times_{\mathrm{Spec}(\mathbb{R})} \mathrm{Spec}(\mathbb{C}) \cong \mathrm{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}).$$

*There are two relevant $\mathbb{R}$-algebra maps:*

*1.*

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \to \mathbb{C}, \qquad a \otimes b \mapsto ab;$$

*2.*

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \to \mathbb{C}, \qquad a \otimes b \mapsto a\sigma(b),$$

*where $\sigma$ is complex conjugation.*

*Their kernels are the two points of*

$$\mathrm{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}) \cong \mathrm{Spec}(\mathbb{C}) \sqcup \mathrm{Spec}(\mathbb{C}).$$

*These two points can be thought of as the "graphs" of* id *and $\sigma$.*

*Another way to see this is that*

$$\mathrm{Spec}(\mathbb{C})(\mathbb{C}) = \mathrm{Hom}_{\mathbb{R}\text{-}alg}(\mathbb{C}, \mathbb{C}) = \{\mathrm{id}, \sigma\} = \mathrm{Gal}(\mathbb{C}/\mathbb{R}).$$

*Thus, each copy of* $\mathrm{Spec}(\mathbb{C})$ *has two $\mathbb{C}$-valued points over $\mathbb{R}$, corresponding to* id *and $\sigma$.*
*Therefore the fiber product has*

$$\mathrm{Spec}(\mathbb{C})(\mathbb{C}) \times \mathrm{Spec}(\mathbb{C})(\mathbb{C}) = \{\mathrm{id}, \sigma\} \times \{\mathrm{id}, \sigma\},$$

*so it has four $\mathbb{C}$-valued points.*

**Remark 10.** *If $f_1, \ldots, f_m \in \mathbb{Z}[X_1, \ldots, X_n]$ and $X = \mathrm{Spec}(\mathbb{Z}[X_1, \ldots, X_n]/(f_1, \ldots, f_m))$, then an $R$-valued point of $X$ is the same as an $n$-tuple $a_1, \ldots, a_n \in R$ such that $f_1(a_1, \ldots, a_n) = \cdots = f_m(a_1, \ldots, a_n) = 0$.*

*A morphism $g : \mathrm{Spec}(R) \to \mathrm{Spec}(\mathbb{Z}[X_1, \ldots, X_n]/(f_1, \ldots, f_m))$ is determined by the $n$-tuple $a_i = g^*(X_i), 1 \leq i \leq n$, and those $n$-tuples that occur are exactly those such that $h \mapsto h(a_1, \ldots, a_n)$ defines a homomorphism $g^* : \mathbb{Z}[X_1, \ldots, X_n]/(f_1, \ldots, f_m) \to R$, i.e. solutions of $f_1, \ldots, f_m$.*

**Remark 11.** *Let $X$ be a prescheme and let $h_X^{(0)}$ be the covariant functor* Rings $\to$ Sets *defined by*

$$h_X^{(0)}(R) = h_X(\mathrm{Spec}(R)) = \mathrm{Hom}(\mathrm{Spec}(R), X).$$

*This deals with the idea of "representability."*

**Proposition 12.** *For any two preschemes $X_1, X_2$,*

$$\mathrm{Hom}(X_1, X_2) \cong \mathrm{Hom}(h_{X_1}^{(0)}, h_{X_2}^{(0)}).$$

*Hence $h^{(0)}$ is a fully faithful functor from the category of preschemes to* $\mathrm{Func}(\mathrm{Rings}, \mathrm{Sets})$.

**Remark 13.** *(Grothendieck's existence problem)*
*Why not identify a prescheme $X$ with its corresponding functor $h_X^{(0)}$ and try to define preschemes as suitable functors $F :$ Rings $\to$ Sets? Problem: to find natural conditions on $F$ to ensure it is isomorphic to a functor of type $h_X^{(0)}$.*

*For example, all $h_X^{(0)}$ have compatibility with faithfully flat descent: if $q : A \to B$ is a homomorphism of rings making $B$ into a faithfully flat $A$-algebra, i.e. for all ideals $I \subseteq A$, $I \otimes_A B \cong I \cdot B$ and $q^{-1}(I \cdot B) = I$. Then if $p_1, p_2 : B \to B \otimes_A B$ are the homomorphisms $\beta \mapsto \beta \otimes 1$ and $\beta \mapsto 1 \otimes \beta$, the induced diagram of sets*

$$F(A) \xrightarrow{F(q)} F(B) \underset{F(p_2)}{\overset{F(p_1)}{\rightrightarrows}} F(B \otimes_A B)$$

*is exact, i.e. $F(q)$ is injective and $\mathrm{Im}(F(q)) = \{x \in F(B) \mid F(p_1)(x) = F(p_2)(x)\}$.*

*Say you want to look at all curves of genus $g$. Can you find a scheme $X$ such that*

$$X(T) = \mathrm{Hom}(T, X)$$

*corresponds to families of genus-$g$ curves parametrized by $T$? Such an $X$ (when it exists) is called a **moduli space**.*

*More generally, start with some geometric objects you want to classify and define a functor*

$$F : \mathrm{Sch}^{\mathrm{op}} \to \mathrm{Set},$$

*where $F(T)$ is the set of families of those objects parametrized by $T$. We then ask: does there exist a scheme $X$ such that*

$$F \cong h_X = \mathrm{Hom}(-, X)?$$

*That is, is $F$ the functor of points of some scheme $X$? If so, we say that $F$ is **representable** by $X$.*

*Thus the moduli problem becomes: define the desired functor $F$, and ask whether it comes from a scheme.*

**Example 14.** *For the case of a field:*
*$\{k$-valued points of $X\} \cong \{$ points $x \in X$, plus injections $\kappa(x) \hookrightarrow k\}$.*

*$\{k$-valued points of $X$ over $k_0\} \cong \{x \in X$ and $\kappa(x) \hookrightarrow k$ that reduce to identity on $k_0\}$.*

*If $k = k_0$ is algebraically closed and $X$ is of finite type over $k$, then*
*$\{k$-valued points of $X/k\} \cong \{x \in X$ such that $k \cong \kappa(x)\} = \{$closed points $x \in X\}$.*

*So if $X_0$ is a prevariety over $k_0$, we recover the underlying set of the old variety over $k$ associated to $X$. Thus when $k$ is an algebraically closed extension field of $k_0$, $k$-valued points of $X_0/k_0$ are often called **geometric points of** $X_0$.*

**Example 15.** $X = \mathbb{A}^1_{\mathbb{R}} = \mathrm{Spec}\,\mathbb{R}[x]$.

*The point $x = 0$ corresponds to the ideal $(x)$, and*

$$\mathbb{R}[x]/(x) \cong \mathbb{R}.$$

*The point $x = 1$ corresponds to the ideal $(x-1)$, and*

$$\mathbb{R}[x]/(x-1) \cong \mathbb{R}.$$

*The ideal $(x^2+1)$ also corresponds to a point of* $\operatorname{Spec}\mathbb{R}[x]$*. Since $x^2+1$ is irreducible over $\mathbb{R}$, this is a closed point of degree 2, with residue field*

$$\mathbb{R}[x]/(x^2+1) \cong \mathbb{C}.$$

*Over $\mathbb{C}$, there are two ways to realize this point, corresponding to the two $\mathbb{R}$-algebra homomorphisms*

$$\varphi_1 : x \mapsto i \qquad \text{and} \qquad \varphi_2 : x \mapsto -i.$$

*These give two $\mathbb{C}$-valued points lying over the same degree-2 point $(x^2+1)$ of $\mathbb{A}^1_{\mathbb{R}}$. The two are related by complex conjugation.*

*From the other perspective, to look at $X(\mathbb{R})$, we look at the morphisms*

$$\operatorname{Spec}\mathbb{R} \to X = \mathbb{A}^1_{\mathbb{R}} = \operatorname{Spec}\mathbb{R}[x],$$

*so we get homomorphisms of $\mathbb{R}$-algebras in the other direction:*

$$\mathbb{R}[x] \to \mathbb{R}.$$

*Here we can map $x$ to any element of $\mathbb{R}$. Thus the $\mathbb{R}$-valued points are exactly the usual real points of the real line.*

*To look at $X(\mathbb{C})$, we instead look at morphisms*

$$\operatorname{Spec}\mathbb{C} \to \operatorname{Spec}\mathbb{R}[x],$$

*which correspond to homomorphisms of $\mathbb{R}$-algebras*

$$\mathbb{R}[x] \to \mathbb{C}.$$

*Now we may still send $x$ to, for example, $0$ or $1$, but we can also send $x$ to $i$ or $-i$. The two choices*

$$x \mapsto i \qquad \text{and} \qquad x \mapsto -i$$

*give two $\mathbb{C}$-valued points, but both lie over the same point $(x^2+1) \in \operatorname{Spec}\mathbb{R}[x]$. Its residue field is $\mathbb{C}$, which has degree 2 over the base field $\mathbb{R}$. The two $\mathbb{C}$-valued points are related by complex conjugation, i.e. by the nontrivial element of* $\operatorname{Gal}(\mathbb{C}/\mathbb{R})$.

**Example 16.** $X = \mathbb{A}^2_{\mathbb{R}} = \operatorname{Spec}\mathbb{R}[x,y]$
$Z = \mathbb{A}^1_{\mathbb{R}} = \operatorname{Spec}\mathbb{R}[t]$

$$X(Z) = \operatorname{Hom}_{\mathbb{R}\text{-}alg}(\mathbb{R}[x,y], \mathbb{R}[t])$$

*We can send e.g. $x \mapsto 3, y \mapsto 4$ and get a point, or*
*$x \mapsto t, y \mapsto t$ and get a line, or*
*$x \mapsto t, y \mapsto t^2$ and get a parametrized parabola.*
*Thus*

$$X(Z) = \{\text{polynomially parametrized maps } \mathbb{A}^1_{\mathbb{R}} \to \mathbb{A}^2_{\mathbb{R}}\}.$$

**Example 17.** $X = \mathbb{A}^1_{\mathbb{R}} = \operatorname{Spec} \mathbb{R}[x]$
$Z = \operatorname{Spec} \mathbb{R}(x)$
$X(Z) = \operatorname{Hom}_{\mathbb{R}\text{-}alg}(\mathbb{R}[X], \mathbb{R}(x))$
*Distinguished element:* $x \mapsto x$
*Generic point before was* $[(0)]$. *The residue field of* $X$ *at the generic point is*

$$\mathbb{R}[x]_{(0)}/(0) = \mathbb{R}(x)/(0) \cong \mathbb{R}(x)$$

*so the generic point is an* $\mathbb{R}(x)$*-valued point.*

**Example 18.** *If* $R$ *is a local ring, if* $M$ *is the maximal ideal of* $R$*, then if* $f : \operatorname{Spec}(R) \to X$ *is a morphism, then any open set on* $X$ *containing* $f([M])$ *contains the whole image of* $f$*. So the image always lies in one affine piece of* $X$*.*

*This is because if* $M$ *is maximal, then if* $D(g) \supseteq M$*,* $g$ *is a unit so the preimage is everything.*

**Proposition 19.** *Let* $x \in X$*. Then there is a bijection between the set of* $R$*-valued points of* $X$ *such that* $f([M]) = x$ *and the set of local homomorphisms* $q : \mathcal{O}_{X,x} \to R$*.*

**Definition 20.** *For* $X$ *and* $K$ *preschemes, if* $f, g$ *are two* $K$*-valued points of* $X$*, then we say* $f(x) \equiv g(x)$ *if* $f \cdot i_x = g \cdot i_x$ *where* $i_x : \operatorname{Spec}(\kappa(x)) \to K$ *is the canonical morphism. Equivalently, this means that* $f(x) = g(x)$*, and that the two maps* $f_x^*, g_x^* : \kappa(f(x)) \to \kappa(x)$ *are equal.*

If $K$ is reduced, then $f = g \iff f(x) \equiv g(x)$ for all $x \in K$.

**Proposition 21.** *For all* $f, g : K \to X$*,* $\{x \in K \mid f(x) \equiv g(x)\}$ *is locally closed.*

**Definition 22.** *A prescheme* $X$ *is a* ***scheme*** *if for all preschemes* $K$ *and all* $K$*-valued points* $f, g$ *of* $X$*,* $\{x \in K \mid f(x) \equiv g(x)\}$ *is closed.*

**Proposition 23.** *If* $X$ *is a prescheme over a ring* $R$*, then the criterion for* $X$ *to be a scheme is satisfied for all* $K, f, g$ *if it is satisfied in the case:*

$$K = X \times_{\operatorname{Spec}(R)} X, \quad f = p_1, \quad g = p_2.$$

In modern language, Prop 21 says $\Delta : X \to X \times_R X$ is always an immersion, and Prop 23 says $X/R$ is separated $\iff$ $\Delta$ is a closed immersion with image exactly $\{p_1 \equiv p_2\}$.

**Corollary 24.** *If* $k$ *is an algebraically closed field, then a prevariety over* $k$ *is a variety in the sense of Ch. I* $\iff$ *it is a scheme.*

**Proposition 25.** *If* $X$ *is a scheme and* $U, V \subseteq X$ *are open affine sets, then* $U \cap V$ *is affine and the canonical homomorphism*

$$\Gamma(U, \mathcal{O}_X) \otimes \Gamma(V, \mathcal{O}_X) \to \Gamma(U \cap V, \mathcal{O}_X)$$

*is surjective.*

**Example 26.** *Let $R$ be a ring. The* ***functor of points*** *of a scheme $X$ is*

$$R \longmapsto X(R), \qquad X(R) := \mathrm{Hom}_{\mathrm{Sch}}(\mathrm{Spec}\, R, X).$$

*If $X = \mathrm{Spec}\, A$ is affine, then*

$$X(R) \cong \mathrm{Hom}_{\mathrm{Ring}}(A, R).$$

*Let's find the functor of points in some examples:*

1. $\mathbb{A}^n_{\mathbb{Z}} = \mathrm{Spec}(\mathbb{Z}[x_1, \ldots, x_n])$.

   *A morphism*

   $$\mathrm{Spec}\, R \to \mathbb{A}^n_{\mathbb{Z}}$$

   *corresponds to a ring homomorphism in the other direction*

   $$\mathbb{Z}[x_1, \ldots, x_n] \to R.$$

   *Such a homomorphism is completely determined by where we send $x_1, \ldots, x_n$, and each $x_i$ can be sent to any element of $R$. Therefore*

   $$\mathrm{Hom}(\mathbb{Z}[x_1, \ldots, x_n], R) \cong R^n,$$

   *so*

   $$\boxed{\mathbb{A}^n_{\mathbb{Z}}(R) = R^n.}$$

   *In particular,*

   $$\mathbb{G}_a = \mathbb{A}^1_{\mathbb{Z}}$$

   *represents the additive group, since*

   $$\mathbb{G}_a(R) = R$$

   *with its additive group structure.*

2. $\mathbb{G}_m = \mathrm{Spec}\, \mathbb{Z}[t, t^{-1}]$.

   *A ring homomorphism*

   $$\mathbb{Z}[t, t^{-1}] \to R$$

   *must send $t$ to an invertible element of $R$, since $t$ has an inverse. Conversely, any $r \in R^\times$ determines such a homomorphism by $t \mapsto r$. Thus*

   $$\mathrm{Hom}(\mathbb{Z}[t, t^{-1}], R) \cong R^\times,$$

   *so*

   $$\boxed{\mathbb{G}_m(R) = R^\times.}$$

   *Thus $\mathbb{G}_m$ represents the multiplicative group.*

3. $\mu_n = \operatorname{Spec}(\mathbb{Z}[t]/(t^n - 1))$.

*A ring homomorphism*

$$\mathbb{Z}[t]/(t^n - 1) \to R$$

*is determined by the image* $r$ *of* $t$*, but the relation* $t^n = 1$ *forces*

$$r^n = 1.$$

*Therefore*

$$\boxed{\mu_n(R) = \{r \in R \mid r^n = 1\}.}$$

*Thus* $\mu_n$ *represents the group of* $n$*th roots of unity.*

4. $X = \operatorname{Spec}(\mathbb{Z}[x]/(x^2))$.

*A ring homomorphism*

$$\mathbb{Z}[x]/(x^2) \to R$$

*is determined by the image* $r$ *of* $x$*, and the relation* $x^2 = 0$ *forces* $r^2 = 0$*. Hence*

$$\boxed{X(R) = \{r \in R \mid r^2 = 0\}.}$$

*Thus the functor of points detects nilpotent elements of* $R$*.*

5.

$$\mathrm{GL}_2 = \operatorname{Spec}(\mathbb{Z}[a, b, c, d, t]/((ad - bc)t - 1)).$$

*An* $R$*-point corresponds to elements*

$$a, b, c, d, t \in R$$

*satisfying*

$$(ad - bc)t = 1.$$

*Thus* $ad - bc$ *must be invertible in* $R$*, and* $t$ *is then its inverse. Therefore an* $R$*-point is exactly a matrix*

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

*with invertible determinant. Hence*

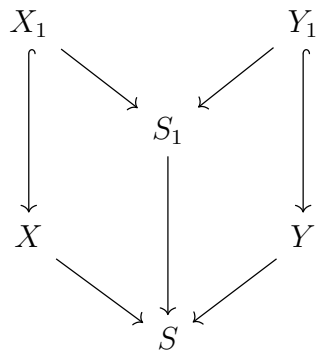
$$\boxed{\mathrm{GL}_2(R) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(R) \;\middle|\; ad - bc \in R^\times \right\} = \mathrm{GL}_2(R).}$$

*Thus the scheme* $\mathrm{GL}_2$ *represents the functor sending a ring* $R$ *to the group of invertible* $2 \times 2$ *matrices over* $R$*.*

**Problem 2.4.** *1.* ***Check that a fiber product of schemes is a scheme.***

*2.* ***Prove that*** $\mathbb{P}^n_{\mathbb{Z}}$ ***is a scheme***

*3.* ***Given a diagram of schemes***




***where*** $X_1, Y_1$ ***are closed subschemes of*** $X$ ***and*** $Y$***, show that*** $X_1 \times_{S_1} Y_1$ ***is a closed subscheme of*** $X \times_S Y$ ***(Hint: You must use the fact that*** $S_1$ ***is a scheme)***

*1) Let*

$$X \longrightarrow S, \qquad Y \longrightarrow S$$

*be morphisms of schemes, and let*

$$P = X \times_S Y.$$

*We already know that the fiber product exists as a prescheme. We must check that* $P$ *is a scheme. Let* $K$ *be any prescheme, and let*

$$f, g : K \longrightarrow P$$

*be two morphisms. Write*

$$p_X : P \longrightarrow X, \qquad p_Y : P \longrightarrow Y$$

*for the two projections. Consider the set*

$$E = \{z \in K \mid f(z) \equiv g(z)\}.$$

*By the universal property of the fiber product, two maps to* $P$ *agree at* $z$ *if and only if their compositions with both projections agree at* $z$*. Hence*

$$E = \{z \in K \mid p_X f(z) \equiv p_X g(z)\} \cap \{z \in K \mid p_Y f(z) \equiv p_Y g(z)\}.$$

*Since* $X$ *and* $Y$ *are schemes, both sets on the right are closed in* $K$*. Therefore* $E$ *is closed.*

*Thus $P = X \times_S Y$ satisfies the defining condition for a scheme, and hence*

$$\boxed{X \times_S Y \text{ is a scheme.}}$$

*2) Let*

$$P = \mathbb{P}^n_{\mathbb{Z}}.$$

*By the criterion for a prescheme to be a scheme, it is enough to show that the diagonal is closed in*

$$P \times_{\operatorname{Spec}\mathbb{Z}} P.$$

*Use homogeneous coordinates*

$$[X_0 : \cdots : X_n]$$

*on the first factor and*

$$[Y_0 : \cdots : Y_n]$$

*on the second factor. Consider the equations*

$$X_i Y_j - X_j Y_i = 0, \qquad 0 \le i, j \le n.$$

*These equations define a closed subprescheme of*

$$P \times_{\operatorname{Spec}\mathbb{Z}} P.$$

*Indeed, on a standard affine open*

$$U_r \times U_s, \qquad U_r = \{X_r \neq 0\}, \quad U_s = \{Y_s \neq 0\},$$

*they become the ordinary polynomial equations*

$$\frac{X_i}{X_r}\frac{Y_j}{Y_s} - \frac{X_j}{X_r}\frac{Y_i}{Y_s} = 0.$$

*Thus their zero sets are closed on every standard affine piece and agree on the overlaps.*

*We claim that the closed subprescheme defined by these equations is the diagonal. To see this, let $z$ be a point of*

$$P \times_{\operatorname{Spec}\mathbb{Z}} P$$

*and let $\kappa(z)$ be its residue field. Every point $z$ of a prescheme comes with a morphism $\operatorname{Spec}\kappa(z) \to Z = P \times_{\operatorname{Spec}\mathbb{Z}} P$. So we can compose the morphisms with the projections $p_1, p_2 : Z \to \mathbb{P}^n_{\mathbb{Z}}$, and thus the two projections give two morphisms $\operatorname{Spec}\kappa(z) \to \mathbb{P}^n_{\mathbb{Z}}$, i.e. two $\kappa(z)$-valued points*

$$[x_0 : \cdots : x_n], \qquad [y_0 : \cdots : y_n]$$

*of projective space. They represent the same projective point if and only if the two vectors are proportional, which is equivalent to*

$$x_i y_j - x_j y_i = 0 \qquad \text{for every } i, j.$$

*Hence the locus on which the two projections agree is precisely*

$$V(X_iY_j - X_jY_i \mid 0 \leq i, j \leq n),$$

*which is closed.*

*Therefore the diagonal of* $\mathbb{P}^n_{\mathbb{Z}}$ *is closed, and so*

$$\boxed{\mathbb{P}^n_{\mathbb{Z}} \text{ is a scheme.}}$$

*3) We will do this in two steps.*
*Step 1:* $X_1 \times_S Y_1$ *is a closed subscheme of* $X \times_S Y$*.*
*Since*

$$X_1 \hookrightarrow X$$

*is a closed immersion, base-changing it by* $Y \to S$ *gives a closed immersion*

$$X_1 \times_S Y \hookrightarrow X \times_S Y.$$

*Similarly, since*

$$Y_1 \hookrightarrow Y$$

*is a closed immersion, base-changing it by* $X_1 \to S$ *gives a closed immersion*

$$X_1 \times_S Y_1 \hookrightarrow X_1 \times_S Y.$$

*Thus the composite*

$$X_1 \times_S Y_1 \hookrightarrow X_1 \times_S Y \hookrightarrow X \times_S Y$$

*is a closed immersion. Therefore*

$$X_1 \times_S Y_1$$

*is already a closed subscheme of* $X \times_S Y$*.*

*So it remains only to see that*

$$X_1 \times_{S_1} Y_1$$

*is a closed subscheme of*

$$X_1 \times_S Y_1.$$

*Step 2: Impose the condition that the two maps to* $S_1$ *agree.*
*Set*

$$Z = X_1 \times_S Y_1.$$

*A point of* $Z$ *consists, roughly speaking, of a point of* $X_1$ *and a point of* $Y_1$ *which have the same image in* $S$*.*

*There are two natural maps*

$$a, b : Z \longrightarrow S_1.$$

*The first is obtained by projecting to $X_1$ and then using $X_1 \to S_1$:*

$$a : Z \longrightarrow X_1 \longrightarrow S_1.$$

*The second is obtained by projecting to $Y_1$ and then using $Y_1 \to S_1$:*

$$b : Z \longrightarrow Y_1 \longrightarrow S_1.$$

*Because $Z = X_1 \times_S Y_1$, the two maps $a$ and $b$ have the same image after composing with $S_1 \to S$. Hence together they define a map*

$$(a, b) : Z \longrightarrow S_1 \times_S S_1.$$

*Now,*

$$X_1 \times_{S_1} Y_1$$

*is exactly the part of $Z$ on which the two maps $a$ and $b$ are equal. In other words, it is the inverse image of the diagonal*

$$\Delta_{S_1/S} : S_1 \longrightarrow S_1 \times_S S_1.$$

*Thus we have a cartesian square*

$$\begin{array}{ccc} X_1 \times_{S_1} Y_1 & \longrightarrow & S_1 \\ \downarrow & & \downarrow{\scriptstyle \Delta_{S_1/S}} \\ X_1 \times_S Y_1 & \xrightarrow{(a,b)} & S_1 \times_S S_1. \end{array}$$

*This is the point where we use the hypothesis that $S_1$ is a* scheme. *Since $S_1$ is a scheme, its diagonal is closed. Therefore*

$$\Delta_{S_1/S} : S_1 \longrightarrow S_1 \times_S S_1$$

*is a closed immersion.*

*Since closed immersions remain closed immersions after base change, the left vertical map in the cartesian square is also a closed immersion. Hence*

$$X_1 \times_{S_1} Y_1 \hookrightarrow X_1 \times_S Y_1$$

*is a closed immersion.*

*Finally, we already know from Step 1 that*

$$X_1 \times_S Y_1 \hookrightarrow X \times_S Y$$

*is a closed immersion. Composing the two closed immersions gives*

$$X_1 \times_{S_1} Y_1 \hookrightarrow X \times_S Y.$$

*Therefore*

$$\boxed{X_1 \times_{S_1} Y_1 \text{ is a closed subscheme of } X \times_S Y.}$$